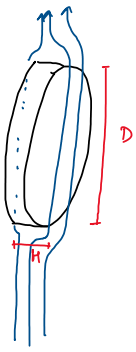
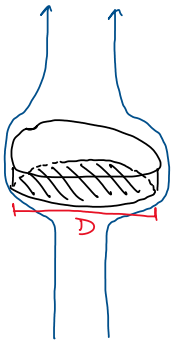
Drag coefficient

Monday, 28 August 2023 10:11 AM



Characteristic Area: $H \cdot D$

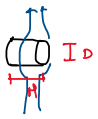


Characteristic Area: $\pi \frac{D^2}{4}$

If the disk was very small:
w/ a much large H
(i.e. a cylinder)

↓

Then characteristic area of circumference would make the most sense



Characteristic area: $\pi D \cdot H$

Either way, providing a C_D w/ reference area taken for calculated is the convention

Week 3

Monday, September 4, 2023 4:08 PM

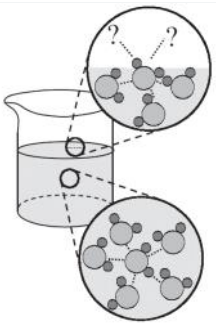
Recitation 09/05 draft

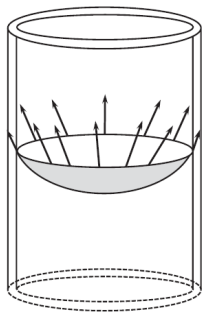
Monday, September 4, 2023 3:47 PM

Concept Review

Surface Tension

- Now we are dealing with interactions b/w two fluid at their interface (e.g. bubbles)
- Phase boundary = "free surface" of the flow
- Surface tension is a fluid property (like density and viscosity; denoted by σ $[\frac{N}{m}]$ and is a result of unbalanced molecular forces at a surface
- Force is tangent to the surface and normal to the boundary

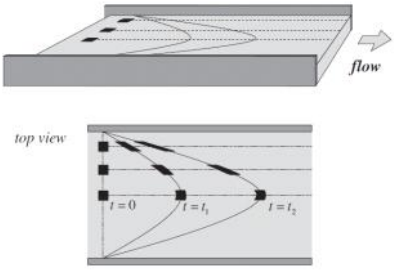




the downward force on surface particles is not balanced by an upward force of fluid on the other side of the interface, the net effect is that the *free surface behaves like a thin massless film under tension.*

Reynolds Transport Theorem

- We could apply balance laws on each fluid element (box) --> but have to deal with awkward shapes
- We can define a control volume - an arbitrary shape region that is fixed within the flow and sees fluid coming and going out



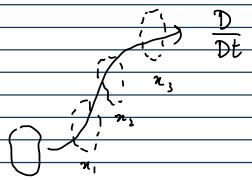
Extensive property : Depends on the amount of matter in a sample Ψ

Intensive property : Independent of the size of sample $\psi = \frac{\partial \Psi}{\partial m}$ ← mass

Total material Volume corresponding to a mass w/ the property Ψ is written as: $\Psi = \iiint_{V(t)} \rho \cdot \psi \, dV$ (arbitrary material volume)

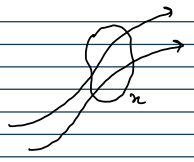
Frames of Reference :

Material Volume vs. Control volume



+ Flux of a property across a material volume is 0
  - i.e. Volume whose boundaries are moving w/ time

+ Rate of change of a fluid property given by: $\left[\frac{D}{Dt}\right]$



+ Fixed x-position; there is a non-zero flux

+ Rate of change of a fluid property given by: $\left[\frac{\partial}{\partial t}\right]$

$$\boxed{\frac{D}{Dt}[FP] = \frac{\partial}{\partial t}[FP] + v \cdot \nabla(FP)}$$

FP = Fluid property
v = velocity of fluid

We know: $\Psi = \iiint_{V(t)} \rho \Psi \, dV$ → This is w/ respect to material frame of reference

- However, the volume is changing so we cannot evaluate (as we don't know limits of integration)
- But we know we can go from material to control frame of reference via the green equation

$$\frac{D\Psi}{Dt} = \frac{\partial}{\partial t}\Psi + v \cdot \nabla\Psi$$

$$= \frac{\partial}{\partial t}\left[\iiint_{V_0}\right.$$

Momentum Balance

Momentum (p) = Mass (m) · Velocity (V)

$$\underline{p} = m \cdot \underline{v}$$

Mass = (Density) · Volume

$$(m) = \rho \cdot \int V$$

$$\underline{p} = (\rho \cdot V) \cdot \underline{v}$$

m^3 term

$m^2 \cdot m$

Rate of Momentum Change: Momentum per unit time

"dividing by time"

$$\underline{p} = (\rho \cdot \text{Area} \cdot l) \cdot \underline{v}$$

$$\approx \frac{\underline{p}}{\Delta t} = \rho \cdot \text{Area} \cdot \frac{l}{\Delta t} \cdot v$$

($\frac{l}{\Delta t} \approx v$)

That is why momentum balance has two velocity terms!

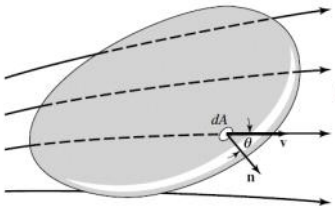
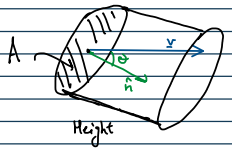


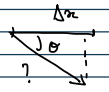
Figure 5.2 Fluid flow through a control volume.

We care about fluid velocity in the direction of the normal to the surface



→ Volume of fluid passed through cross-section A:

$V_{cylinder} = A \cdot \text{Height}$ ← unknown



$\frac{?}{\Delta x} = \cos\theta$

$? = \Delta x \cos\theta$

$\therefore V_{cylinder} = A \cdot \Delta x \cdot \cos\theta$

$\Delta V = A \cdot \Delta x \cos\theta$

$\frac{\Delta V}{\Delta t} = A \cdot \frac{\Delta x}{\Delta t} \cos\theta = A \cdot \underline{v} \cdot \cos\theta$

$\frac{\Delta x}{\Delta t} = \underline{v}$

$\hat{n} \cdot \underline{v} = |\hat{n}|\,|\underline{v}| \cos\theta = v\cos\theta$ (since $|\hat{n}| = 1$)

Volumetric flow rate through A : $Q = v \cdot \cos\theta \, A = (\hat{n} \cdot \underline{v}) A$

↓

Mass flow = Density · Volumetric flow rate

$m = \rho \cdot Q$

$m = \rho \cdot (\hat{n} \cdot \underline{v}) A$

Momentum flow of liquid through A $\Rightarrow \underline{P} = m \cdot \underline{v}$

$= \rho (\hat{n} \cdot \underline{v}) A \, \underline{v}$

↳ Now we chose arbitrary area → Taking a differential area & integrating momentum quantity over the entire control surface

Net momentum change from CV : $\iint_S \underline{v}\, \rho (\hat{n} \cdot \underline{v})\, dA$

$\Rightarrow$ (In − out) = change

Recitation 09/05 Master

Monday, September 4, 2023 3:47 PM

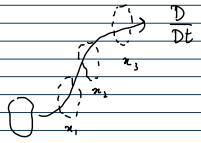
Concept Review

Reynolds Transport Theorem

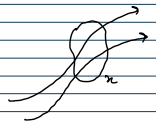
- Has different forms depending on the frame of reference we are in

Frames of Reference:

Material Volume vs. Control volume



+ Flux of a property across a material volume is 0
  - i.e. Volume whose boundaries are moving w/ time
+ Rate of change of a fluid property given by: $\left[\frac{D}{Dt}\right]$



+ Fixed x-position; there is a non-zero flux
+ Rate of change of a fluid property given by: $\left[\frac{\partial}{\partial t}\right]$

$$\frac{D}{Dt}[FP] = \frac{\partial}{\partial t}[FP] + v \cdot \nabla(FP)$$

FP = Fluid property
v = velocity of fluid

Material Coordinates

$$\frac{D\Psi}{Dt} = \int_{V_a(t)} \frac{\partial \psi}{\partial t} \rho dV + \int_{A_a(t)} \psi \rho \underline{v} \cdot \hat{n} dA_a$$

Control Volume coordinates

Reynolds transport theorem (momentum balance on CV)

$$\sum_{\text{on CV}} \underline{f} = \frac{d\mathbf{P}}{dt} + \iint_{CS} (\hat{n} \cdot \underline{v})\, \rho \underline{v}\, dS \qquad (3.137)$$

↳ General form (Can be applied to a particular fluid property such as mass, energy, momentum)

Momentum Balance

Momentum (p) = Mass (m) · Velocity (V)

$\underline{p} = m \cdot \underline{v}$

Mass = (Density) · Volume

$m = \rho \cdot \forall$

$\underline{p} = (\rho \cdot \forall) \cdot \underline{v}$

m^3 term

$m^2 \cdot m$

Rate of Momentum Change: Momentum per unit time

"dividing by time" ↝ $\underline{p} = (\rho \cdot \text{Area} \cdot \ell) \cdot \underline{v}$

$\underline{p} = \rho \cdot \text{Area} \cdot \frac{\ell}{t} \cdot v$

"dividing by time" ↳ $\underline{F} = (p \cdot \text{Area} \cdot l) \cdot \underline{v}$

$\approx \frac{\underline{F}}{\Delta t} = p \cdot \text{Area} \cdot \frac{l}{\Delta t} \cdot \underline{v}$ ($\frac{\Delta t}{v}$)

That is why momentum balance has two velocity terms!

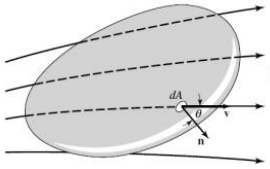
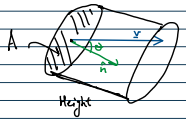


Figure 5.2 Fluid flow through a control volume.

We care about fluid velocity in the direction of the normal to the surface



→ Volume of fluid passed through cross-section A:

$V_{cylinder} = A \cdot \text{Height}$ (unknown)

$\frac{?}{\Delta x} = \cos\theta$

$? = \Delta x \cos\theta$

$\therefore V_{cylinder} = A \cdot \Delta x \cdot \cos\theta$

$\Delta V = A \cdot \Delta x \cos\theta$

$\frac{\Delta V}{\Delta t} = A \cdot \frac{\Delta x}{\Delta t} \cos\theta = A \cdot \underline{v} \cdot \cos\theta$ ($\frac{\Delta x}{\Delta t} = \underline{v}$)

$\hat{n} \cdot \underline{v} = |\hat{n}|\,|\underline{v}| \cos\theta = v\cos\theta$ ($|\hat{n}| = 1$)

Volumetric flow rate through A: $Q = v \cdot \cos\theta \, A = (\hat{n} \cdot \underline{v}) A$

Mass flow = Density · Volumetric flow rate

$m = \rho \cdot Q$

$m = \rho \cdot (\hat{n} \cdot \underline{v}) A$

Momentum flow of liquid through A $\Rightarrow \underline{F} = m \cdot \underline{v}$

$= \rho (\hat{n} \cdot \underline{v}) A \, \underline{v}$

↳ Now we choose arbitrary area → Taking a differential area & integrating momentum quantity over the entire control surface

Net momentum change from CV: $\iint_S \underline{v}\, \rho (\hat{n} \cdot \underline{v})\, dA$ $\Rightarrow$ (In − out) = change

Problems

PROBLEM: 3.15 What is the flow of momentum through the hemispherical surface described in Problem 14?

PROBLEM: 3.14 A uniform flow $\underline{v} = U\hat{e}_z$ of an incompressible fluid of density ρ passes through a volume that is in the shape of a half sphere of radius R. The outwardly pointing unit normal of the flat surface of the half sphere is $\hat{n} = -\hat{e}_z$. What is the mass flow rate through the hemispherical surface of this volume? Show that you can obtain the correct answer by integrating the formal expression for Q (Equation 3.87).

PROBLEM: 3.21 The y-component of a velocity field in flow through a slit (Cartesian coordinates) is given here. What is the average value of the velocity? $2H$ is the gap between the plates. At what location is the velocity a maximum? The units on velocity and A are m/s and the units of x and H are m.

$$\underline{v} = \begin{pmatrix} 0 \\ v_y(x) \\ 0 \end{pmatrix}_{xyz}$$

$$v_y(x) = A\left(1 - \frac{(x-H)^2}{H^2}\right)$$

PROBLEM: 3.31 For the Cartesian coordinate system shown in Figure 3.38, what is a unit vector in the direction of gravity? What is the component of gravity in the flow direction?

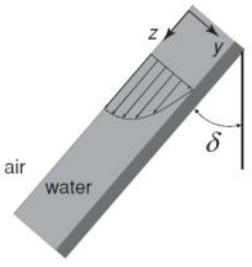


Figure 3.38 Flow coordinate system for Problem 31.

Figure 3.38 Flow coordinate system for Problem 31.

Recitation 09/05

Monday, September 4, 2023 3:47 PM

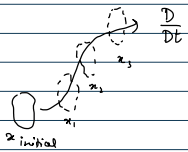
Concept Review

Reynolds Transport Theorem

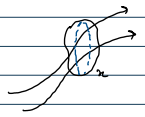
- Has different forms depending on the frame of reference we are in

Frames of Reference:

Material Volume vs. Control volume



+ Flux of a property across a material volume is 0
  - ↳ i.e. Volume whose boundaries are moving w/ time
+ Rate of change of a fluid property given by: $\left[\frac{D}{Dt}\right]$



+ Fixed x-position; there is a non-zero flux
+ Rate of change of a fluid property given by: $\left[\frac{\partial}{\partial t}\right]$

$$\frac{D}{Dt}[FP] = \frac{\partial}{\partial t}[FP] + v \cdot \nabla(FP)$$

FP = Fluid property
v = velocity of fluid

$= 0$

Extensive var: Ψ ≠ Intensive var: ψ (Independent of mass)

$\Psi = \frac{\partial \psi}{\partial m}$

Material Coordinates

$$\frac{D\Psi}{Dt} = \int_{V_a(t)} \frac{\partial \psi}{\partial t} \rho dV + \int_{A_a(t)} \psi \rho \underline{v} \cdot \hat{n} dA_a$$

(ρ: density)

↳ General form (Can be applied to a particular fluid property such as mass, energy, momentum)

Reynolds transport theorem (momentum balance on CV)

Control Volume coordinates

$$\sum_{\text{on CV}} \underline{f} = \frac{d\mathbf{P}}{dt} + \iint_{CS} (\hat{n} \cdot \underline{v}) \rho \underline{v} \, dS \qquad (3.137)$$

(P: momentum; f: force; CS: Control surface)

Momentum Balance

Momentum (p) = Mass (m) · velocity (V)

$p = m \cdot \underline{v}$

Mass = (Density) · Volume

$m = \rho \cdot V$

$p = (\rho \cdot V) \cdot \underline{v}$ (density, volume, velocity)

m^3 term

$m^2 \cdot m \rightarrow$ Area · length

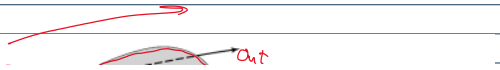
Rate of Momentum Change: Momentum per unit time

"dividing by time"

$\underline{f} = (\rho \cdot \text{Area} \cdot l) \cdot \underline{v}$

$\approx \frac{\underline{f}}{\Delta t} = \rho \cdot \text{Area} \cdot \frac{l}{\Delta t} \cdot v$ ($\frac{l}{\Delta t} = v$)

That is why momentum balance has two velocity terms!



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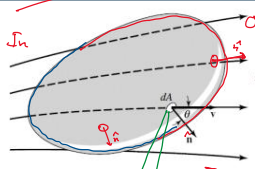
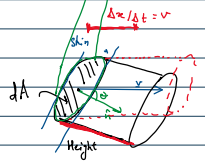


Figure 5.2 Fluid flow through a control volume.

In − out (eqn) = 0

We care about fluid velocity in the direction of the normal to the surface



→ Volume of fluid passed through cross-section A:

$V_{cylinder} = A \cdot \text{Height}$ (unknown)

SOH CAH TOA

$\frac{?}{\Delta x} = \cos\theta$

$? = \Delta x \cos\theta$

$\therefore V_{cylinder} = A \cdot \Delta x \cos\theta$ → Change in Volume of cylinder

$\Delta V = A \cdot \Delta x \cos\theta$

$\frac{\Delta V}{\Delta t} = A \cdot \frac{\Delta x}{\Delta t} \cos\theta = A \cdot \underline{v} \cdot \cos\theta$ ($\frac{\Delta x}{\Delta t} = \underline{v}$)

$V = \bar{V}$

$\underline{v} = \bar{v} = v$

$a \cdot b = |a||b| \cos\theta$

$\hat{n} \cdot \underline{v} = |\hat{n}||\underline{v}| \cos\theta = v\cos\theta$

Volumetric flow rate through A: $\frac{\Delta V}{\Delta t} = Q = v \cdot \cos\theta \, A = (\hat{n} \cdot \underline{v}) A$

Mass flow = Density · Volumetric flow rate

$m = \rho \cdot Q$

$m = \rho \cdot (\hat{n} \cdot \underline{v}) dA$

Momentum flow of liquid through A $\Rightarrow$ $d\underline{p} = m\underline{v}$

$d\underline{p} = \rho (\hat{n} \cdot \underline{v}) A \, \underline{v}$

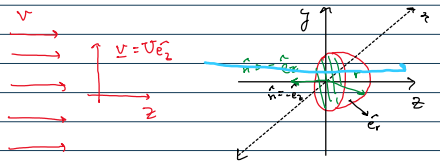
↳ Now we chose arbitrary area → Taking a differential area & integrating momentum quantity over the entire control surface

Net momentum change from CV: $\iint_S \underline{v} \rho (\hat{n} \cdot \underline{v}) \, dA$ $\Rightarrow$ (In − out) = change

Efflux [In − out] | Surface | Area

Problems

PROBLEM: 3.15 What is the flow of momentum through the hemispherical surface described in Problem 14?



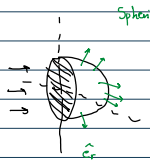
PROBLEM: 3.14 A uniform flow $\underline{v} = V\hat{e}_z$ of an incompressible fluid of density ρ passes through a volume that is in the shape of a half sphere of radius R. The outwardly pointing unit normal of the flat surface of the half sphere is $\hat{n} = -\hat{e}_z$. What is the mass flow rate through the hemispherical surface of this volume? Show that you can obtain the correct answer by integrating the formal expression for Q (Equation 3.87).

$\hat{e}_z$ Acceleration

Mass balance

Spherical coordinates

Momentum change / flow rate $= \iint_A \rho \underline{v} (\hat{n} \cdot \underline{v}) \, dS$ (uniform, constant)



$\frac{dp}{dt} = \rho \underline{v} \iint_A (\hat{n} \cdot \underline{v}) \, dS$

$(\hat{n} \cdot \underline{v})$

$\hat{n} = \hat{e}_r$

$\therefore (\hat{e}_r \cdot \underline{v})$ $V\hat{e}_z$

$(\hat{e}_r \cdot V\hat{e}_z)$

(Unit basis commutes) representation of

$\hat{n} = \hat{e}_r$, $v\hat{e}_z$

$\therefore (\hat{e}_r \cdot \underline{v})$

$(\hat{e}_r \cdot v\hat{e}_z)$

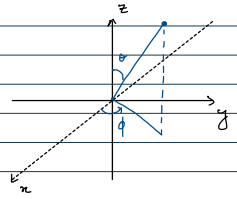
(Unit basis conversion) representation of spherical in cartesian

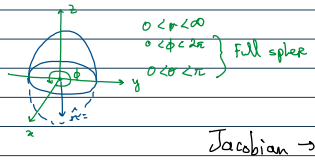
$\hat{e}_r = \sin\theta\cos\phi\, e_x + \sin\theta\sin\phi\, \hat{e}_y + \cos\theta\, \hat{e}_z$

$\begin{pmatrix} \sin\theta\cos\phi \\ \sin\theta\sin\phi \\ \cos\theta \end{pmatrix}_{xyz} \cdot \begin{pmatrix} 0 \\ 0 \\ v \end{pmatrix}_{xyz} = v\cos\theta$

$\frac{dp}{dt} = \rho \underline{v} \iint_A (\hat{n}\cdot\underline{v})\, dS$, $\hat{n}\cdot\underline{v} = v\cos\theta$

$= \rho \underline{v} \iint_A v\cos\theta\, dS$; $\underline{v}$ constant





$0 < r < \infty$
$0 < \phi < 2\pi$
$0 < \theta < \pi$
} Full sphere

Jacobian → $R^2 \sin\theta\, d\theta\, d\phi$

momentum / density

$\frac{\Delta p}{\Delta t} = \frac{dp}{dt} = \rho v \int_0^{2\pi}\int_0^{\pi/2} (\hat{e}_r \cdot \underline{v})\, dS$, $d\theta\, d\phi$, $v\cos\theta$

$= \rho v \int_0^{2\pi}\int_0^{\pi/2} v\cos\theta\,(R^2\sin\theta)\, d\theta\, d\phi$ } Solve this !

PROBLEM: 3.21 The y-component of a velocity field in flow through a slit (Cartesian coordinates) is given here. What is the average value of the velocity? $2H$ is the gap between the plates. At what location is the velocity a maximum? The units on velocity and A are m/s and the units of x and H are m.

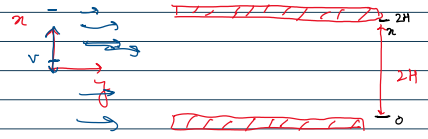
$\underline{v} = \begin{pmatrix} 0 \\ v_y(x) \\ 0 \end{pmatrix}_{xyz}$, $\begin{pmatrix} v_x(x,y,z) \\ 0 \\ 0 \end{pmatrix}$

$v_y(x) = A\left(1 - \frac{(x-H)^2}{H^2}\right)$

$v_y(x)$ — dependence — flow

Average of $f(z) = \langle f \rangle = \frac{\int_{z_{min}}^{z_{max}} f(z)\, dz}{(z_{max} - z_{min})}$ (3.41)

Total / # ; ⟨Ashu⟩



$\langle v_y \rangle = \frac{\int_0^{2H} v_y(x)\cdot dx}{2H - 0} = \frac{1}{2H}\int_0^{2H} A\left(1 - \frac{(x-H)^2}{H^2}\right) dx$

$\langle v_y \rangle = \frac{A}{2H}\int_0^{2H}\left(1 - \frac{(x-H)^2}{H^2}\right) dx$, A, H

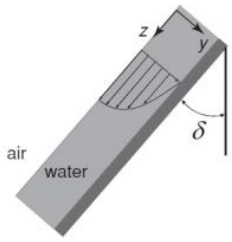
$\boxed{\langle v_y \rangle = \frac{2}{3}A}$

$\frac{d v_y}{dx} = \frac{d}{dx}\left(A\left(1 - \frac{(x-H)^2}{H^2}\right)\right) = 0$

$x = H$, $\frac{d v_y}{dx} = 0$

$\boxed{v_y(x=H) = A}$

PROBLEM: 3.31 For the Cartesian coordinate system shown in Figure 3.38, what is a unit vector in the direction of gravity? What is the component of gravity in the flow direction?



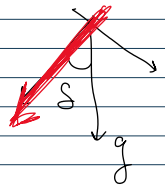


Figure 3.38 Flow coordinate system for Problem 31.

Recitation 09/07

Monday, September 4, 2023 3:47 PM

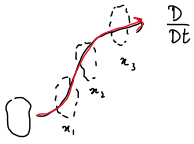
Concept Review

Reynolds Transport Theorem

- Has different forms depending on the frame of reference we are in

Frames of Reference:

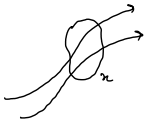
Material Volume → Experiments



+ Flux of a property across a material volume is 0
  - i.e. Volume whose boundaries are moving w/ time
+ Rate of change of a fluid property given by: $\left[\frac{D}{Dt}\right]$

vs.

Control volume



+ Fixed x-position; there is a non-zero flux
+ Rate of change of a fluid property given by: $\left[\frac{\partial}{\partial t}\right]$

$$\frac{D}{Dt}[FP] = \frac{\partial}{\partial t}[FP] + v \cdot \nabla(FP)$$

FP = Fluid property
v = velocity of fluid

$\Psi \neq \psi$

- Ψ: Extensive
- ψ: Intensive → Independent of mass or size

Material Coordinates

Ψ →

$$\frac{D\Psi}{Dt} = \int_{V_a(t)} \frac{\partial \psi}{\partial t} \rho dV + \int_{A_a(t)} \psi \rho \underline{v} \cdot \hat{n} dA_a$$

↳ General form (Can be applied to a particular fluid property such as mass, energy, momentum)

via Newton's second law

Control Volume coordinates

Reynolds transport theorem (momentum balance on CV)

$$\sum_{\text{on CV}} \underline{f} = \frac{d\mathbf{P}}{dt} + \iint_{CS} (\hat{n} \cdot \underline{v}) \rho \underline{v} \, dS \qquad (3.137)$$

- $\sum \underline{f}$: forces on CV
- $\frac{d\mathbf{P}}{dt}$: momentum
- $\iint_{CS}$: In - out

Momentum Balance

Momentum (p) = Mass (m) · velocity ($\underline{v}$)

$\underline{p} = m \cdot \underline{v}$

Mass (m) = (Density) · Volume = $\rho \cdot V$

$\underline{p} = (\rho \cdot V) \cdot \underline{v}$

m^3 term

$m^2 \cdot m$

Rate of Momentum Change: Momentum per unit time

"dividing by time" ↳

$\underline{p} = (\rho \cdot \text{Area} \cdot l) \cdot \underline{v}$

$\approx \frac{\Delta \underline{p}}{\Delta t} = \rho \cdot \text{Area} \cdot \frac{l}{\Delta t} \cdot v$

"dividing by time" ↳ $\underline{F} = (\rho \cdot \text{Area} \cdot l)\cdot \underline{v}$

$\approx \frac{\Delta \underline{F}}{\Delta t} = \rho \cdot \text{Area} \cdot \frac{l}{\Delta t} \cdot \underline{v}$, with $\frac{l}{\Delta t} = \underline{v}$

That is why momentum balance has two velocity terms!

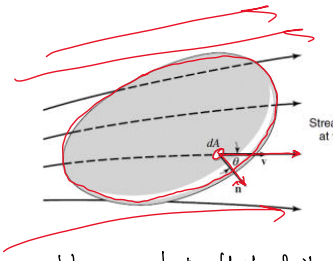
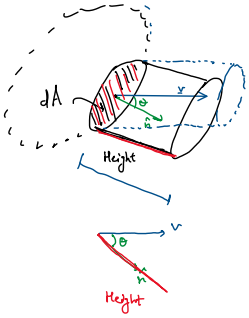


Figure 5.2 Fluid flow through a control volume.

We care about fluid velocity in the direction of the normal to the surface



→ Volume of fluid passed through cross-section A:

$V_{cylinder} = A \cdot \text{Height}$ ← unknown

$v = \frac{\Delta x}{\Delta t}$ → direction of fluid

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$\frac{?}{\Delta x} = \cos\theta$

$? = \Delta x \cos\theta$

$\therefore V_{cylinder} = dA \cdot \Delta x \cdot \cos\theta$

$\Delta V = dA \cdot \Delta x \cos\theta$

$\underline{a}\cdot\underline{b} = |\underline{a}||\underline{b}|\cos\theta$

$\frac{\Delta V}{\Delta t} = dA \cdot \frac{\Delta x}{\Delta t}\cos\theta = A \cdot \underline{v} \cdot \cos\theta$

$\hat{n}\cdot\underline{v} = |\hat{n}||\underline{v}|\cos\theta = \underline{v}\cos\theta$

Volumetric flow rate through A:

$$dQ = v \cdot \cos\theta \, dA = (\hat{n}\cdot\underline{v})\, dA$$

diff. volumetric flow rate

Mass flow = Density · Volumetric flow rate

$dm = \rho \times Q$

$dm = \rho \times (\hat{n}\cdot\underline{v})\, dA$

Momentum flow of liquid through A ⇒ $\underline{F} = (m\underline{v})$

$d\underline{p} = \rho(\hat{n}\cdot\underline{v})dA(\underline{v})$

(ρ: scalar)

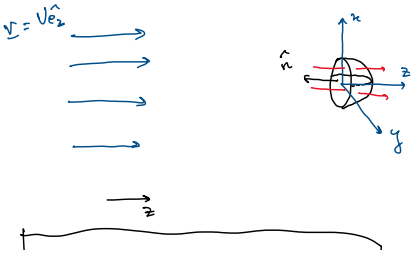
↳ Now we chose arbitrary area → Taking a differential area & integrating momentum quantity over the entire control surface

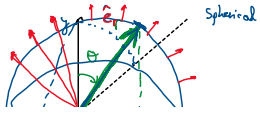
Net momentum change from CV: $\iint_{CS} \underline{v}\,\rho(\hat{n}\cdot\underline{v})\, dA$ ⇒ (In − out) = change (Acc.)

control Surface

Problems

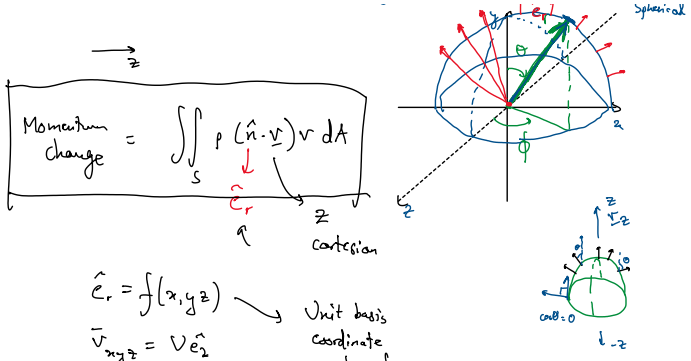
PROBLEM: 3.15 What is the flow of momentum through the hemispherical surface described in Problem 14?





PROBLEM: 3.14 A uniform flow $\underline{v} = U\hat{e}_z$ of an incompressible fluid of density ρ passes through a volume that is in the shape of a half sphere of radius R. The outwards pointing unit normal of the flat surface of the half sphere is $\hat{n} = -\hat{e}_z$. What is the mass flow rate through the hemispherical surface of this volume? Show that you can obtain the correct answer by integrating the formal expression for ∫∫ (Equation 3.87).

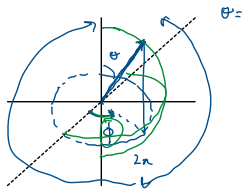
R ρ



$$\hat{e}_r = \sin\theta\cos\phi\,\hat{e}_x + \sin\theta\sin\phi\,\hat{e}_y + \cos\theta\,\hat{e}_z$$

$$\hat{e}_r = \begin{pmatrix} \sin\theta\cos\phi \\ \sin\theta\sin\phi \\ \cos\theta \end{pmatrix}_{xyz}$$

$$(\hat{n}\cdot\underline{v}) = \downarrow \cdot \begin{pmatrix} 0 \\ 0 \\ v \end{pmatrix}_{xyz} = v\cos\theta$$



$0 < \theta < \frac{\pi}{2}$

$0 < \phi < 2\pi$

$$\frac{\Delta p}{\Delta t} = \rho \iint (v\cos\theta)\,\hat{v}\, dA$$

$R^2\sin\theta\, d\theta\, d\phi$ } Jacobian

$$\frac{\Delta p}{\Delta t} = \rho \cdot v\hat{e}_z \int_0^{2\pi}\int_0^{\pi/2} (\hat{e}_r\cdot v)\, R^2\sin\theta\, d\theta\, d\phi$$

$(\hat{e}_r\cdot v) = v\cos\theta$; v constant

$$= \text{—} \int\int \cos\theta\sin\theta\, d\theta\, d\phi$$

U-sub
Int. by parts

$$\text{charge in } \oint = \int d\phi \int \sin\theta\cos\theta\, d\theta$$

→ Solve for HW

average velocity

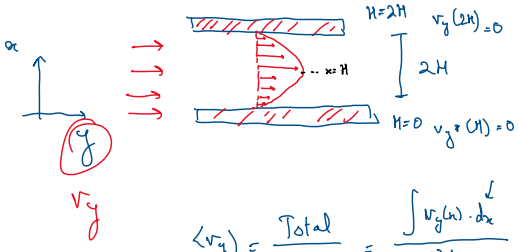
PROBLEM: 3.21 The y-component of a velocity field in flow through a slit (Cartesian coordinates) is given here. What is the average value of the velocity? $2H$ is the gap between the plates. At what location is the velocity a maximum? The units on velocity and A are m/s and the units of x and H are m.

$$\underline{v} = \begin{pmatrix} 0 \\ v_y(x) \\ 0 \end{pmatrix}_{xyz}$$

$$v_y(x) = A\left(1 - \frac{(x-H)^2}{H^2}\right)$$

A: m/s; $x-H$: m; H: m

$v_y(x)$ only in the y direction



$$\langle v_y\rangle = \frac{\text{Total}}{\#} = \frac{\int v_y(x)\,dx}{\int_0^{2H} dx} = \frac{\int_0^{2H} v_y(x)\,dx}{2H} = \frac{1}{2H}\int_0^{2H} A\left(1 - \frac{(x-H)^2}{H^2}\right) dx$$

$$\boxed{\langle v_y\rangle = \frac{2}{3}A}$$

Maximum velocity

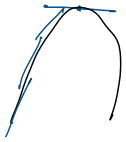
Maximum velocity

$$\frac{dv_y}{dx} = \frac{d}{dx}\left[A\left(1-\frac{(x-H)^2}{H^2}\right)\right] = 0$$

where the max is

$$x = H$$

$$v_y(x=H) = A$$



PROBLEM: 3.31 For the Cartesian coordinate system shown in Figure 3.38, what is a unit vector in the direction of gravity? What is the component of gravity in the flow direction?

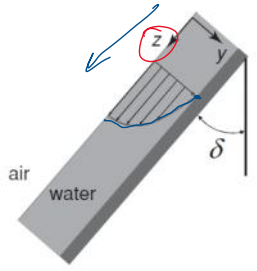
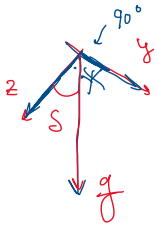


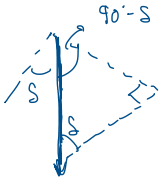
Figure 3.38 Flow coordinate system for Problem 31.



$F = mg$

$$g = \begin{pmatrix} \cdot \\ \cdot \\ g \end{pmatrix}_{xyz}$$

Component in flow direction



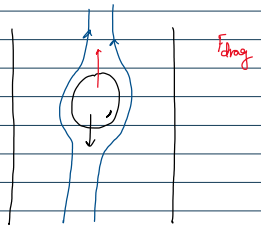
SOH CAH TOA

Week 4

Monday, 11 September 2023 7:04 PM

Fundamental Lab (FL): Experiment 1 → Viscosity of fluids: Stoke's law

Drag force exerted on a spherical object in a viscous fluid



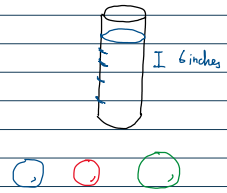
Empirically defined for fully laminar flow conditions ($Re < 0.1$)

$$F_D = 3\pi \mu D v$$

- D: diameter of sphere
- v: velocity of sphere
- μ: Fluid viscosity → Dependent on temperature!

Lab:

+ 3 columns filled w/ liquid (of varying viscosities)
+ Spherical particles of different materials, diameter
+ Electronic balance, tweezers, calipers, stop watch



Determine estimates for viscosity

Viscosity and Stokes Law

Logistics:

+ No Make up sessions! [1 hr 50 min] → 5 min early, Lockers provided, Proper PPE! (You will be sent home otherwise!)
  - PPE: + Safety glasses + Long pants + Closed toe shoes
+ Lab groups → established via Brightspace
+ Complete the trainings → material uploaded on Brightspace

Assignment:

+ Lab Report (due 2 weeks after your Lab → uploaded on brightspace/gradescope)
  - ↳ Introduction
  - ↳ Theory
  - ↳ Methods and Materials
  - ↳ Results and Discussion (w/ graphs, error calculation)
  - ↳ Conclusion & Next steps
  - ↳ Appendix (Derivations, error propagation)

↳ References

+ Group Work expectation:

↳ Meet up before the lab to discuss responsibilities

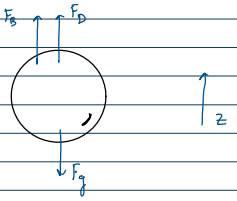
↳ Keep each other accountable

↳ Divide up work to utilize each other's strengths (help out on deficiencies)

Pre-lab assignment

→ How can we go from measurable quantities to determine μ?

$F_D = 3\pi\mu D v$



Terminal velocity → $\Delta F = 0$

$F_B + F_D = F_g$

F_B: Archimedes → Upward force related to the volume of water displaced (equal to the volume of submerged object)

F_D: Stoke's Law

F_g: $mg = (\rho_{ball} V) g$

→ Write viscosity as a function of ρ_{liq} & ρ_{ball}?

$F_B = \rho_{liq} V g$ (m_{liq})

$F_B + F_D = F_g$

$[(\rho_l V) g] + [3\mu\pi D v] = (\rho_{ball} V) g$

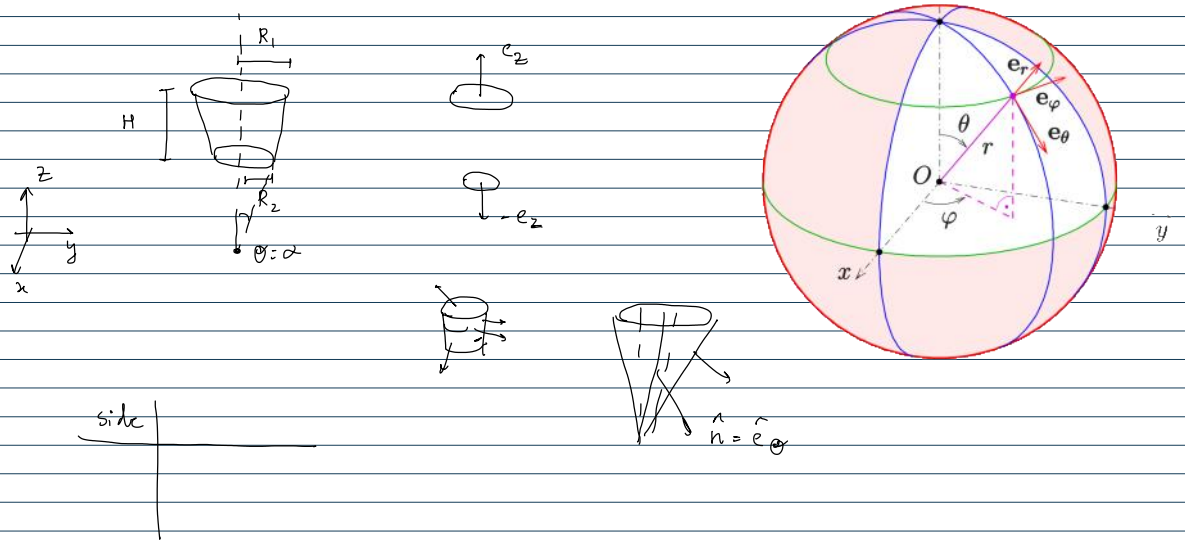
$\mu = \dfrac{(\rho_l V g - \rho_{ball} V g)}{3\pi D v}$

$$\mu = \frac{(\rho_l - \rho_b) V g}{3\pi D v}$$

measure, measure, vary

Interesting analysis: μ varying w/ v
μ varying w/ D
v varying w/ D

PROBLEM: 3.39 For a particular flow problem, the control volume is chosen to be a truncated cone of height H, bottom widest radius R_1, and top smaller radius R_2. The cone is truncated a distance l from the tip, and the cone angle is $\theta = \alpha$, where θ is the coordinate variable for a spherical coordinate system with origin at the cone tip. For each enclosing control surface, write the unit vectors that are normal to each control surface. For a uniform flow $\underline{v} = -U\hat{e}_z$ down the axis of the control volume, what is the mass flow rate through each control surface? The fluid has constant density ρ, and the flow first passes through the bottom of the control volume.

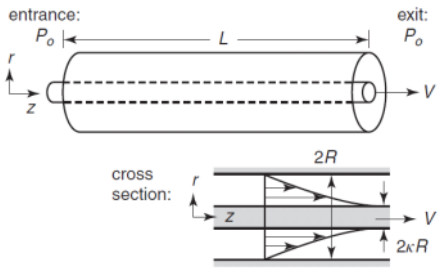


$$\hat{\boldsymbol{r}} = (\sin\theta\cos\phi)\hat{\boldsymbol{x}} + (\sin\theta\sin\phi)\hat{\boldsymbol{y}} + (\cos\theta)\hat{\boldsymbol{z}}$$
$$\hat{\boldsymbol{\theta}} = (\cos\theta\cos\phi)\hat{\boldsymbol{x}} + (\cos\theta\sin\phi)\hat{\boldsymbol{y}} - (\sin\theta)\hat{\boldsymbol{z}}$$
$$\hat{\boldsymbol{\phi}} = -(\sin\phi)\hat{\boldsymbol{x}} + (\cos\phi)\hat{\boldsymbol{y}}.$$

PROBLEM: 3.23 For the velocity-profile function $v_z(r)$ given here, what is the average value $\langle v_z \rangle$ of the function between $r = 5$ and $r = 10$? The function is written in cylindrical coordinates. The units on velocity are m/s and the units of r are m.

$$v_z(r) = 8\ln\left(\frac{r}{3}\right)$$

PROBLEM: 3.32 For the cylindrical coordinate system shown in Figure 3.39 for the axial flow in a wire-coating operation, what is a unit vector in the direction of gravity? What is the component of gravity in the flow direction?



PROBLEM: 3.32 For the cylindrical coordinate system shown in Figure 3.39 for the axial flow in a wire-coating operation, what is a unit vector in the direction of gravity? What is the component of gravity in the flow direction?

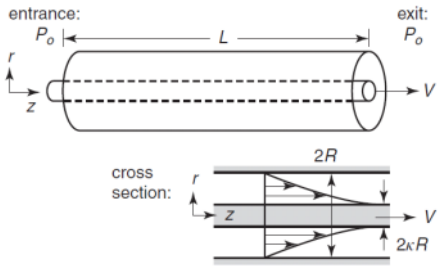


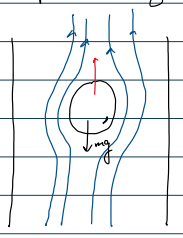
Figure 3.39 Axial annular flow that occurs in wire coating (Problem 32).

Recitation 09/12

Monday, 11 September 2023 7:02 PM

Fundamental Lab (FL): Experiment 1 → Viscosity of fluids: Stoke's law

Drag force exerted on a spherical object in a viscous fluid



non-dimensional

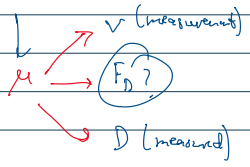
$(Re < 2000)$

Laminar

Empirically defined for fully laminar flow conditions ($Re < 0.1$) creeping flow

$$F_D = 3\pi \mu D v$$

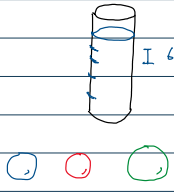
- v: velocity of sphere
- μ: viscosity of fluid
- D: diameter of sphere



Lab:

- 3 columns filled w/ liquid (of varying viscosities) — unknown
- Spherical particles of different materials, diameter
- Electronic balance, tweezers, calipers, stop watch

Demo



Determine estimate for viscosity (μ)

Viscosity and Stokes Law

Logistics:

- No Make up sessions! [4 hr 50 min] → 5 min early, Lockers provided, Proper PPE!
  - Safety glasses
  - Long pants
  - Closed toe shoes

  } You will be sent home otherwise!
- Lab groups → established via Brightspace
- Complete the trainings → material uploaded on Brightspace

Assignment:

- Lab Report (due 2 weeks after your Lab → uploaded on brightspace/gradescope)
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  - Theory
  - Methods and Materials
  - Results and Discussion
  - Conclusion & Next steps
  - Appendix (Derivations, error propogation)
  - References

+ Group Work expectation:

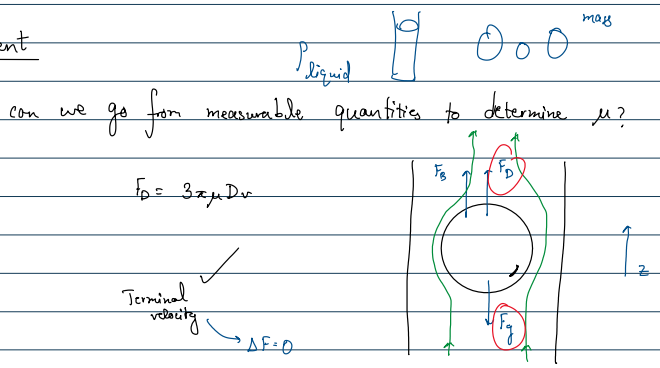
↳ Meet up before the lab to discuss responsibilities

↳ Keep each other accountable

↳ Divide up work to utilize each other's strengths (help out on deficiencies)

Pre-lab assignment

→ How can we go from measurable quantities to determine μ?



$F_D = 3\pi\mu D v$

Terminal velocity ✓ → $\Delta F = 0$

$F_B + F_D = F_g$, $mg = (\rho_{ball} V) g$

Archimedes: Upward force related to the volume of water displaced (equal to the volume of submerged object)

Stoke's Law: ($F_D = 3\pi\mu D v$)

→ Write viscosity as a function of ρ_{liq}, ρ_{ball}? → remove F_D

$F_B = \rho_{liq} V g$, m_{liq}

$F_B + F_D = F_g$

$(\rho_{liq} V g) + 3\pi\mu D v = (\rho_{ball} V) \cdot g$ (m)

$\mu = \dfrac{(\rho_{liq} V g - \rho_{ball} V g)}{3\pi D v}$

$\mu = \dfrac{(\rho_{liq} - \rho_{ball}) V g}{3\pi D v}$

given; measure; measure; mass; Vol → $\frac{4}{3}\pi r^3$; cart; ball; measure

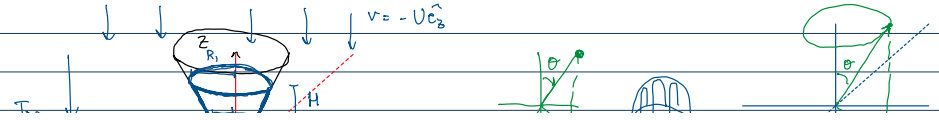
Interesting analysis:

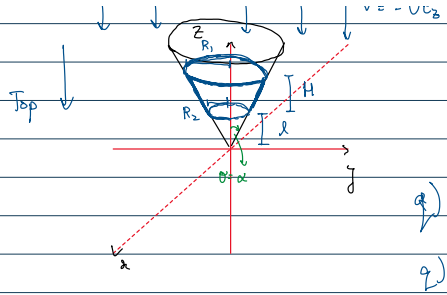
μ varies w/ v

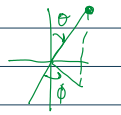
μ " D

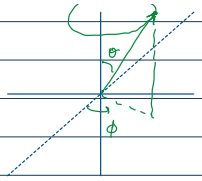
v " D

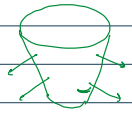
PROBLEM: 3.39 For a particular flow problem, the control volume is chosen to be a truncated cone of height H, bottom widest radius R_1, and top smaller radius R_2. The cone is truncated a distance l from the tip, and the cone angle is $\theta = \alpha$, where θ is the coordinate variable for a spherical coordinate system with origin at the cone tip. For each enclosing control surface, write the unit vectors that are normal to each control surface. For a uniform flow $v = -U\hat{e}_z$ down the axis of the control volume, what is the mass flow rate through each control surface? The fluid has constant density ρ, and the flow first passes through the bottom of the control volume.











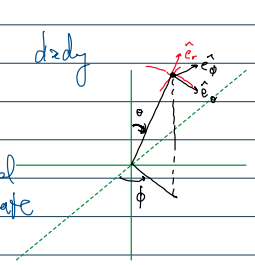
q) How many surfaces

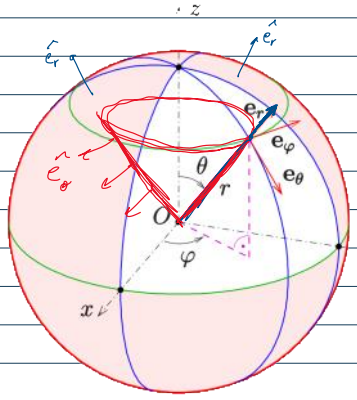
q) Draw them out & find $\hat{n}$

q) what is mass flow rate formula

$$\iint \hat{n}\cdot\underline{v}\; dA \quad dz\,dy$$

original coordinate





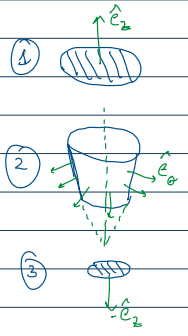
$\hat{e}_\theta \rightarrow xyz$? dot w/ $\underline{v} \rightarrow$ how? some unit basis transformation

$\theta \rightarrow xyz$

$$\hat{e}_\theta = \cos\theta\cos\phi\,\hat{e}_x + \cos\theta\sin\phi\,\hat{e}_y - \sin\theta\,\hat{e}_z \rightarrow \text{from geometry}$$

Mass flow rate $\Rightarrow \rho \iint (\hat{n}\cdot\underline{v})\,dA$

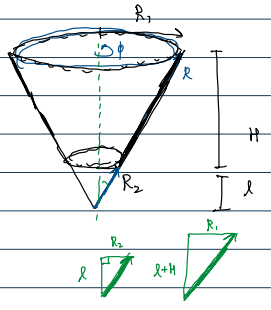
$\hat{n}\cdot\underline{v}$ → "Volume"



$$\begin{pmatrix}\cos\theta\cos\phi\\ \cos\theta\sin\phi\\ -\sin\theta\end{pmatrix}\cdot\begin{pmatrix}0\\0\\-v\end{pmatrix} = v\sin\theta \qquad \theta=\alpha$$

$$\rightarrow \int_0^{2\pi}\int_{\sqrt{R_2^2+l^2}}^{\sqrt{(l+H)^2+R_1^2}} \sin\alpha\; dA \qquad \text{(Constant)}$$

$dA = r^2\sin\theta\; d\phi\, dr$, $\theta=\alpha$



ϕ $dr\,d\phi\,d\theta$ $\quad 0<\phi<2\pi$

PROBLEM: 3.23 For the velocity-profile function $v_z(r)$ given here, what is the average value $\langle v_z\rangle$ of the function between $r = 5$ and $r = 10$? The function is written in cylindrical coordinates. The units on velocity are m/s and the units of r are m.

$$v_z(r) = 8\ln\left(\frac{r}{3}\right)$$

PROBLEM: 3.32 For the cylindrical coordinate system shown in Figure 3.39 for the axial flow in a wire-coating operation, what is a unit vector in the direction of gravity? What is the component of gravity in the flow direction?

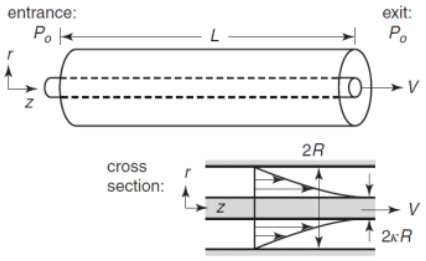
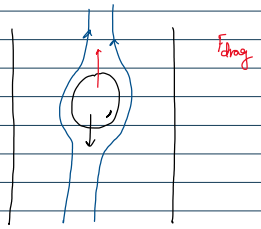


Figure 3.39 Axial annular flow that occurs in wire coating (Problem 32).

Fundamental Lab (FL): Experiment 1 → Viscosity of fluids: Stoke's law

Drag force exerted on a spherical object in a viscous fluid



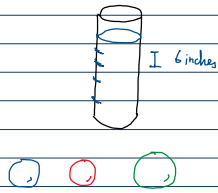
Empirically defined for fully laminar flow conditions ($Re < 0.1$)

$$F_D = 3\pi \mu D v$$

- D: diameter of sphere
- v: velocity of sphere
- μ: Fluid viscosity → Dependent on temperature!

Lab:

+ 3 columns filled w/ liquid (of varying viscosities)
+ Spherical particles of different materials, diameter
+ Electronic balance, tweezers, calipers, stop watch



Determine estimates for viscosity

Viscosity and Stokes Law

Logistics:

+ No Make up sessions! [1 hr 50 min] → 5 min early, Lockers provided, Proper PPE! (You will be sent home otherwise!)
  - PPE: + Safety glasses + Long pants + Closed toe shoes
+ Lab groups → established via Brightspace
+ Complete the trainings → material uploaded on Brightspace

Assignment:

+ Lab Report (due 2 weeks after your Lab → uploaded on brightspace/gradescope)
  - Introduction
  - Theory
  - Methods and Materials
  - Results and Discussion (w/ graphs, error calculation)
  - Conclusion & Next steps
  - Appendix (Derivations, error propagation)

↳ References

+ Group Work Expectation:

↳ Meet up before the lab to discuss responsibilities

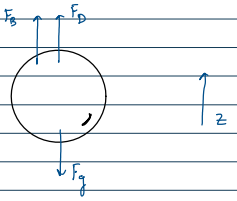
↳ Keep each other accountable

↳ Divide up work to utilize each other's strengths (help out on deficiencies)

Pre-lab assignment

→ How can we go from measurable quantities to determine μ?

$F_D = 3\pi\mu D v$



Terminal velocity → $\Delta F = 0$

$F_B + F_D = F_g$

F_B: Archimedes → Upward force related to the volume of water displaced (equal to the volume of submerged object)

F_D: Stoke's Law

F_g: $mg = (\rho_{ball} V) g$

→ Write viscosity as a function of ρ_{liq} & ρ_{ball}?

$F_B = \rho_{liq} V g$ (m_{liq})

$F_B + F_D = F_g$

$[(\rho_l V) g] + [3\mu\pi D v] = (\rho_{ball} V) g$

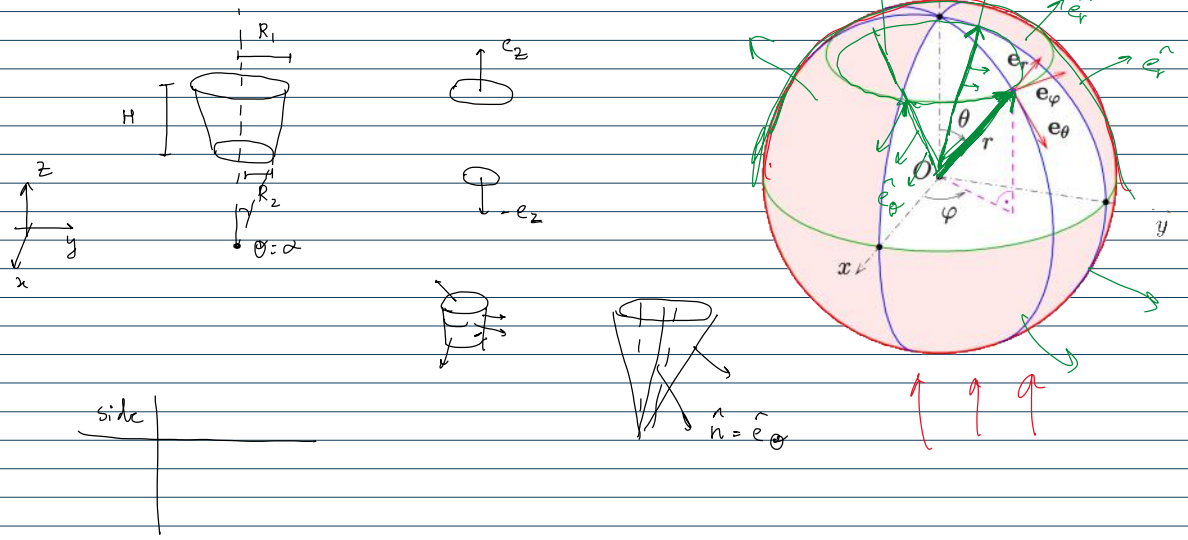
$\mu = \dfrac{(\rho_l V g - \rho_{ball} V g)}{3\pi D v}$

$\mu = \dfrac{(\rho_l - \rho_b) V g}{3\pi D v}$

measure measure vary

Interesting analysis: μ varying w/ v
μ varying w/ D
v varying w/ D

PROBLEM: 3.39 For a particular flow problem, the control volume is chosen to be a truncated cone of height H, bottom widest radius R_1, and top smaller radius R_2. The cone is truncated a distance l from the tip, and the cone angle is $\theta \equiv \alpha$, where θ is the coordinate variable for a spherical coordinate system with origin at the cone tip. For each enclosing control surface, write the unit vectors that are normal to each control surface. For a uniform flow $v = -U\hat{e}_z$ down the axis of the control volume, what is the mass flow rate through each control surface? The fluid has constant density ρ, and the flow first passes through the bottom of the control volume.

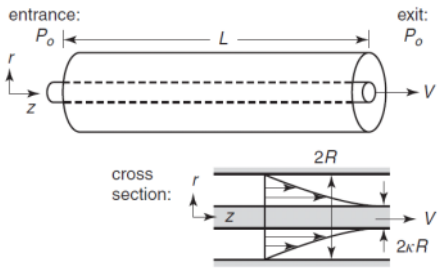


$$\hat{\boldsymbol{r}} = (\sin\theta\cos\phi)\hat{\boldsymbol{x}} + (\sin\theta\sin\phi)\hat{\boldsymbol{y}} + (\cos\theta)\hat{\boldsymbol{z}}$$
$$\hat{\boldsymbol{\theta}} = (\cos\theta\cos\phi)\hat{\boldsymbol{x}} + (\cos\theta\sin\phi)\hat{\boldsymbol{y}} - (\sin\theta)\hat{\boldsymbol{z}}$$
$$\hat{\boldsymbol{\phi}} = -(\sin\phi)\hat{\boldsymbol{x}} + (\cos\phi)\hat{\boldsymbol{y}}.$$

PROBLEM: 3.23 For the velocity-profile function $v_z(r)$ given here, what is the average value $\langle v_z \rangle$ of the function between $r = 5$ and $r = 10$? The function is written in cylindrical coordinates. The units on velocity are m/s and the units of r are m.

$$v_z(r) = 8\ln\left(\frac{r}{3}\right)$$

PROBLEM: 3.32 For the cylindrical coordinate system shown in Figure 3.39 for the axial flow in a wire-coating operation, what is a unit vector in the direction of gravity? What is the component of gravity in the flow direction?



PROBLEM: 3.32 For the cylindrical coordinate system shown in Figure 3.39 for the axial flow in a wire-coating operation, what is a unit vector in the direction of gravity? What is the component of gravity in the flow direction?

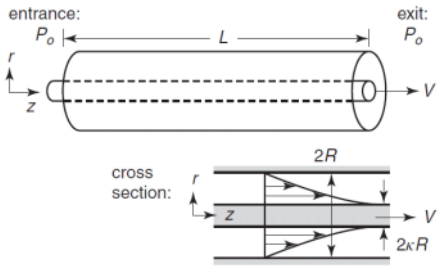


Figure 3.39 Axial annular flow that occurs in wire coating (Problem 32).

Week 25th Sept

Monday, 25 September 2023 9:15 PM

+ Accounts for intermolecular forces during fluid motion
+ requires mag., direction & orientation

Viscosity (μ) = $\frac{\text{shear stress}}{\text{rate of shear strain}}$ ~ rate of deformation

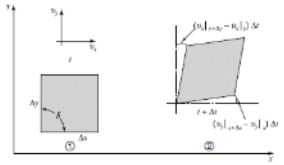


Figure 7.7 Shear strain in the xy plane.

Deformation occuring in two directions due to fluid flow

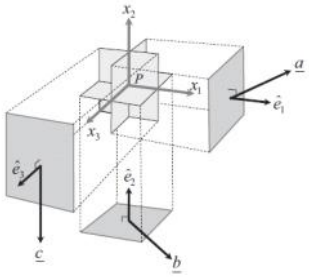
$$-\frac{dS_{xy}}{dt} = \frac{\partial v_x}{\partial y} + \frac{\partial v_y}{\partial x}$$

$-\frac{dS_{xy}}{dt}$ ~ deformation

$\frac{\partial v_x}{\partial y}$ or $\dot{\gamma}_0$

Shear stress = (viscosity) (rate of shear strain)

μ $\left(-\frac{dS}{dt}\right)$



Similar analysis on each plane (xy, yz, xz) } For stress at a point P in fluid motion

+

Normal stresses

gives 9 elements to describe all forces at a point in a fluid. (moving or stationary)

$$\therefore \quad \tau_{xy} = \tau_{yx} = \mu\left(\frac{\partial v_x}{\partial y} + \frac{\partial v_y}{\partial x}\right)$$

$$\tau_{yz} = \tau_{zy} = \mu\left(\frac{\partial v_y}{\partial z} + \frac{\partial v_z}{\partial y}\right)$$

$$\tau_{zx} = \tau_{xz} = \mu\left(\frac{\partial v_z}{\partial x} + \frac{\partial v_x}{\partial z}\right)$$

Laminar flow ✓

Normal stress: $\sigma = \tau_{ii}$ (based on Hooke's law)

$\tau_{ij} = \tau_{ji}$ } Always true ✓ (angular momentum conservation)

$$\sigma_{xx} = \mu\left(2\frac{\partial v_x}{\partial x} - \frac{2}{3}(\nabla \cdot \underline{v})\right) - p$$

$$\sigma_{xx} = \mu\left(2\frac{\partial v_x}{\partial x} - \frac{2}{3}(\nabla\cdot\underline{v})\right) - p$$

$$\sigma_{yy} = \mu\left(2\frac{\partial v_y}{\partial y} - \frac{2}{3}(\nabla\cdot\underline{v})\right) - p$$

$$\sigma_{zz} = \mu\left(2\frac{\partial v_z}{\partial z} - \frac{2}{3}(\nabla\cdot\underline{v})\right) - p$$

Newtonian fluid

$\underline{\underline{\Pi}} = -p\underline{\underline{I}}$

Fluid Stress

→ Isotropic [same mag. of stress exerted on any surface & stress acts ~ Normal stress] → Independent of surface orientation & always normal to any surface

→ Anisotropic (shear stress → different values of stress on different surfaces) } $\underline{\underline{\tau}}$

↓

symmetric; 6 components

$$\underline{\underline{\Pi}} = \underline{\underline{\tau}} - p\underline{\underline{I}}$$

Tension Stress Tensor in Flow

- Molecular contact stress
- Symmetric; i.e., $\tilde{\Pi}_{ik} = \tilde{\Pi}_{ki}$, $\tilde{\tau}_{pm} = \tilde{\tau}_{mp}$
- $\tilde{\Pi}_{jn}$ is stress on a j-surface in the n-direction
- Infinitesimal surface: $\underline{f} = \Delta A\, \hat{n}\cdot\underline{\underline{\tilde{\Pi}}}$
- Finite surface:

$$\underline{f} = \iint_S \left[\hat{n}\cdot\underline{\underline{\tilde{\Pi}}}\right] dS$$

$$\underline{\underline{\tilde{\Pi}}} = \underline{\underline{\tilde{\tau}}} - p\underline{\underline{I}}$$

$$\begin{pmatrix} \tilde{\Pi}_{11} & \tilde{\Pi}_{12} & \tilde{\Pi}_{13} \\ \tilde{\Pi}_{21} & \tilde{\Pi}_{22} & \tilde{\Pi}_{23} \\ \tilde{\Pi}_{31} & \tilde{\Pi}_{32} & \tilde{\Pi}_{33} \end{pmatrix}_{123} = \begin{pmatrix} \tilde{\tau}_{11} - p & \tilde{\tau}_{12} & \tilde{\tau}_{13} \\ \tilde{\tau}_{21} & \tilde{\tau}_{22} - p & \tilde{\tau}_{23} \\ \tilde{\tau}_{31} & \tilde{\tau}_{32} & \tilde{\tau}_{33} - p \end{pmatrix}_{123}$$

Figure 4.52 The stress tensor in matrix form can be used to calculate the tension on a surface in a flow.

Manometer Review

Pascal's principle: Pressure exerted on a confined liquid is transmitted equally to every part of the liquid & to the walls of the container } Idea of incompressibility

↓

applicable to

liquid ⊥ to the walls of the container ∪ ∪

↓

applicable to confined fluids at rest

Manometer → device to accurately measure pressure differences

Momentum balance (RTT) → $\frac{dp}{dz} = -\rho g$ → $\boxed{p(z) = -\rho g z + p_0}$

Pressure in a stationary fluid at elevation z (upward)

i.e. only dependent on the vertical height! (Not shape)

Pressures at same elevation (of same fluid) are eq

PROBLEM: 4.16 A manometer is configured as shown in Figure 4.67. A heavy fluid (Fluid B) has been placed in the bottom of the manometer. A light fluid (Fluid A) has been added to the left side only. The left side of the manometer has been connected to a process stream in which water (25°C, Fluid C) is flowing, while the right side of the manometer is open to the atmosphere (Fluid D). The manometer is being used to measure the pressure difference $p_1 - p_2$. The density of the two fluids in the manometer are 1.75 g/cm^3 and 13.6 g/cm^3. What is the pressure difference $p_1 - p_2$ in terms of fluid heights and fluid densities? For $h_1 = 2.3\ cm$, $h_2 = 2.3\ cm$, $h_3 = 1.0\ cm$, what is the pressure difference in psi?

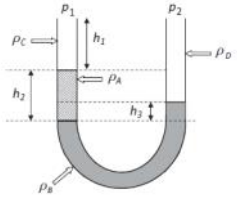
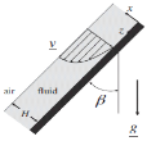
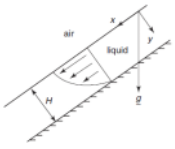


Figure 4.67 A schematic of a manometer that contains two different measurement fluids (Problem 16).

PROBLEM: 5.35 We solved the incline problem using a coordinate system that sits on the solid surface (Figure 5.12). Here, choose instead to solve the same problem with a coordinate system in which the coordinate position $x = 0$ is located at the free surface (i.e., the top of the film) (Figure 5.42). What is the gravity acceleration vector in this coordinate system? What are the flow boundary conditions? Are there any advantages or disadvantages to this choice of coordinate system? Discuss your observations.



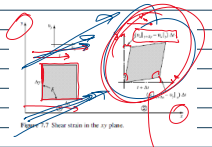


Review

+ Stress tensor, viscosity & static fluid problems

+ Accounts for intermolecular forces during fluid motion
+ requires mag., direction & orientation

Viscosity (μ) = $\frac{\text{shear stress}}{\text{rate of shear strain}}$ ~ rate of deformation → $\frac{dv}{dz} \sim \frac{dv}{dz}$



Deformation occuring in two directions due to fluid flow

$-\frac{d\delta_{xy}}{dt} = \frac{\partial v_x}{\partial y} + \frac{\partial v_y}{\partial x}$

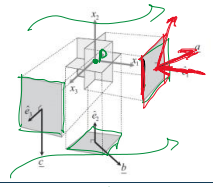
deformation

$\frac{dv}{dy}$ or $\dot{\gamma}$

Shear stress = (viscosity) (rate of shear strain)

$\mu \left(\frac{d\delta}{dt}\right)$

τ_{xy} τ_{yz} τ_{xz}



Similar analysis on each plane for stress at a point P in fluid motion

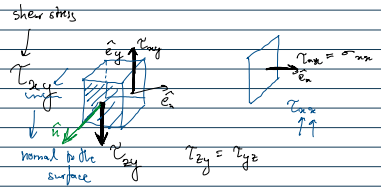
$(\tau_{xy}, \tau_{yz}, \tau_{xz})$

$\frac{\partial v_x}{\partial y}$ $\frac{\partial v_x}{\partial z}$ $\frac{\partial v_y}{\partial z}$ $\frac{1}{2}\frac{\partial v_z}{\partial x}$

Normal stresses

gives 9 elements to describe all forces at a point in a fluid. (moving or stationary)

Shear stress



$\tau_{zy} = \tau_{yz}$

stress μ

Cartesian coordinate system $\dot{\gamma}$

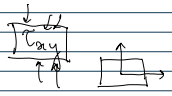
$\therefore$

$\tau_{xy} = \tau_{yx} = \mu\left(\frac{\partial v_x}{\partial y} + \frac{\partial v_y}{\partial x}\right)$ ✓

$\tau_{yz} = \tau_{zy} = \mu\left(\frac{\partial v_y}{\partial z} + \frac{\partial v_z}{\partial y}\right)$ ✓

$\tau_{zx} = \tau_{xz} = \mu\left(\frac{\partial v_z}{\partial x} + \frac{\partial v_x}{\partial z}\right)$ ✓

Laminar flow ✓



Normal stress $\sigma = \tau_{ii}$ (based on Hooke's law)

$\tau_{ij} = \tau_{ji}$ (Always true (Angular momentum conservation))

normal point of face

pressure

$\sigma_{xx} = \mu\left(2\frac{\partial v_x}{\partial x} - \frac{2}{3}(\nabla\cdot\underline{v})\right) - p$

$\sigma_{yy} = \mu\left(2\frac{\partial v_y}{\partial y} - \frac{2}{3}(\nabla\cdot\underline{v})\right) - p$

$\sigma_{zz} = \mu\left(2\frac{\partial v_z}{\partial z} - \frac{2}{3}(\nabla\cdot\underline{v})\right) - p$

$\left(2\mu\frac{\partial v_x}{\partial x} - \frac{2}{3}\right)$

$(\nabla\cdot\underline{v}) = 0$ → Incompressible

Newtonian fluid

Fluid Stress

→ Isotropic [same mag. of stress exerted on any surface & stress acts normal] → Normal stress

→ Anisotropic (shear stress → different values of stress on different surfaces)

Independent of surface orientation & always normal to any surface

$\underline{\underline{\Pi}} = (-p\underline{\underline{I}})$

$\underline{\underline{\Pi}}$ → Symmetric; 6 components

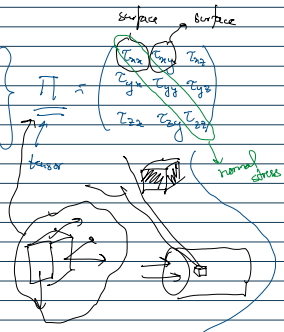
stress shear normal

$\underline{\underline{\Pi}} = \underline{\underline{\tau}} - p\underline{\underline{I}}$

Normal stress

Shear stress

$\underline{\underline{\Pi}}$ = tensor



$-p\begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix} = \begin{pmatrix}-p&0&0\\0&-p&0\\0&0&-p\end{pmatrix} + \begin{pmatrix}0&-1&0\\-1&0&0\\0&0&0\end{pmatrix}$

normal shear

$\underline{\underline{\Pi}}$

Tension Stress Tensor in Flow

- Molecular contact stress
- Symmetric; i.e., $\tilde{\Pi}_{ij} = \tilde{\Pi}_{ji}$, $\tilde{\tau}_{pm} = \tilde{\tau}_{mp}$
- $\tilde{\Pi}_{jn}$ is stress on a j-surface in the n-direction
- Infinitesimal surface: $\underline{f} = \Delta A\, \hat{n} \cdot \tilde{\Pi}$
- Finite surface:

$$\underline{f} = \iint_S \left[\hat{n} \cdot \tilde{\Pi}\right] dS$$

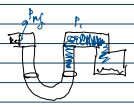
surface analogy

$$\tilde{\Pi} = \tilde{\tau} - pI$$

$$\begin{pmatrix} \tilde{\Pi}_{11} & \tilde{\Pi}_{12} & \tilde{\Pi}_{13} \\ \tilde{\Pi}_{21} & \tilde{\Pi}_{22} & \tilde{\Pi}_{23} \\ \tilde{\Pi}_{31} & \tilde{\Pi}_{32} & \tilde{\Pi}_{33} \end{pmatrix}_{ij} = \begin{pmatrix} \tilde{\tau}_{11} - p & \tilde{\tau}_{12} & \tilde{\tau}_{13} \\ \tilde{\tau}_{21} & \tilde{\tau}_{22} - p & \tilde{\tau}_{23} \\ \tilde{\tau}_{31} & \tilde{\tau}_{32} & \tilde{\tau}_{33} - p \end{pmatrix}_{ij}$$

Figure 4.02 The stress tensor in matrix form can be used to calculate the tension on a surface in a flow.

Manometer Review



Pascal's principle: Pressure exerted on a confined liquid is transmitted equally to every part of the liquid & to the walls of the container } Idea of incompressibility

↓

applicable to confined fluids at rest

Manometer → device to accurately measure pressure differences

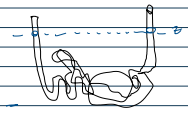
Manometer balance (RTT) → $\frac{dp}{dz} = -\rho g$ → Integrate → $p(z) = -\rho g z + p_0$

Pressure in a stationary fluid at elevation z (upward)

i.e. only dependent on the vertical height! (Not shape)

Pressures at same elevation (of same fluid) are eq

$p = f(z)$ ✓



PROBLEM: 4.16 A manometer is configured as shown in Figure 4.67. A heavy fluid (Fluid B) has been placed in the bottom of the manometer. A light fluid (Fluid A) has been added to the left side only. The left side of the manometer has been connected to a process stream in which water (25°C, Fluid C) is flowing, while the right side of the manometer is open to the atmosphere (Fluid D). The manometer is being used to measure the pressure difference $p_1 - p_2$. The density of the two fluids in the manometer are 1.75 g/cm^3 and 13.6 g/cm^3. What is the pressure difference $p_1 - p_2$ in terms of fluid heights and fluid densities? For $h_1 = 2.3$ cm, $h_2 = 2.3$ cm, $h_3 = 1.0$ cm, what is the pressure difference in psi?

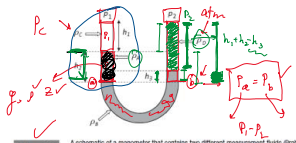
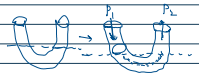


Figure 4.67 A schematic of a manometer that contains two different measurement fluids (Problem 16).

+ Pascals → more incomp



$+ \; p(z) = -\rho g z + p_0$ ✓

$P_{bottom} = P_{top} + \rho g z$

$P_a = P_B$ ✓ $\quad P_{bot} = P_{top} + \rho g z$

P_1 + fluid 1 + fluid 2

$l_A g(h_1 + h_2)$

$$P_a = \rho_A g h_2 + \rho_C g h_1 + P_1$$

$$P_b = \rho_B g h_3 + \rho_D g (h_1 + h_2 - h_3) + P_2$$

$P_a = P_B$

$$\Rightarrow \; P_1 + \rho_C g h_1 + \rho_A g h_2 = P_2 + \rho_D g h_1 + \rho_D g h_2 - \rho_D g h_3 + \rho_B h_3$$

$$\Rightarrow \; \boxed{(P_1 - P_2) = (\rho_D - \rho_C) g h_1 + (\rho_D - \rho_A) g h_2 + (\rho_B - \rho_D) g h_3}$$

PROBLEM: 5.35 We solved the incline problem using a coordinate system that sits on the solid surface (Figure 5.12). Here, choose instead to solve the same problem with a coordinate system in which the coordinate position $x = 0$ is located at the free surface (i.e., the top of the film) (Figure 5.42). What is the gravity acceleration vector in this coordinate system? What are the flow boundary conditions? Are there any advantages or disadvantages to this choice of coordinate system? Discuss your observations.

Week 25th Sept

Monday, 25 September 2023 9:15 PM

+ Accounts for intermolecular forces during fluid motion
+ requires mag., direction & orientation

Viscosity (μ) = $\frac{\text{shear stress}}{\text{rate of shear strain}}$ ~ rate of deformation

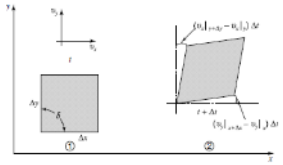


Figure 7.7 Shear strain in the xy plane.

Deformation occuring in two directions due to fluid flow

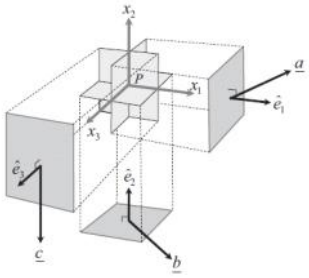
$$-\frac{dS_{xy}}{dt} = \frac{\partial v_x}{\partial y} + \frac{\partial v_y}{\partial x}$$

$-\frac{dS_{xy}}{dt}$ ~ deformation

$\frac{\partial v_x}{\partial y}$ or $\dot{\gamma}_{yx}$

Shear stress = (viscosity) (rate of shear strain)

μ $\left(-\frac{dS}{dt}\right)$



Similar analysis on each plane (xy, yz, xz) } For stress at a point P in fluid motion

+

Normal stresses

gives 9 elements to describe all forces at a point in a fluid. (moving or stationary)

$$\therefore \quad \tau_{xy} = \tau_{yx} = \mu\left(\frac{\partial v_x}{\partial y} + \frac{\partial v_y}{\partial x}\right)$$

$$\tau_{yz} = \tau_{zy} = \mu\left(\frac{\partial v_y}{\partial z} + \frac{\partial v_z}{\partial y}\right)$$

$$\tau_{zx} = \tau_{xz} = \mu\left(\frac{\partial v_z}{\partial x} + \frac{\partial v_x}{\partial z}\right)$$

Laminar flow ✓

Normal stress: $\sigma = \tau_{ii}$ (based on Hooke's law)

$\tau_{ij} = \tau_{ji}$ } Always true ✓ (angular momentum conservation)

$$\sigma_{xx} = \mu\left(2\frac{\partial v_x}{\partial x} - \frac{2}{3}(\nabla \cdot \underline{v})\right) - p$$

$$\sigma_{xx} = \mu\left(2\frac{\partial v_x}{\partial x} - \frac{2}{3}(\nabla\cdot\underline{v})\right) - p$$

$$\sigma_{yy} = \mu\left(2\frac{\partial v_y}{\partial y} - \frac{2}{3}(\nabla\cdot\underline{v})\right) - p$$

$$\sigma_{zz} = \mu\left(2\frac{\partial v_z}{\partial z} - \frac{2}{3}(\nabla\cdot\underline{v})\right) - p$$

Newtonian fluid

$\underline{\underline{\Pi}} = -p\underline{\underline{I}}$

Fluid Stress → Isotropic [same mag. of stress exerted on any surface & stress acts ~ Normal stress] → Independent of surface orientation & always normal to any surface

→ Anisotropic (shear stress → different values of stress on different surfaces) } $\underline{\underline{\tau}}$

↓

symmetric; 6 components

$$\underline{\underline{\Pi}} = \underline{\underline{\tau}} - p\underline{\underline{I}}$$

Tension Stress Tensor in Flow

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$$\underline{\underline{\tilde{\Pi}}} = \underline{\underline{\tilde{\tau}}} - p\underline{\underline{I}}$$

$$\begin{pmatrix} \tilde{\Pi}_{11} & \tilde{\Pi}_{12} & \tilde{\Pi}_{13} \\ \tilde{\Pi}_{21} & \tilde{\Pi}_{22} & \tilde{\Pi}_{23} \\ \tilde{\Pi}_{31} & \tilde{\Pi}_{32} & \tilde{\Pi}_{33} \end{pmatrix}_{123} = \begin{pmatrix} \tilde{\tau}_{11} - p & \tilde{\tau}_{12} & \tilde{\tau}_{13} \\ \tilde{\tau}_{21} & \tilde{\tau}_{22} - p & \tilde{\tau}_{23} \\ \tilde{\tau}_{31} & \tilde{\tau}_{32} & \tilde{\tau}_{33} - p \end{pmatrix}_{123}$$

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↓

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↓

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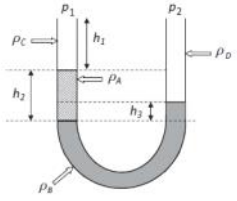

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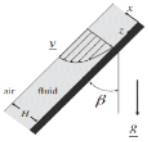
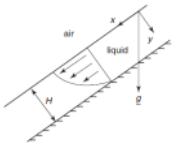




Week of 23rd

Monday, 23 October 2023 5:14 PM

"Macro"

Control Volume Balance

$$0 = \frac{dm}{dt} + \iint_{CS} (\hat{n}\cdot\underline{v})\rho\, dS$$

$$\sum_{on\,CV} \underline{f} = \frac{d(m\underline{v})_{CV}}{dt} + \iint_{CS} (\hat{n}\cdot\underline{v})\rho\,\underline{v}\, dS$$

→ Macro → Micro
- arbitrary CV
- Express terms as integrals
- Leibniz rule & divergence theorem

Microscopic Balance

$$\frac{\partial\rho}{\partial t} + \underline{v}\cdot\nabla\rho = -\rho(\nabla\cdot\underline{v})$$

$$\rho\left(\frac{\partial\underline{v}}{\partial t} + \underline{v}\cdot\nabla\underline{v}\right) = -\nabla p + \nabla\cdot\underline{\underline{\tau}} + \rho\underline{g}$$

ρ = density
$\underline{v}$ = velocity
p = pressure
$\underline{g}$ = gravity

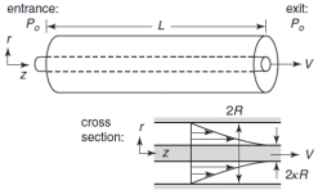
1) Choosing CV (can be arbitrary)
↓
writing out equations specific to that CV
↳ Have to rewrite & recalculate if CV is changed

→

1) Already written for an arbitrary, microscopic CV
↳ applicable to any problem ✓
↳ Just need to change boundary conditions & get velocity & pressure profiles!

PROBLEM: 6.49 Flow Problem: Axial annular drag with pressure drop, wire coating. Repeat Problem 43 with an imposed pressure gradient $-\Delta p/L$ in the flow direction. Calculate the velocity field only.

PROBLEM: 6.43 Flow Problem: Axial annular drag, wire coating. An incompressible Newtonian fluid fills the annular gap between a cylindrical wire of radius κR and an outer shell of inner radius R (Figure 6.22). The wire is pulled to the right at speed V. There is no pressure variation throughout the apparatus. Calculate the steady state velocity profile, the flow rate, and the force on the wire. This geometry occurs in wire-coating. Answer: [redacted]



→ Flow in z ✓
→ gradients in r ✓
→ Symmetric around θ ✓

In wire-coating, a wire is drawn through a bath. This axial drag flow is addressed in Problem 43.

r-direction

$$\rho\left(\frac{\partial v_r}{\partial t} + v_r\frac{\partial v_r}{\partial r} + \frac{v_\theta}{r}\frac{\partial v_r}{\partial\theta} - \frac{v_\theta^2}{r} + v_z\frac{\partial v_r}{\partial z}\right) = -\frac{\partial P}{\partial r} + \rho g_r + \mu\left[\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}(rv_r)\right) + \frac{1}{r^2}\frac{\partial^2 v_r}{\partial\theta^2} - \frac{2}{r^2}\frac{\partial v_\theta}{\partial\theta} + \frac{\partial^2 v_r}{\partial z^2}\right]$$

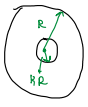
θ-direction

$$\rho\left(\frac{\partial v_\theta}{\partial t} + v_r\frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r}\frac{\partial v_\theta}{\partial\theta} + \frac{v_r v_\theta}{r} + v_z\frac{\partial v_\theta}{\partial z}\right) = -\frac{1}{r}\frac{\partial P}{\partial\theta} + \rho g_\theta + \mu\left[\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}(rv_\theta)\right) + \frac{1}{r^2}\frac{\partial^2 v_\theta}{\partial\theta^2} + \frac{2}{r^2}\frac{\partial v_r}{\partial\theta} + \frac{\partial^2 v_\theta}{\partial z^2}\right]$$

z-direction

$$\rho\left(\frac{\partial v_z}{\partial t} + v_r\frac{\partial v_z}{\partial r} + \frac{v_\theta}{r}\frac{\partial v_z}{\partial\theta} + v_z\frac{\partial v_z}{\partial z}\right) = -\frac{\partial P}{\partial z} + \rho g_z + \mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 v_z}{\partial\theta^2} + \frac{\partial^2 v_z}{\partial z^2}\right]$$

6.43: No pressure variation ; $\underline{v} = \begin{pmatrix} 0 \\ 0 \\ v_z \end{pmatrix}_{r\theta z}$; $\underline{g} = \begin{pmatrix} g_r \\ g_\theta \\ 0 \end{pmatrix}_{r\theta z}$



Mass balance / Continuity equation:

SS, Incompressible

$$\frac{\partial\rho}{\partial t} + v\cdot\nabla\rho = -\rho(\nabla\cdot\underline{v})$$

$$(\nabla\cdot\underline{v}) = 0$$

$$\frac{1}{r}\frac{\partial}{\partial r}(rv_r) + \frac{1}{r}\frac{\partial v_\theta}{\partial\theta} + \frac{\partial v_z}{\partial z} = 0$$

$$\boxed{\frac{\partial v_z}{\partial z} = 0}$$

Start w/ Navier stokes:

r-direction : $0 = \rho g_r$

θ-direction : $0 = \rho g_\theta$

θ- direction : $0 = \rho g_\theta$

z - direction : $\rho\left(v_z \frac{\partial v_z}{\partial z}\right) = \mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right) + \frac{\partial^2 v_z}{\partial z^2}\right]$ (terms: $v_z\frac{\partial v_z}{\partial z}$ and $\frac{\partial^2 v_z}{\partial z^2}$ both $= 0$)

$$\mu\left(\frac{1}{r}\right)\frac{\partial}{\partial r}\left[r\frac{\partial v_z}{\partial r}\right] = 0$$

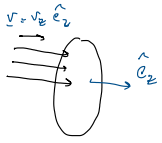
Solve for velocity profile

Boundary conditions: (always have to be about the variable you need to solve in the PDE! (its value or its derivatives value))

@ $r = kR$, $v_z = V$

@ $r = R$, $v_z = 0$

$$Q = \iint_S \hat{n}\cdot\underline{v}\, dS$$



$$= \int_0^{2\pi}\int_{kR}^{R} v_z\, r\, dr\, d\theta$$

$\hookrightarrow \int \underbrace{x}_{u} \underbrace{\ln x}_{dv}\cdot dx$ (easier to calculate derivative of !)

Int. by parts: $\int u\, dv = uv - \int v\, du$ $\quad u = \ln x$ $\quad du = \frac{1}{x}\cdot dx$ $\quad v = \frac{x^2}{2}$ $\quad dv = x$

$$\int d(uv) = \int v\, du + \int u\, dv$$

$$uv = \int v\, du + \int u\, dv$$

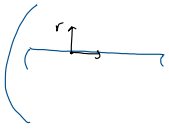
$$\int u\, dv = uv - \int v\, du$$

$$\int x \ln x = \frac{x^2}{2}\ln x - \int \frac{x}{2}\cdot dx$$

Force on wall?

$$F = \iint \left(\hat{n}\cdot\underline{\Pi}\right)\Big|_{@\, r = kR} dS \qquad \Pi = -pI + \underline{\tau}$$

$$=$$



$\hat{e}_r$

$$F = \iint \hat{e}_r \cdot \begin{pmatrix} -p & 0 & \mu\frac{\partial v_z}{\partial r} \\ 0 & -p & 0 \\ \mu\frac{\partial v_z}{\partial r} & 0 & -p \end{pmatrix}_{r\theta z} (r)\, dr\, d\theta$$

kR Constant

$$\begin{pmatrix} \tau_{rr} & \tau_{r\theta} & \tau_{rz} \\ \tau_{\theta r} & \tau_{\theta\theta} & \tau_{\theta z} \\ \tau_{zr} & \tau_{z\theta} & \tau_{zz} \end{pmatrix}_{r\theta z} = \mu \begin{pmatrix} 2\frac{\partial v_r}{\partial r} & r\frac{\partial}{\partial r}\left(\frac{v_\theta}{r}\right) + \frac{1}{r}\frac{\partial v_r}{\partial \theta} & \frac{\partial v_r}{\partial z} + \frac{\partial v_z}{\partial r} \\ r\frac{\partial}{\partial r}\left(\frac{v_\theta}{r}\right) + \frac{1}{r}\frac{\partial v_r}{\partial \theta} & 2\left(\frac{1}{r}\frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r}\right) & \frac{1}{r}\frac{\partial v_z}{\partial \theta} + \frac{\partial v_\theta}{\partial z} \\ \frac{\partial v_r}{\partial z} + \frac{\partial v_z}{\partial r} & \frac{1}{r}\frac{\partial v_z}{\partial \theta} + \frac{\partial v_\theta}{\partial z} & 2\frac{\partial v_z}{\partial z} \end{pmatrix}_{r\theta z}$$

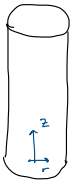
(6.49) Pressure variation

PROBLEM: 6.52 Flow Problem: Upward pressure-driven flow in a tube, power-law non-Newtonian fluid. Repeat problem 6.44 with a power-law, generalized Newtonian fluid with parameters m and n.

PROBLEM: 6.44 Flow Problem: Upward, pressure-driven flow of a Newtonian fluid in a pipe. Calculate the velocity profile and flow rate for pressure-driven flow of an incompressible, Newtonian liquid in a vertical pipe of radius R. The pressure at the bottom entrance to the tube is P_0; at a point a distance L upward the pressure is P_L. Assume that the flow is well developed and at steady state. Do not neglect gravity.

HW problem

$$\underline{v} = \begin{pmatrix} 0 \\ 0 \\ v_z \end{pmatrix} \qquad \underline{g} = \begin{pmatrix} 0 \\ 0 \\ g_z \end{pmatrix}$$



r-direction

$$\rho\left(\frac{\partial v_r}{\partial t}+v_r\frac{\partial v_r}{\partial r}+\frac{v_\theta}{r}\frac{\partial v_r}{\partial \theta}-\frac{v_\theta^2}{r}+v_z\frac{\partial v_r}{\partial z}\right) = -\frac{\partial P}{\partial r}+\rho g_r+\mu\left[\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}(rv_r)\right)+\frac{1}{r^2}\frac{\partial^2 v_r}{\partial \theta^2}-\frac{2}{r^2}\frac{\partial v_\theta}{\partial \theta}+\frac{\partial^2 v_r}{\partial z^2}\right]$$

θ-direction

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z-direction

$$\rho\left(\frac{\partial v_z}{\partial t}+v_r\frac{\partial v_z}{\partial r}+\frac{v_\theta}{r}\frac{\partial v_z}{\partial \theta}+v_z\frac{\partial v_z}{\partial z}\right) = -\frac{\partial P}{\partial z}+\rho g_z+\mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right)+\frac{1}{r^2}\frac{\partial^2 v_z}{\partial \theta^2}+\frac{\partial^2 v_z}{\partial z^2}\right]$$

Power Law : Non-newtonian behavior :

Start from Cauchy balance: $\rho\left(\frac{\partial \underline{v}}{\partial t}+\underline{v}\cdot\nabla\underline{v}\right) = -\nabla p + \nabla\cdot\underline{\underline{\tau}} + \rho\underline{g}$

$$\underline{\underline{\tau}} = \eta(\dot\gamma)\,\underline{\underline{\dot\gamma}} = \eta(\dot\gamma)\left[\nabla\underline{v}+(\nabla\underline{v})^T\right] =$$

$$\mu\begin{pmatrix} 2\frac{\partial v_r}{\partial r} & r\frac{\partial}{\partial r}\left(\frac{v_\theta}{r}\right)+\frac{1}{r}\frac{\partial v_r}{\partial \theta} & \frac{\partial v_r}{\partial z}+\frac{\partial v_z}{\partial r} \\ r\frac{\partial}{\partial r}\left(\frac{v_\theta}{r}\right)+\frac{1}{r}\frac{\partial v_r}{\partial \theta} & 2\left(\frac{1}{r}\frac{\partial v_\theta}{\partial \theta}+\frac{v_r}{r}\right) & \frac{1}{r}\frac{\partial v_z}{\partial \theta}+\frac{\partial v_\theta}{\partial z} \\ \frac{\partial v_r}{\partial z}+\frac{\partial v_z}{\partial r} & \frac{1}{r}\frac{\partial v_z}{\partial \theta}+\frac{\partial v_\theta}{\partial z} & 2\frac{\partial v_z}{\partial z} \end{pmatrix}_{r\theta z}$$

where $\dot\gamma = |\underline{\underline{\dot\gamma}}|$

$$= +\sqrt{\frac{1}{2}\left(\begin{array}{c}\text{Sum of Squares}\\ \text{of each term}\end{array}\right)}$$

$$\eta(\dot\gamma) = m\,\dot\gamma^{n-1} \quad \Big\} \text{ Power law}$$

$$\therefore \tau = m\left(-\frac{\partial v_z}{\partial r}\right)^{n-1}\begin{pmatrix} 0 & 0 & \frac{\partial v_z}{\partial r} \\ 0 & 0 & 0 \\ \frac{\partial v_z}{\partial r} & 0 & 0 \end{pmatrix} = \begin{pmatrix} \tau_{rr} & \tau_{r\theta} & \tau_{rz} \\ \tau_{\theta r} & \tau_{\theta\theta} & \tau_{\theta z} \\ \tau_{zr} & \tau_{z\theta} & \tau_{zz} \end{pmatrix}$$

Momentum equation :

$$\rho\left(\frac{\partial \underline{v}}{\partial t}+\underline{v}\cdot\nabla\underline{v}\right) = -\nabla p + \nabla\cdot\underline{\underline{\tau}} + \rho\underline{g}$$

$$\begin{pmatrix} 0 \\ 0 \\ \cdot \end{pmatrix} \qquad \nabla\cdot\underline{\underline{\tau}} = \begin{pmatrix} \frac{\partial \tau_{zr}}{\partial z} \\ 0 \\ \frac{1}{r}\frac{\partial}{\partial z}(r\tau_{rz}) \end{pmatrix}$$

$$\nabla\cdot\underline{\underline{A}} = \begin{pmatrix} \frac{1}{r}\frac{\partial}{\partial r}(rA_{rr}) + \frac{1}{r}\frac{\partial A_{\theta r}}{\partial \theta} + \frac{\partial A_{zr}}{\partial z} - \frac{A_{\theta\theta}}{r} \\ \frac{1}{r}\frac{\partial}{\partial r}(r^2A_{r\theta}) + \frac{1}{r}\frac{\partial A_{\theta\theta}}{\partial \theta} + \frac{\partial A_{z\theta}}{\partial z} \\ \frac{1}{r}\frac{\partial}{\partial r}(rA_{rz}) + \frac{1}{r}\frac{\partial A_{\theta z}}{\partial \theta} + \frac{\partial A_{zz}}{\partial z} \end{pmatrix}_{rtz} \quad \text{B.3-10}$$

$$\begin{pmatrix} 0 \\ 0 \\ \rho \end{pmatrix}_{z\delta r} \qquad \nabla \cdot \overleftrightarrow{\tau} = \begin{pmatrix} \frac{\partial \tau_{zr}}{\partial z} \\ 0 \\ \frac{1}{r}\frac{\partial}{\partial r}(r\tau_{rz}) \end{pmatrix}$$

$$\downarrow$$

$$\tau_{rz} = \tau_{zr} = m\left(-\frac{\partial v_z}{\partial r}\right)^{n-1}\left(\frac{\partial v_z}{\partial r}\right) = \frac{(-1)}{(-1)}\, m\left(-\frac{\partial v_z}{\partial r}\right)^{n-1}\left(\frac{\partial v_z}{\partial r}\right)$$

$$= \frac{m}{(-1)}\left(-\frac{\partial v_z}{\partial r}\right)^{n-1}\left(-\frac{\partial v_z}{\partial r}\right)$$

$$\tau_{zr} = -m\left(-\frac{\partial v_z}{\partial r}\right)^{n}$$

Macro

Control Volume Balance

$$0 = \frac{dm}{dt} + \iint_{CS} (\hat{n}\cdot\underline{v})\rho\, dS$$

$$\sum_{on\,CV} \underline{f} = \frac{d(m\underline{v})_{CV}}{dt} + \iint_{CS} (\hat{n}\cdot\underline{v})\rho\,\underline{v}\, dS$$

Macro → Micro
- arbitrary CV
- Express terms as integrals
- Leibniz rule & divergence theorem

Microscopic Balance

$$\frac{\partial\rho}{\partial t} + \underline{v}\cdot\nabla\rho = -\rho(\nabla\cdot\underline{v})$$ — Mass

$$\rho\left(\frac{\partial \underline{v}}{\partial t} + \underline{v}\cdot\nabla\underline{v}\right) = -\nabla p + \nabla\cdot\underline{\underline{\tau}} + \rho\underline{g}$$ — Cauchy Momentum

ρ = density
$\underline{v}$ = velocity
p = pressure
$\underline{g}$ = gravity

Navier Stokes

Newtonian + Incom

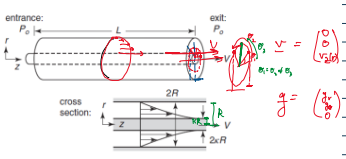
1) Choosing CV (can be arbitrary)
↓
writing out equations specific to that CV
↳ Have to rewrite & recalculate if CV is changed

→

1) Already written for an arbitrary, microscopic CV
↳ applicable to any problem ✓
↳ Just need to change boundary conditions & get velocity & pressure profile!

PROBLEM 6.49 Flow Problem: Axial annular drag with pressure drop, wire coating. Repeat Problem 43 with an imposed pressure gradient $-\Delta p/L$ in the flow direction. Calculate the velocity field only.

PROBLEM: 6.43 Flow Problem: Axial annular drag, wire coating. An incompressible, Newtonian fluid fills the annular gap between a cylindrical wire of radius κR and an outer shell of inner radius R (Figure 6.22). The wire is pulled to the right at speed V. There is no pressure variation throughout the apparatus. Calculate the steady state velocity profile, the flow rate, and the force on the wire. This geometry occurs in wire-coating. Answer: [redacted]



In wire-coating, a wire is drawn through a bath. This axial drag flow is addressed in Problem 43.

1) Coord
2) $v = f(r)$

$\underline{v} = \begin{pmatrix} 0 \\ 0 \\ v_z \end{pmatrix}$; $\underline{g} = \begin{pmatrix} g_r \\ g_\theta \\ g_z \end{pmatrix}$

$v_r = v_\theta = 0$; $\frac{\partial p}{\partial \theta} = 0$

r-direction

$$\rho\left(\frac{\partial v_r}{\partial t} + v_r\frac{\partial v_r}{\partial r} + \frac{v_\theta}{r}\frac{\partial v_r}{\partial \theta} - \frac{v_\theta^2}{r} + v_z\frac{\partial v_r}{\partial z}\right) = \rho g_r - \frac{\partial p}{\partial r} + \mu\left[\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}(r v_r)\right) + \frac{1}{r^2}\frac{\partial^2 v_r}{\partial \theta^2} - \frac{2}{r^2}\frac{\partial v_\theta}{\partial \theta} + \frac{\partial^2 v_r}{\partial z^2}\right]$$

$\frac{\partial p}{\partial r} = 0$

θ-direction

$$\rho\left(\frac{\partial v_\theta}{\partial t} + v_r\frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r}\frac{\partial v_\theta}{\partial \theta} + v_z\frac{\partial v_\theta}{\partial z}\right) = \rho g_\theta - \frac{1}{r}\frac{\partial p}{\partial \theta} + \mu\left[\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}(r v_\theta)\right) + \frac{1}{r^2}\frac{\partial^2 v_\theta}{\partial \theta^2} + \frac{2}{r^2}\frac{\partial v_r}{\partial \theta} + \frac{\partial^2 v_\theta}{\partial z^2}\right]$$

$\frac{\partial p}{\partial \theta} = 0$

z-direction

$$\rho\left(\frac{\partial v_z}{\partial t} + v_r\frac{\partial v_z}{\partial r} + \frac{v_\theta}{r}\frac{\partial v_z}{\partial \theta} + v_z\frac{\partial v_z}{\partial z}\right) = \rho g_z - \frac{\partial p}{\partial z} + \mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 v_z}{\partial \theta^2} + \frac{\partial^2 v_z}{\partial z^2}\right]$$

Constit. Eqn

$\frac{\partial v_z}{\partial z} = 0$; $\frac{\partial (\cdots)}{\partial z} = 0$

$0 = \frac{\partial p}{\partial z} + \mu\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right)$

$P(z)$; $\frac{\partial p}{\partial z} \approx \frac{\Delta p}{\Delta z} = \frac{\Delta p}{L}$

$\frac{\partial p}{\partial z} = \frac{p_L - p_0}{(z_L - z_0)}$; $@\, z_0 = p_0$; $@\, z_L = p_L$

6.43: No pressure variation ; $\underline{v} = \begin{pmatrix} 0 \\ 0 \\ v_z \end{pmatrix}$; $\underline{g} = \begin{pmatrix} g_r \\ g_\theta \\ g_z \end{pmatrix}$

Mass balance / Continuity equation: Incompressible

$$\frac{\partial\rho}{\partial t} + \underline{v}\cdot\nabla\rho = -\rho(\nabla\cdot\underline{v})$$

ss.

$$\nabla\cdot\underline{v} = 0$$

$$\frac{1}{r}\frac{\partial}{\partial r}(r v_r) + \frac{1}{r}\frac{\partial v_\theta}{\partial \theta} + \frac{\partial v_z}{\partial z} = 0$$

$\frac{\partial v_z}{\partial z} = 0$

Start w/ Navier stake:

r-direction : $0 = \rho g_r$

θ-direction : $0 = \rho g_\theta$

z-direction : $0 = \mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right)\right]$

Solve for velocity profile

$$\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right) = 0$$

$$r\frac{\partial v_z}{\partial r} = c_1$$

$$\frac{\partial v_z}{\partial r} = \frac{c_1}{r}$$

$v_z = c_1 \ln r + c_2$

Boundary conditions: always have to be about the variable you need to solve in the PDE! (its value or its derivatives value)

@ $r = \kappa R$; $v_z = V$

@ $r = R$; $v_z = 0$

@ $r = \kappa R$ $v_z = V$ → $V = c_1 \ln \kappa R + c_2$ — 2 eq, 2 unk

@ $r = R$ $v_z = 0$ ⇒ $0 = c_1 \ln R + c_2$

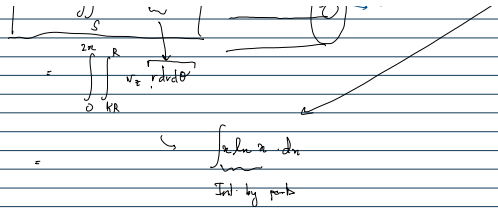
c_1 & c_2

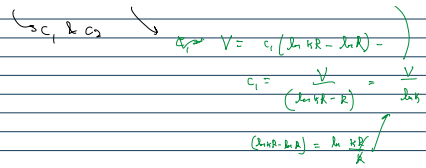
$V = c_1(\ln \kappa R - \ln R)$

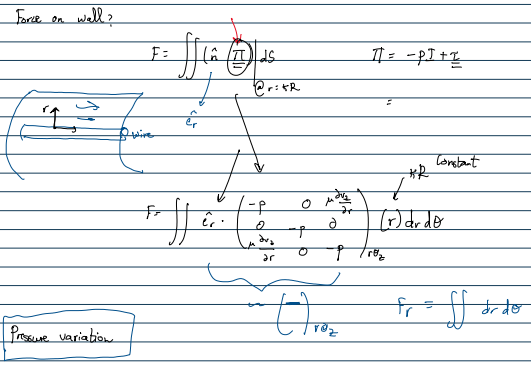
$c_1 = \frac{V}{(\ln \kappa R - \ln R)} = \frac{V}{\ln \kappa}$

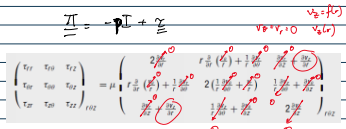
$Q = \iint_S \hat{n}\cdot\underline{v}\, dS$; $\underline{v} = v_z\hat{e}_z$; $\hat{n} = \hat{e}_z$

$= \int_0^{2\pi}\int_\kappa^R v_z\, r\, dr\, d\theta$



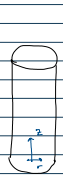


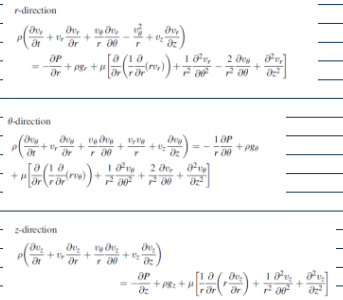




PROBLEM: 6.52 Flow Problem: Upward pressure-driven flow in a tube, power-law non-Newtonian fluid. Repeat problem 6.44 with a power-law, generalized Newtonian fluid with parameters m and n.

PROBLEM: 6.44 Flow Problem: Upward, pressure-driven flow of a Newtonian fluid in a pipe. Calculate the velocity profile and flow rate for pressure-driven flow of an incompressible, Newtonian liquid in a vertical pipe of radius R. The pressure at the bottom entrance to the tube is P_0; at a point a distance L upward the pressure is P_L. Assume that the flow is well developed and at steady state. Do not neglect gravity.





Power Law : Non-newtonian behavior :

Start from Cauchy balance: $\rho\left(\frac{\partial \underline{v}}{\partial t} + \underline{v}\cdot\nabla\underline{v}\right) = -\nabla p + \nabla\cdot\underline{\underline{\tau}} + \rho\underline{g}$

$\downarrow$

$\underline{\underline{\tau}} = \eta(\dot{\gamma})\,\underline{\underline{\dot{\gamma}}}$

$= \eta(\dot{\gamma})\left[\nabla\underline{v} + (\nabla\underline{v})^T\right]$ =

$$\mu\begin{pmatrix} 2\frac{\partial v_r}{\partial r} & r\frac{\partial}{\partial r}\left(\frac{v_\theta}{r}\right) + \frac{1}{r}\frac{\partial v_r}{\partial \theta} & \frac{\partial v_r}{\partial z} + \frac{\partial v_z}{\partial r} \\ r\frac{\partial}{\partial r}\left(\frac{v_\theta}{r}\right) + \frac{1}{r}\frac{\partial v_r}{\partial \theta} & 2\left(\frac{1}{r}\frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r}\right) & \frac{1}{r}\frac{\partial v_z}{\partial \theta} + \frac{\partial v_\theta}{\partial z} \\ \frac{\partial v_r}{\partial z} + \frac{\partial v_z}{\partial r} & \frac{1}{r}\frac{\partial v_z}{\partial \theta} + \frac{\partial v_\theta}{\partial z} & 2\frac{\partial v_z}{\partial z} \end{pmatrix}_{r\theta z}$$

where $\dot{\gamma} = \left|\underline{\underline{\dot{\gamma}}}\right|$

$$= +\sqrt{\frac{1}{2}\left(\begin{matrix}\text{Sum of Squares}\\ \text{of each term}\end{matrix}\right)}$$

$$\eta(\dot{\gamma}) = m\,\dot{\gamma}^{n-1}$$

Momentum equation :

$$\nabla\cdot\underline{\underline{A}} = \begin{pmatrix} \frac{1}{r}\frac{\partial}{\partial r}(rA_{rr}) + \frac{1}{r}\frac{\partial A_{\theta r}}{\partial \theta} + \frac{\partial A_{zr}}{\partial z} - \frac{A_{\theta\theta}}{r} \\ \frac{1}{r}\frac{\partial}{\partial r}(r^2A_{r\theta}) + \frac{1}{r}\frac{\partial A_{\theta\theta}}{\partial \theta} + \frac{\partial A_{z\theta}}{\partial z} + \frac{A_{\theta r}-A_{r\theta}}{r} \\ \frac{1}{r}\frac{\partial}{\partial r}(rA_{rz}) + \frac{1}{r}\frac{\partial A_{\theta z}}{\partial \theta} + \frac{\partial A_{zz}}{\partial z} \end{pmatrix}_{rtz} \qquad \text{B.3-10}$$

$$\tau_{rz} = \tau_{zr} = m\left(-\frac{\partial v_z}{\partial r}\right)^{n-1}\left(\frac{\partial v_z}{\partial r}\right) = \frac{(-1)}{(-1)}\,m\left(-\frac{\partial v_z}{\partial r}\right)^{n-1}\left(\frac{\partial v_z}{\partial r}\right)$$

$$= \frac{m}{(-1)}\left(-\frac{\partial v_z}{\partial r}\right)^{n-1}\left(-\frac{\partial v_z}{\partial r}\right)^{1}$$

$$\tau_{zr} = -m\left(-\frac{\partial v_z}{\partial r}\right)^{n}$$

Macro

Control Volume Balance

$$0 = \frac{dm}{dt} + \iint_{CS} (\hat{n}\cdot\underline{v})\rho\, dS$$

$$\sum_{on\,CV} f = \frac{d(m\underline{v})_{CV}}{dt} + \iint_{CS} (\hat{n}\cdot\underline{v})\rho\underline{v}\, dS$$

Macro → Micro
- arbitrary CV
- Express terms as integrals
- Leibniz rule & divergence theorem

Microscopic Balance

$$\frac{\partial\rho}{\partial t} + \underline{v}\cdot\nabla\rho = -\rho(\nabla\cdot\underline{v})$$ (Mass)

$$\rho\left(\frac{\partial\underline{v}}{\partial t} + \underline{v}\cdot\nabla\underline{v}\right) = -\nabla p + \nabla\cdot\underline{\underline{\tau}} + \rho\underline{g}$$ (Cauchy momentum)

+ Newton, incom. → Navier Stokes

ρ = density
$\underline{v}$ = velocity
p = pressure
$\underline{g}$ = gravity

1) Choosing CV (can be arbitrary)
↓
writing out equations specific to that CV
↳ Have to rewrite & recalculate if CV is changed

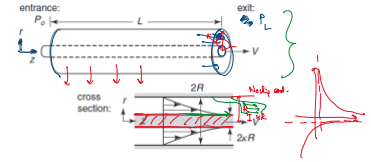
→

2) Already written for an arbitrary, microscopic CV
↳ applicable to any problem ✓
↳ Just need to change boundary conditions & get velocity & pressure profiles!

P → flow direction

PROBLEM: 6.49 Flow Problem: Axial annular drag with pressure drop, wire coating. Repeat Problem 43 with an imposed pressure gradient $-\Delta p/L$ in the flow direction. Calculate the velocity field only.

PROBLEM: 6.43 Flow Problem: Axial annular drag, wire coating. An incompressible Newtonian fluid fills the annular gap between a cylindrical wire of radius kR and an outer shell of inner radius R (Figure 6.22). The wire is pulled to the right at speed V_0. There is no pressure variation throughout the apparatus. Calculate the steady state velocity profile, the flow rate, and the force on the wire. This geometry occurs in wire-coating. Answer: [redacted]



In wire-coating, a wire is drawn through a bath. This axial drag flow is addressed in Problem 43.

r-direction

$$\rho\left(\frac{\partial v_r}{\partial t} + v_r\frac{\partial v_r}{\partial r} + \frac{v_\theta}{r}\frac{\partial v_r}{\partial\theta} - \frac{v_\theta^2}{r} + v_z\frac{\partial v_r}{\partial z}\right) = -\frac{\partial P}{\partial r} + \rho g_r + \mu\left[\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}(rv_r)\right) + \frac{1}{r^2}\frac{\partial^2 v_r}{\partial\theta^2} - \frac{2}{r^2}\frac{\partial v_\theta}{\partial\theta} + \frac{\partial^2 v_r}{\partial z^2}\right]$$

r_c

$$0 = -\frac{\partial p}{\partial r} + \rho g_r \;\Rightarrow\; \frac{\partial p}{\partial r} = 0$$

θ-direction

$$\rho\left(\frac{\partial v_\theta}{\partial t} + v_r\frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r}\frac{\partial v_\theta}{\partial\theta} + \frac{v_r v_\theta}{r} + v_z\frac{\partial v_\theta}{\partial z}\right) = -\frac{1}{r}\frac{\partial P}{\partial\theta} + \rho g_\theta + \mu\left[\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}(rv_\theta)\right) + \frac{1}{r^2}\frac{\partial^2 v_\theta}{\partial\theta^2} + \frac{2}{r^2}\frac{\partial v_r}{\partial\theta} + \frac{\partial^2 v_\theta}{\partial z^2}\right]$$

$$\frac{\partial p}{\partial\theta} = 0$$

$p = f(r,\theta,z)$ $\quad p_L = f(z)$

$$\frac{\partial p}{\partial z} \sim \frac{\Delta p}{\Delta z} \sim \frac{\Delta p}{L}$$

z-direction

$$\rho\left(\frac{\partial v_z}{\partial t} + v_r\frac{\partial v_z}{\partial r} + \frac{v_\theta}{r}\frac{\partial v_z}{\partial\theta} + v_z\frac{\partial v_z}{\partial z}\right) = -\frac{\partial P}{\partial z} + \rho g_z + \mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 v_z}{\partial\theta^2} + \frac{\partial^2 v_z}{\partial z^2}\right]$$

$$0 = -\frac{\partial p}{\partial z} + \mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right)\right]$$

$\frac{\Delta p}{L}$ = constant

Navier Stokes

1) Characterize the flow ✓
$\underline{v}$, $\underline{g}$ = ?
2) Mass / Continuity Balance ✓
3) Navier Stokes ✓
4) BC's

6.43: No pressure variation ; $\underline{v} = \begin{pmatrix} 0 \\ 0 \\ v_z(r) \end{pmatrix}_{rtz}$; $\underline{g} = \begin{pmatrix} g_r \\ g_\theta \\ 0 \end{pmatrix}_{rtz}$

Mass balance / Continuity equation:

$$\frac{\partial\rho}{\partial t} + \underline{v}\cdot\nabla\rho = -\rho(\nabla\cdot\underline{v})$$ (SS, incompressible)

$v_r = v_\theta = 0$, v_z

$$(\nabla\cdot\underline{v}) = 0$$ (coordinate system)

$$\Rightarrow \frac{1}{r}\frac{\partial(rv_r)}{\partial r} + \frac{1}{r}\frac{\partial v_\theta}{\partial\theta} + \frac{\partial v_z}{\partial z} = 0$$

$$\frac{\partial v_z}{\partial z} = 0$$

Newtonian, incomp.

Start w/ Navier stokes:

r-direction : $0 = \rho g_r$

θ-direction : $0 = \rho g_\theta$

z-direction : $\rho\left(v_z\frac{\partial v_z}{\partial z}\right) = \mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right) + \frac{\partial^2 v_z}{\partial z^2}\right]$

$$\mu\left(\frac{1}{r}\right)\frac{\partial}{\partial r}\left[r\frac{\partial v_z}{\partial r}\right] = 0$$ PDE

$$\frac{\partial}{\partial r}\left[r\frac{\partial v_z}{\partial r}\right] = 0$$

$$r\frac{\partial v_z}{\partial r} = c_1$$

$$\frac{\partial v_z}{\partial r} = \frac{c_1}{r}$$

Integrate

$$v_z = c_1\ln r + c_2$$

Solve for velocity profile

Boundary conditions: @ $r = R$, $v_z = 0$

@ $r = kR$, $v_z = V$

always have to be about the variable you need to solve in the PDE! (its value or its derivatives value)

Flow rate → $v = v_z\hat{e}_z$

(its value or its derivatives value)

Flow rate

$$Q = \iint_S (\hat{n}\cdot\underline{v})\,dS$$

$\underline{v} = v_z \hat{e}_z$

$$= \int_0^{2\pi}\int_0^R v_z\, r\,dr\,d\theta$$

$v_z = c_1 \ln r + c_2$

easier to calculate derivative of !

$$\int \underbrace{x}_{u}\,\underbrace{\ln x}_{dv}\,dx$$

Int. by parts.

$$\int u\,dv = uv - \int v\,du \qquad u = \ln x \quad du = \frac{1}{x}dx \qquad v = \frac{x^2}{2} \quad dv = x$$

$$\int d(uv) = \int v\,du + \int u\,dv$$

$$uv = \int v\,du + \int u\,dv$$

$$\int u\,dv = uv - \int v\,du$$

$$\int x\ln x = \frac{x^2}{2}\ln x - \int \frac{x}{2}\,dx$$

Force on wire ?

$$F = \iint_{\text{solid object}} (\hat{n}\cdot\underline{\underline{\Pi}})\,dS \qquad \underline{\underline{\Pi}} = -p\underline{\underline{I}} + \underline{\underline{\tau}}$$

$\hat{e}_r$

$r = R$

$$\underline{\underline{\Pi}} = -p\underline{\underline{I}} + \underline{\underline{\tau}}$$

$$\begin{pmatrix} \tau_{rr} & \tau_{r\theta} & \tau_{rz} \\ \tau_{\theta r} & \tau_{\theta\theta} & \tau_{\theta z} \\ \tau_{zr} & \tau_{z\theta} & \tau_{zz} \end{pmatrix}_{r\theta z} = \mu \begin{pmatrix} 2\frac{\partial v_r}{\partial r} & r\frac{\partial}{\partial r}\left(\frac{v_\theta}{r}\right) + \frac{1}{r}\frac{\partial v_r}{\partial \theta} & \frac{\partial v_r}{\partial z} + \frac{\partial v_z}{\partial r} \\ r\frac{\partial}{\partial r}\left(\frac{v_\theta}{r}\right) + \frac{1}{r}\frac{\partial v_r}{\partial \theta} & 2\left(\frac{1}{r}\frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r}\right) & \frac{1}{r}\frac{\partial v_z}{\partial \theta} + \frac{\partial v_\theta}{\partial z} \\ \frac{\partial v_r}{\partial z} + \frac{\partial v_z}{\partial r} & \frac{1}{r}\frac{\partial v_z}{\partial \theta} + \frac{\partial v_\theta}{\partial z} & 2\frac{\partial v_z}{\partial z} \end{pmatrix}_{r\theta z}$$

1×3 $(1\ 0\ 0)$

3×3

3×3

Constant

$$F = \iint \hat{e}_r \cdot \begin{pmatrix} -p & 0 & \mu\frac{\partial v_z}{\partial r} \\ 0 & -p & \partial \\ \mu\frac{\partial v_z}{\partial r} & 0 & -p \end{pmatrix}_{r\theta z} (r)\,dr\,d\theta$$

(6.49) Pressure variation

$p \rightarrow$ has to be included.

$$F = \iint (\quad)\,R\,dr\,d\theta$$

$r\theta_z \rightarrow f_\theta$, f_r, f_z

1) $\underline{v} = \begin{pmatrix} 0 \\ 0 \\ v_z \end{pmatrix}_{r\theta z}$; $\underline{g} = \begin{pmatrix} 0 \\ 0 \\ g \end{pmatrix}_{r\theta z}$

2) Mass Balance : $\nabla\cdot\underline{v} = 0 \Rightarrow \frac{\partial v_z}{\partial z} = 0$

3) Navier stokes :

PROBLEM: 6.52 Flow Problem: Upward pressure-driven flow in a tube, power-law non-Newtonian fluid. Repeat problem 6.44 with a power-law, generalized Newtonian fluid with parameters m and n.

PROBLEM: 6.44 Flow Problem: Upward, pressure-driven flow of a Newtonian fluid in a pipe. Calculate the velocity profile and flow rate for pressure-driven flow of an incompressible, Newtonian liquid in a vertical pipe of radius R. The pressure at the bottom entrance to the tube is P_0; at a point a distance L upward the pressure is P_L. Assume that the flow is well developed and at steady state. Do not neglect gravity.

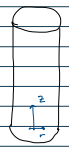
HW problem

$$\underline{v} = \begin{pmatrix} 0 \\ 0 \\ v_z \end{pmatrix} \qquad \underline{g} = \begin{pmatrix} 0 \\ 0 \\ g_z \end{pmatrix}$$

r-direction

$$\rho\left(\frac{\partial v_r}{\partial t} + v_r\frac{\partial v_r}{\partial r} + \frac{v_\theta}{r}\frac{\partial v_r}{\partial \theta} - \frac{v_\theta^2}{r} + v_z\frac{\partial v_r}{\partial z}\right) = -\frac{\partial P}{\partial r} + \rho g_r + \mu\left[\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}(rv_r)\right) + \frac{1}{r^2}\frac{\partial^2 v_r}{\partial \theta^2} - \frac{2}{r^2}\frac{\partial v_\theta}{\partial \theta} + \frac{\partial^2 v_r}{\partial z^2}\right]$$

$$\underline{v} = \begin{pmatrix} 0 \\ 0 \\ v_z \end{pmatrix} \qquad \underline{g} = \begin{pmatrix} 0 \\ 0 \\ g_z \end{pmatrix}$$



r-direction

$$\rho\left(\frac{\partial v_r}{\partial t} + v_r\frac{\partial v_r}{\partial r} + \frac{v_\theta}{r}\frac{\partial v_r}{\partial \theta} - \frac{v_\theta^2}{r} + v_z\frac{\partial v_r}{\partial z}\right) = -\frac{\partial P}{\partial r} + \rho g_r + \mu\left[\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}(rv_r)\right) + \frac{1}{r^2}\frac{\partial^2 v_r}{\partial \theta^2} - \frac{2}{r^2}\frac{\partial v_\theta}{\partial \theta} + \frac{\partial^2 v_r}{\partial z^2}\right]$$

θ-direction

$$\rho\left(\frac{\partial v_\theta}{\partial t} + v_r\frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r}\frac{\partial v_\theta}{\partial \theta} + \frac{v_r v_\theta}{r} + v_z\frac{\partial v_\theta}{\partial z}\right) = -\frac{1}{r}\frac{\partial P}{\partial \theta} + \rho g_\theta + \mu\left[\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}(rv_\theta)\right) + \frac{1}{r^2}\frac{\partial^2 v_\theta}{\partial \theta^2} + \frac{2}{r^2}\frac{\partial v_r}{\partial \theta} + \frac{\partial^2 v_\theta}{\partial z^2}\right]$$

z-direction

$$\rho\left(\frac{\partial v_z}{\partial t} + v_r\frac{\partial v_z}{\partial r} + \frac{v_\theta}{r}\frac{\partial v_z}{\partial \theta} + v_z\frac{\partial v_z}{\partial z}\right) = -\frac{\partial P}{\partial z} + \rho g_z + \mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 v_z}{\partial \theta^2} + \frac{\partial^2 v_z}{\partial z^2}\right]$$

Power Law : Non-newtonian behavior :

Start from Cauchy balance : $\rho\left(\frac{\partial \underline{v}}{\partial t} + \underline{v}\cdot\nabla\underline{v}\right) = -\nabla p + \nabla\cdot\underline{\underline{\tau}} + \rho\underline{g}$

$$\underline{\underline{\tau}} = \eta(\dot\gamma)\,\underline{\underline{\dot\gamma}}$$

$$= \eta(\dot\gamma)\left[\nabla\underline{v} + (\nabla\underline{v})^T\right] = \mu\begin{pmatrix} 2\frac{\partial v_r}{\partial r} & r\frac{\partial}{\partial r}\left(\frac{v_\theta}{r}\right) + \frac{1}{r}\frac{\partial v_r}{\partial \theta} & \frac{\partial v_r}{\partial z} + \frac{\partial v_z}{\partial r} \\ r\frac{\partial}{\partial r}\left(\frac{v_\theta}{r}\right) + \frac{1}{r}\frac{\partial v_r}{\partial \theta} & 2\left(\frac{1}{r}\frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r}\right) & \frac{1}{r}\frac{\partial v_z}{\partial \theta} + \frac{\partial v_\theta}{\partial z} \\ \frac{\partial v_r}{\partial z} + \frac{\partial v_z}{\partial r} & \frac{1}{r}\frac{\partial v_z}{\partial \theta} + \frac{\partial v_\theta}{\partial z} & 2\frac{\partial v_z}{\partial z} \end{pmatrix}_{r\theta z}$$

where $\dot\gamma = |\underline{\underline{\dot\gamma}}|$

$$= +\sqrt{\frac{1}{2}\left(\begin{array}{c}\text{Sum of Squares} \\ \text{of each term}\end{array}\right)}$$

$$\eta(\dot\gamma) = m\,\dot\gamma^{n-1} \quad \Big\} \text{ Power law}$$

$$\therefore \tau = m\left(-\frac{\partial v_z}{\partial r}\right)^{n-1}\begin{pmatrix} 0 & 0 & \frac{\partial v_z}{\partial r} \\ 0 & 0 & 0 \\ \frac{\partial v_z}{\partial r} & 0 & 0 \end{pmatrix} = \begin{pmatrix} \tau_{rr} & \tau_{r\theta} & \tau_{rz} \\ \tau_{\theta r} & \tau_{\theta\theta} & \tau_{\theta z} \\ \tau_{zr} & \tau_{z\theta} & \tau_{zz} \end{pmatrix}$$

Momentum equation :

$$\rho\left(\frac{\partial \underline{v}}{\partial t} + \underline{v}\cdot\nabla\underline{v}\right) = -\nabla p + \nabla\cdot\underline{\underline{\tau}} + \rho\underline{g}$$

$$\begin{pmatrix} 0 \\ 0 \\ g \end{pmatrix}_{r\theta z} \qquad \nabla\cdot\underline{\underline{\tau}} = \begin{pmatrix} \frac{\partial \tau_{zr}}{\partial z} \\ 0 \\ \frac{1}{r}\frac{\partial}{\partial r}(r\tau_{rz}) \end{pmatrix}$$

$$\nabla\cdot\underline{\underline{A}} = \begin{pmatrix} \frac{1}{r}\frac{\partial}{\partial r}(rA_{rr}) + \frac{1}{r}\frac{\partial A_{\theta r}}{\partial \theta} + \frac{\partial A_{zr}}{\partial z} - \frac{A_{\theta\theta}}{r} \\ \frac{1}{r}\frac{\partial}{\partial r}(rA_{r\theta}) + \frac{1}{r}\frac{\partial A_{\theta\theta}}{\partial \theta} + \frac{\partial A_{z\theta}}{\partial z} + \frac{A_{r\theta}}{r} \\ \frac{1}{r}\frac{\partial}{\partial r}(rA_{rz}) + \frac{1}{r}\frac{\partial A_{\theta z}}{\partial \theta} + \frac{\partial A_{zz}}{\partial z} \end{pmatrix}_{rtz} \qquad \text{B.3-10}$$

$$\tau_{rz} = \tau_{zr} = m\left(-\frac{\partial v_z}{\partial r}\right)^{n-1}\left(\frac{\partial v_z}{\partial r}\right) = \frac{(-1)}{(-1)}\,m\left(-\frac{\partial v_z}{\partial r}\right)^{n-1}\left(\frac{\partial v_z}{\partial r}\right)$$

$$= \frac{m}{(-1)}\left(\frac{\partial v_z}{\partial r}\right)^{n-1}\left(-\frac{\partial v_z}{\partial r}\right)^{1}$$

$$\tau_{zr} = -m\left(-\frac{\partial v_z}{\partial r}\right)^{n}$$

Week of 31st

Monday, 30 October 2023 4:51 PM

Realistic, & practical flow:

Turbulent flow: can use the same equations (microscopic balances)

↳ However, flow is no longer unidirectional & SS } time dependent and 3D

$v_\theta, v_r \neq 0$

↓

Cannot solve for velocity profile!

↓

Use experiments to fill in the gaps! → Empirical correlations

Pressure drop vs. flow rate relationships (F_{drag} vs. Q)

Fluid force on a surface S: $\underline{F} = \iint_S [\hat{n} \cdot \underline{\underline{\Pi}}]\big|_{@surface}\, dS$

This force is being exerted by the fluid on the surface

↓

This is the cause of pressure drop in pipe flow!!

$\underline{F}$ ⇒ total force

$\underline{F}_{drag} = \underline{F}_z$ (i.e. z^{th} component of force)

$\therefore \underline{F}_{drag} = \int_0^L \int_0^{2\pi} \hat{e}_z \cdot (-\hat{e}_r \cdot \underline{\underline{\Pi}}\big|_R)\, R\, d\theta\, dz$

↓ non-dimensionalize

$$\underline{F}_{drag} = \int_0^L \int_0^{2\pi} -\mu \left(\frac{\partial v_z}{\partial r} + \frac{\partial v_r}{\partial z} \right)\Big|_R R\, d\theta\, dz$$

Fluid drag in turbulent flow on a pipe surface of unit normal $-\hat{e}_r$

Practice Questions

Laminar flow in a pipe: (Unidirectional, SS) } Continuity + Navier → BC → v profile

$\underline{v} = \begin{pmatrix} 0 \\ 0 \\ v_z \end{pmatrix}_{r\theta z}$; $\underline{g} = \begin{pmatrix} 0 \\ 0 \\ g \end{pmatrix}_{rθz}$ $\quad p = p(z)$

↓

+ Pressure profile: $p(z) = -\left(\frac{p_0 - p_L}{L}\right) z + p_0$

+ Velocity profile: $v_z(r) = \frac{(p_0 - p_L + \rho g L) R^2}{4\mu L} \left[1 - \left(\frac{r}{R}\right)^2 \right]$

$\underline{\underline{\Pi}}(r,z) = -p\underline{\underline{I}} + \underline{\underline{\tau}}$

$= -pI + \begin{pmatrix} 0 & 0 & \mu \frac{\partial v_z}{\partial r} \\ \cdots & \cdots & \cdots \end{pmatrix}$

& $Q = \langle v \rangle \pi R^2 = \frac{\pi (p_0 - p_L) R^4}{8\mu L}$

$$\underline{\underline{\Pi}}(r,z) = -p\underline{\underline{I}} + \underline{\underline{\tau}}$$

$$= -p\underline{\underline{I}} + \begin{pmatrix} 0 & 0 & \mu\frac{\partial v_z}{\partial r} \\ 0 & 0 & 0 \\ \mu\frac{\partial v_z}{\partial r} & 0 & 0 \end{pmatrix}_{rθz}$$

$$= \begin{pmatrix} -p(z) & 0 & (p_L - p_0)\frac{r}{2L} \\ 0 & -p(z) & 0 \\ (p_L - p_0)\frac{r}{2L} & 0 & -p(z) \end{pmatrix}$$

$$\Delta \quad Q = \langle v \rangle \pi R^2 = \frac{\pi (p_0 - p_L) R^4}{8\mu L}$$

$$\langle v \rangle = \frac{(p_0 - p_L) R^2}{8\mu L}$$

→ Neglect gravity:

$$v_z(r) = \frac{(p_0 - p_L + \rho g L)^{0} R^2}{4\mu L}\left[1 - \left(\frac{r}{R}\right)^2\right]$$

$$= \frac{(p_0 - p_L) R^2}{4\mu L} \cdot \frac{2}{2}\left[1 - \left(\frac{r}{R}\right)^2\right]$$

$$v_z(r) = 2\langle v \rangle \left[1 - \left(\frac{r}{R}\right)^2\right]$$

$$\frac{\partial v_z}{\partial r} = \frac{-4\langle v \rangle r}{R^2}$$

$$\underline{F}_{drag} = \int_0^L \int_0^{2\pi} -\mu\left(\frac{\partial v_z}{\partial r} + \frac{\partial v_r}{\partial z}^{0}\right)\Big|_R R\, d\theta\, dz$$

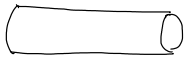
$$= \int_0^L \int_0^{2\pi} -\mu\left(\frac{-4\langle v \rangle r}{R^2}\right)\Big|_R R\, d\theta\, dz$$

$$= (4\mu\langle v \rangle)(2\pi \cdot L)$$

$$= \cancel{4}\cancel{\mu}\cancel{2\pi}\cancel{L} \cdot \left(\frac{(p_0 - p_L) R^2}{\cancel{8}\cancel{\mu}\cancel{L}}\right)$$

$$\boxed{\underline{F}_{drag} = \pi R^2 (p_0 - p_L)}$$

Turbulent flow in smooth pipes:



Prandtl correlation: $\frac{1}{\sqrt{f}} = 4\log\left[Re\sqrt{f}\right] - 0.4$ (Smooth pipes)

7.156

$$Re = \frac{\rho\langle v \rangle D}{\mu}$$

$f = \frac{\Delta p D}{2L\rho\langle v \rangle^2}$ (Pipe friction factor from experiments)

Algorithm 1:

1) Guess Q_{guess} (will give $\langle v \rangle$)
2) Find Re
3) Find f_1 ← Prandtl correlation
4) Find Δp_1 ← friction factor
5) Calculate Δp_2 = known value − Δp_1
6) Solve for "LHS" (given Δp_2) → gives f_2 → $\langle v \rangle$ (using friction factor)

$$\Rightarrow 4\log\left[\frac{\rho\cancel{\langle v \rangle} D}{\mu}\sqrt{\frac{\Delta p D}{2L\rho\cancel{\langle v \rangle^2}}}\right] - 0.4 = \frac{1}{\sqrt{f}}$$

$$\Rightarrow 4\log\left[\frac{\rho D}{\mu}\sqrt{\frac{\Delta p D}{2L\rho}}\right] - 0.4 = \frac{1}{\sqrt{f}}$$

$$\Rightarrow 4\log\left[\sqrt{\frac{\rho \Delta p D^3}{2L\mu^2}}\right] - 0.4 = \frac{1}{\sqrt{f}}$$

5) Calculate Δp_2 = known value − Δp_1

6) Solve for "LHS" (given Δp_2) → gives f_2 → $\langle v \rangle$ (friction factor) using

7) Solve for Q_{calc} [$Q_{calc} = \langle v \rangle \pi R^2$]

8] Compare this to step 1 Q_{guess} → if close then stop
↳ else iterate

$$\left[\frac{\mu}{L} \sqrt{\frac{2}{\rho \Delta p}} \right] \quad \text{if}$$

$$\Rightarrow \underbrace{4 \log \left[\sqrt{\frac{\rho \Delta p D^3}{2L\mu^2}} \right] - 0.4}_{\text{"LHS"}} = \frac{1}{\sqrt{f}}$$

Turbulent flow in connected smooth pipes

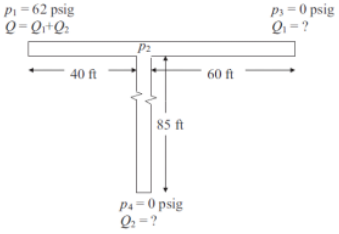
$$\Delta p_2 = P_2 - P_3 = P_2 - P_4$$

1) Guess Δp_2
↓ "LHS"
find f
↓ friction factor
$\langle v \rangle$
↓ ↘
Q_1 Q_2

2) Total Q ($Q_1 + Q_2 = Q$)
↓
Now use Algorithm 1 !
(until step 5)
↳ $\Delta p_1 = p_1 - p_2$

3) $(\Delta p_2)_{calc} = 62 - \Delta p_1$

4) Iterate until $(\Delta p_2)_{calc} \approx (\Delta p_2)_{guess}$



EXAMPLE 7.9. *A 40-foot section of 1.0-inch ID piping branches into two pipes of the same diameter, one of which is 60.0 feet long and one of which is 85.0 feet long (Figure 7.23). The main pipe is connected to the municipal water supply, which supplies a constant water pressure of 62 psig at the pipe entrance. What are the flow rates through the two pipe exits? What is the pressure at the splitting point? Assume smooth pipes.*

① Since we don't know Q
↳ we first need to solve for Q_1 & Q_2 by guessing Δp_2 value

② Once we know Q → then focus on small initial 40 ft pipe and calculate Δp_2 value independtly
↓
See if it matches?
↳ if not we iterate!

Turbulent flow in connected rough pipes

Colebrook correlation: $\frac{1}{\sqrt{f}} = -4 \log \left(\frac{\varepsilon}{D} + \frac{4.67}{Re\sqrt{f}} \right) + 2.28$

$g(f)$

↓

$f = \frac{\Delta p D}{2 L \rho \langle v \rangle^2}$; $Re = \frac{\rho \langle v \rangle D}{\mu}$

↓

$$\frac{1}{\sqrt{f}} = -4 \log \left(\frac{\varepsilon}{D} + \frac{4.67}{\left(\sqrt{\frac{\rho \Delta p D^3}{2L\mu^2}} \right)} \right) + 2.28$$

"LHS" (when we only know pressure drop & no average velocity)

PROBLEM: 7.16

For steady pipe flow, repeat the branched-piping example, Example 7.9, for pipes with roughness $\varepsilon = 0.005$.

Week of 31st

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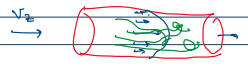
Realistic & practical flow:

Turbulent flow: can use the same equations (microscopic balances)

↳ However, flow is no longer unidirectional & SS } time dependent and 3D PDE

($v_r, v_\theta \neq 0$)



Cannot solve for velocity profile!

Use experiments to fill in the gaps! → Empirical correlations

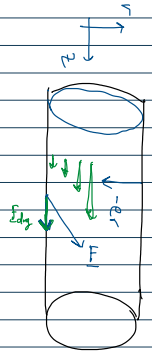
{Pressure drop (F_{drag}) vs. flow rate (Q) relationships}

Fluid force on a surface S: $\underline{F} = \iint_S [\hat{n} \cdot \underline{\Pi}]\big|_{@surface} \, dS$

This force is being exerted by the fluid on the surface

↓

This is the cause of pressure drop in pipe flow!!



$\underline{F}$ ⇒ total force

$\underline{F}_{drag} = \underline{F}_z$ (i.e. z^{th} component of force)

$$\therefore \underline{F}_{drag} = \int_0^L \int_0^{2\pi} \hat{e}_z \cdot \left(-\hat{e}_r \cdot \underline{\Pi}\big|_R\right) R \, d\theta \, dz$$

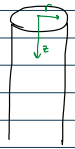
↓ non-dimensionalize

$$\underline{F}_{drag} = \int_0^L \int_0^{2\pi} -\mu \left(\frac{\partial v_z}{\partial r} + \frac{\partial v_r}{\partial z} \right)\Bigg|_R R \, d\theta \, dz$$

Fluid drag in turbulent flow on a piper surface of unit normal $-\hat{e}_r$

Practice Questions

Laminar flow in a pipe: (Unidirectional, SS) } Continuity + Navier → BC → v profile



$\underline{v} = \begin{pmatrix} 0 \\ 0 \\ v_z \end{pmatrix}_{r\theta z}$; $\underline{g} = \begin{pmatrix} 0 \\ 0 \\ g \end{pmatrix}_{r\theta z}$ $\quad p = p(z)$

↓

+ Pressure profile: $p(z) = -\left(\frac{p_0 - p_L}{L}\right) z + p_0$

+ Velocity profile: $v_z(r) = \frac{(p_0 - p_L + \rho g L) R^2}{4 \mu L} \left[1 - \left(\frac{r}{R}\right)^2 \right]$

$$\underline{\underline{\Pi}}(r,z) = -p\underline{\underline{I}} + \underline{\underline{\tau}}$$

$$= -p\underline{\underline{I}} + \begin{pmatrix} 0 & 0 & \mu\frac{\partial v_z}{\partial r} \\ 0 & 0 & 0 \\ \mu\frac{\partial v_z}{\partial r} & 0 & 0 \end{pmatrix}_{rrz}$$

$$= \begin{pmatrix} -p(z) & 0 & (p_L - p_0)\frac{r}{2L} \\ 0 & -p(z) & 0 \\ (p_L - p_0)\frac{r}{2L} & 0 & -p(z) \end{pmatrix}$$

$$\Rightarrow \quad Q = \langle v\rangle \pi R^2 = \frac{\pi (p_0 - p_L) R^4}{8\mu L}$$

$$\langle v\rangle = \frac{(p_0 - p_L) R^2}{8\mu L}$$

→ Neglect gravity:

$$v_z(r) = \frac{(p_0 - p_L + \rho g L) R^2}{4\mu L}\left[1 - \left(\frac{r}{R}\right)^2\right]$$

$$= \frac{(p_0 - p_L) R^2}{4\mu L} \cdot \frac{2}{2}\left[1 - \left(\frac{r}{R}\right)^2\right]$$

$$v_z(r) = 2\langle v\rangle\left[1 - \left(\frac{r}{R}\right)^2\right]$$

$$\frac{\partial v_z}{\partial r} = \frac{-4\langle v\rangle r}{R^2}$$

$$\underline{F}_{drag} = \int_0^L \int_0^{2\pi} -\mu\left(\frac{\partial v_z}{\partial r} + \frac{\partial v_r}{\partial z}\right)\Big|_R R\, d\theta\, dz$$

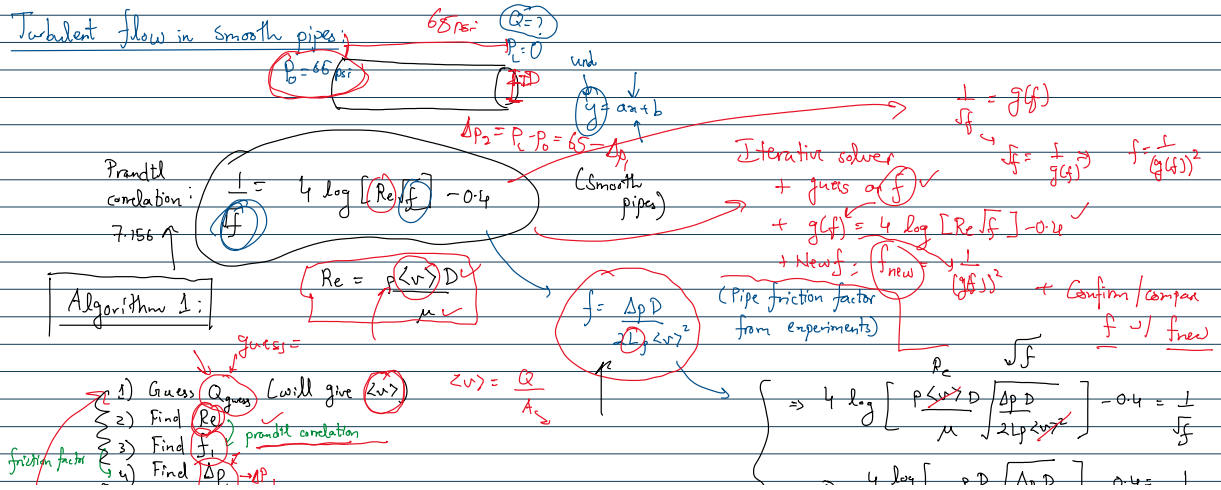
$$= \int_0^L \int_0^{2\pi} -\mu\left(\frac{-4\langle v\rangle r}{R^2}\right)\Big|_R R\, d\theta\, dz$$

$$= (4\mu\langle v\rangle)(2\pi \cdot L)$$

$$= 4\mu 2\pi L \cdot \left(\frac{(p_0 - p_L) R^2}{8\mu L}\right)$$

$$\boxed{\underline{F}_{drag} = \pi R^2 (p_0 - p_L)}$$

Turbulent flow in smooth pipes:



2) Guess Q_{guess} (will give $\langle v \rangle$)
3) Find Re
4) Find f_1 — friction factor — Prandtl correlation
5) Find Δp_1
6) Calculate Δp_2 = known value − Δp_1
7) Solve for "LHS" (given Δp_2) → gives f_2 → $\langle v \rangle$ (friction factor) — using
8) Solve for Q_{calc} [$Q_{calc} = \langle v \rangle \pi R^2$]
9) Compare this to step 1 Q_{guess} → if close then stop
↳ else iterate

Q_{new}

$$\Rightarrow 4 \log\left[\frac{\rho \langle v \rangle D}{\mu}\sqrt{\frac{\Delta p\, D}{2L\rho \langle v \rangle^2}}\right] - 0.4 = \frac{1}{\sqrt{f}}$$

$$\Rightarrow 4 \log\left[\frac{\rho D}{\mu}\sqrt{\frac{\Delta p\, D}{2L\rho}}\right] - 0.4 = \frac{1}{\sqrt{f}}$$

$$\Rightarrow 4 \log\left[\sqrt{\frac{\rho \Delta p D^3}{2L\mu^2}}\right] - 0.4 = \frac{1}{\sqrt{f}}$$

"LHS"

$f(\langle v \rangle)$

Turbulent flow in connected smooth pipes

$\Delta p_2 = p_2 - p_3 = p_2 - p_4$

1) Guess Δp_2 — "LHS"
find f — friction factor
$\langle v \rangle$
Q_1, Q_2

$\langle v_1 \rangle, \langle v_2 \rangle$, $\Delta p_2 \rightarrow Q_1, Q_2$

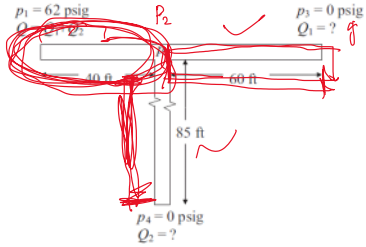
2) Total Q ($Q_1 + Q_2 = Q$)

Now use Algorithm 1 ! (until step 5)

↳ $\Delta p_1 = p_1 - p_2$

3) $(\Delta p_2)_{calc} = 62 - \Delta p_1$

4) Iterate until $(\Delta p_2)_{calc} \approx (\Delta p_2)_{guess}$



EXAMPLE 7.9. *A 40-foot section of 1.0-inch ID piping branches into two pipes of the same diameter, one of which is 60.0 feet long and one of which is 85.0 feet long (Figure 7.23). The main pipe is connected to the municipal water supply, which supplies a constant water pressure of 62 psig at the pipe entrance. What are the flow rates through the two pipe exits? What is the pressure at the splitting point? Assume smooth pipes.*

① Since we don't know Q
↳ we first need to solve for Q_1 & Q_2 by guessing Δp_2 value.

② Once we know Q → then focus on small initial 40 ft pipe and calculate Δp_2 value independtly
↳ See if it matches?
↳ if not we iterate!

Turbulent flow in connected rough pipes

Colebrook correlation: $\frac{1}{\sqrt{f}} = -4 \log\left(\frac{\varepsilon}{D} + \frac{4.67}{Re\sqrt{f}}\right) + 2.28$ → Prandtl correlation

$g(f)$

$f = \frac{\Delta p D}{2L\rho \langle v \rangle^2}$; $Re = \frac{\rho \langle v \rangle D}{\mu}$

$$\frac{1}{\sqrt{f}} = -4 \log\left(\frac{\varepsilon}{D} + \frac{4.67}{\left(\sqrt{\frac{\rho \Delta p D^3}{2L\mu^2}}\right)}\right) + 2.28$$

"LHS" (when we only know Pressure drop)

"LHS" (when we only know Pressure drop & no average velocity)

Week of 31st

Monday, 30 October 2023 4:51 PM

Realistic & practical flow:

Turbulent flow: can use the same equations (microscopic balances)

↳ However, flow is no longer unidirectional & SS } time dependent and 3D

$v_\theta, v_r \neq 0$

3D PDE

CFD

v_z ~~$\neq 0$~~

v_r, v_θ

Cannot solve for velocity profile!

Use experiments to fill in the gaps! → Empirical correlations → darcy friction

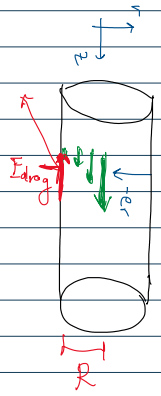
Pressure drop vs. flow rate relationships (F_{drag} vs. Q)

Fluid force on a surface S: $\underline{F} = \iint_S [\hat{n} \cdot \underline{\Pi}]\Big|_{@surface} \, dS$

This force is being exerted by the fluid on the surface

↓

This is the cause of pressure drop in pipe flow!!



$\underline{F}$ ⇒ total force

$\underline{F}_{drag} = \underline{F}_z$ (i.e. z^{th} component of force)

$$\therefore \underline{F}_{drag} = \int_0^L \int_0^{2\pi} \hat{e}_z \cdot \left(-\hat{e}_r \cdot \underline{\Pi}\Big|_R\right) R \, d\theta \, dz$$

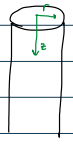
↓ non-dimensionalize

$$\underline{F}_{drag} = \int_0^L \int_0^{2\pi} -\mu \left(\frac{\partial v_z}{\partial r} + \frac{\partial v_r}{\partial z}\right)\Big|_R R \, d\theta \, dz$$

Fluid drag in turbulent flow on a piper surface of unit normal $-\hat{e}_r$

Practice Questions

Laminar flow in a pipe: (Unidirectional, SS) } Continuity + Navier → BC → v profile



$$\underline{v} = \begin{pmatrix} 0 \\ 0 \\ v_z \end{pmatrix}_{r\theta z}, \quad \underline{g} = \begin{pmatrix} 0 \\ 0 \\ g \end{pmatrix}_{rθz} \qquad p = p(z)$$

↓

+ Pressure profile: $p(z) = -\left(\frac{p_0 - p_L}{L}\right) z + p_0$

+ Velocity profile: $v_z(r) = \frac{(p_0 - p_L + \rho g L) R^2}{4\mu L} \left[1 - \left(\frac{r}{R}\right)^2\right]$

$$\underline{\Pi}(r,z) = -p\underline{\underline{I}} + \underline{\underline{\tau}}$$

$$= -p\underline{\underline{I}} + \begin{pmatrix} 0 & 0 & \mu\frac{\partial v_z}{\partial r} \\ \end{pmatrix}$$

⇒ $Q = \langle v \rangle \pi R^2 = \frac{\pi (p_0 - p_L) R^4}{8\mu L}$

$$\underline{\underline{\Pi}}(r,z) = -p\underline{\underline{I}} + \underline{\underline{\tau}}$$

$$= -p\underline{\underline{I}} + \begin{pmatrix} 0 & 0 & \mu\frac{\partial v_z}{\partial r} \\ 0 & 0 & 0 \\ \mu\frac{\partial v_z}{\partial r} & 0 & 0 \end{pmatrix} \quad r\theta z$$

$$= \begin{pmatrix} -p(z) & 0 & (p_L - p_0)\frac{r}{2L} \\ 0 & -p(z) & 0 \\ (p_L - p_0)\frac{r}{2L} & 0 & -p(z) \end{pmatrix}$$

$$\Rightarrow \quad Q = \langle v \rangle \pi R^2 = \frac{\pi (p_0 - p_L) R^4}{8\mu L}$$

$$\langle v \rangle = \frac{(p_0 - p_L) R^2}{8\mu L}$$

→ Neglect gravity:

$$v_z(r) = \frac{(p_0 - p_L + \rho g L) R^2}{4\mu L}\left[1 - \left(\frac{r}{R}\right)^2\right]$$

$$= \frac{(p_0 - p_L) R^2}{4\mu L} \cdot \frac{2}{2}\left[1 - \left(\frac{r}{R}\right)^2\right]$$

$$v_z(r) = 2\langle v \rangle \left[1 - \left(\frac{r}{R}\right)^2\right]$$

$$\frac{\partial v_z}{\partial r} = \frac{-4\langle v \rangle r}{R^2}$$

$$\underline{F}_{drag} = \int_0^L \int_0^{2\pi} -\mu\left(\frac{\partial v_z}{\partial r} + \frac{\partial v_r}{\partial z}\right)\Big|_R R\, d\theta\, dz$$

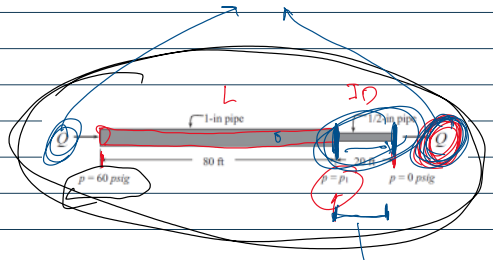
$$= \int_0^L \int_0^{2\pi} -\mu\left(\frac{-4\langle v \rangle r}{R^2}\right)\Big|_R R\, d\theta\, dz$$

$$= (4\mu\langle v \rangle)(2\pi \cdot L)$$

$$= \cancel{4\mu \cdot 2\pi L} \cdot \frac{(p_0 - p_L) R^2}{\cancel{8\mu L}}$$

$$\boxed{\underline{F}_{drag} = \pi R^2 (p_0 - p_L)}$$

Turbulent flow in smooth pipes:



Reynolds

Prandtl correlation 7.1-56

$$\frac{1}{\sqrt{f}} = 4\log\left[Re\sqrt{f}\right] - 0.4$$

friction

$$Re = \frac{\rho\langle v \rangle D}{\mu}$$

(Smooth pipes)

Algorithm 1:

1) Guess Q_{guess} (will give $\langle v \rangle$)
2) Find Re
3) Find f (Prandtl correlation)
4) Find Δp_1 (friction factor)
5) Calculate Δp_2 — known value $= \Delta p_1$ (60 psig)
6) Solve for LHS (given Δp_2) → gives f_2 → $\langle v \rangle$ (using friction factor)

f

$\langle v \rangle$??

$$f = \frac{\Delta p D}{2 L \rho \langle v \rangle^2}$$

(Pipe friction factor from experiments)

Moody friction factor

f in Prandtl

$$\Rightarrow \left\{4\log\left[\frac{\rho\langle v \rangle D}{\mu}\sqrt{\frac{\Delta p D}{2L\rho\langle v \rangle^2}}\right] - 0.4 = \frac{1}{\sqrt{f}}\right.$$

$$\Rightarrow 4\log\left[\frac{\rho D}{\mu}\sqrt{\frac{\Delta p D}{2L\rho}}\right] - 0.4 = \frac{1}{\sqrt{f}}$$

$$\Rightarrow 4\log\left[\sqrt{\frac{\rho \Delta p}{2L\mu^2}}\right] - 0.4 = \frac{1}{\sqrt{f}}$$

Find an Q

Δp, f, New Q

$$\frac{1}{\sqrt{f}} = \left[4\log\left[Re\sqrt{f}\right] - 0.4\right]$$

$g(f)$

→ guess f

→ $g(f)$

→ $f_{new} = \frac{1}{(g(f))^2}$

→ $f \approx f$

6) Solve for "LHS" (given Δp_2) → gives f_2 → $\langle v \rangle$ (friction factor) using

7) Solve for Q_{calc} [$Q_{calc} = \langle v \rangle \pi R^2$]

8] Compare this to step 1 Q_{guess} → if close then stop
↳ else iterate

$$\Rightarrow 4 \log\left[\left(\frac{\rho \Delta p D^3}{2L\mu^2}\right) - 0.4\right] = \frac{1}{\sqrt{f}}$$

"LHS" $\neq f(\langle v_2 \rangle)$

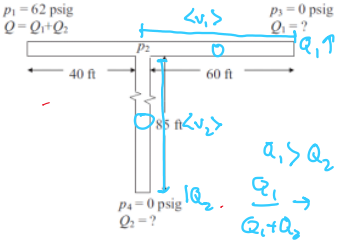
$(g(f))^2$

$\rightarrow f_{new} \approx f$

Rough pipe

Turbulent flow in connected smooth pipes

$$\Delta p_2 = p_2 - p_3 = p_2 - p_4$$



EXAMPLE 7.9. *A 40-foot section of 1.0-inch ID piping branches into two pipes of the same diameter, one of which is 60.0 feet long and one of which is 85.0 feet long (Figure 7.23). The main pipe is connected to the municipal water supply, which supplies a constant water pressure of 62 psig at the pipe entrance. What are the flow rates through the two pipe exits? What is the pressure at the splitting point? Assume smooth pipes.*

1) Guess Δp_2 → "LHS"
↓
find f
↓
friction factor ($f, \Delta p \rightarrow \langle v \rangle$)
↓
$\langle v \rangle$
↓
Q_1 Q_2
$Q_1 + Q_2 = Q$

2) Total Q ($Q_1 + Q_2 = Q$) ↓ Algorithm 1

Now use Algorithm 1 ! (until step 5)

↳ $\Delta p_1 = p_1 - p_2$

3) $(\Delta p_2)_{calc} \neq 62 - \Delta p_1$

4) Iterate until $(\Delta p_2)_{calc} \approx (\Delta p_2)_{guess}$

① Since we don't know Q
↳ we first need to solve for Q_1 & Q_2 by guessing Δp_2 value

② Once we know Q → then focus on small initial 40 ft pipe and calculate Δp_2 value independtly
↓
See if it matches?
↳ if not we iterate!

Turbulent flow in connected rough pipes

Colebrook correlation: $$\frac{1}{\sqrt{f}} = -4 \log\left(\frac{\varepsilon}{D} + \frac{4.67}{Re\sqrt{f}}\right) + 2.28$$

$g(f)$

$$f = \frac{\Delta p D}{2L\rho \langle v \rangle^2} \quad ; \quad Re = \frac{\rho \langle v \rangle D}{\mu}$$

$$\frac{1}{\sqrt{f}} = -4 \log\left(\frac{\varepsilon}{D} + \frac{4.67}{\left(\sqrt{\frac{\rho \Delta p D^3}{2L\mu^2}}\right)}\right) + 2.28$$

"LHS" (when we only know pressure drop & no average velocity)

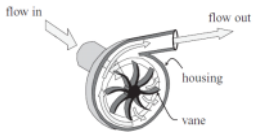
PROBLEM: 7.16

For steady pipe flow, repeat the branched-piping example, Example 7.9, for pipes with roughness $\varepsilon = 0.005$.

Pump Characteristics Curves

Pumps → mechanical devices that allow for fluid transportation via pipes (Section 9.2.4.1)

↳ Centrifugal pumps are a common type → transfer rotational energy (from electric motor) to fluid movement



+ Working principle is energy balance! Electric energy → shaft work (by pump to fluid) → affects speed, pressure & other fluid characteristics
+ Pump manufacturers need to characterize a pump → pump operation under various conditions [Pump curves!]

$\Delta E = Q + W$ (W: work done on system by environment; Q: heat from environment to the system via boundary; ΔE: total energy of system)

Energy balance on CV: $\frac{dE_{cv}}{dt} + \iint_{cs} (\hat{n}\cdot\underline{v})\rho\hat{E}\,dS = \dot{Q}_{in,cv} - \dot{W}_{by,cv}$ $\quad [E_{cv} = U + E_K + E_P]$

- Closed system
- Open system (mass flow in & out)

Some assumptions

$\Delta E_p + \Delta E_K + \Delta U = Q_{in,cv} - W_{by,cv}$ $\quad (W_f = F\cdot d = (P\cdot A)\cdot d)$

$W_{by,cv}$ → W_s shaft work (work performed by fluid due to moving parts) + W_f flow work (work done by fluid as it enters & leaves the system)

$H = U + PV$

SS Macro Energy balance:

$$\Delta E_p + \Delta E_K + \Delta H = Q_{in,cv} - W_{s,by\,cv}$$

Single in, single out, Incompressible fluid

$$\dot{m}\left(\frac{\langle v\rangle_2^2}{2} - \frac{\langle v\rangle_1^2}{2}\right) + \dot{m}g(z_2 - z_1) + \dot{m}\left(\hat{U}_2 - \hat{U}_1 + \frac{P_2}{\rho} - \frac{P_1}{\rho}\right) = \dot{Q}_{in,cv} - \dot{W}_{s,by\,cv}$$

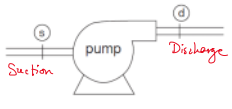
Temp constant, no phase changes/chemical reaction, negligible heat transfer

Mechanical energy Balance (MEB):

$$\frac{\Delta P}{\rho} + \frac{\Delta\langle v\rangle^2}{2\alpha} + g\Delta z + F = -\frac{W_{s,by}}{m}$$

- ρ: density
- α: average velocity ($\alpha = 1$ turbulent flow)
- F: friction term $[\Delta U - Q]$

Pump can add energy via:
+ Increase velocity of fluid
+ Create pressure differential
+ Pump can change fluid height (raise its potential energy)



η = Pump efficiency $= \frac{\text{(useful energy)}}{\text{(total energy)}}$ (energy converted to work (rather than friction))

$$= \frac{\left(\frac{-W_{s,by}}{m} - F\right)}{\left(\frac{-W_{s,by}}{m}\right)}$$

$$\frac{\left(\frac{\dots}{m}\right)}{\left(-\frac{W_{s,by}}{m}\right)}$$

Pump brake horsepower: $bhp = -W_{s,by}$ (Total work delivered by pump)

↓ Dividing by g

$$\therefore \quad \frac{\Delta p}{\rho g} + \frac{\Delta \langle v \rangle^2}{2\alpha g} + \Delta z = \eta \left(-\frac{W_{s,by}}{m}\right) = \eta \cdot (bhp)$$

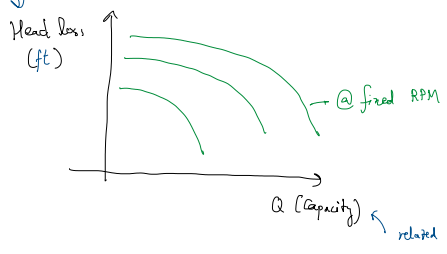
$$\boxed{\left(\frac{P_d}{\rho g} + \frac{\langle v \rangle_d^2}{2g} + z_d\right) - \left(\frac{P_s}{\rho g} + \frac{\langle v \rangle_s^2}{2g} + z_s\right) = \eta \left(-\frac{W_{s,by}}{mg}\right)}$$

Discharge head (H_d) [=] length

Suction head (H_s) [=] length

$$\underbrace{(H_d - H_s)}_{\text{Head loss}} = \eta \left(-\frac{W_{s,by}}{mg}\right) = \eta \left(-\frac{bhp}{mg}\right)$$

gives info about η & bhp



Pumping Ability of a CP:
+ Shaft rotational speed (RPM)
↑ Shape of vanes
+ flow path

Lab: Characterize relationships b/w discharge flow rate, net head, bhp, efficiency in a CP

↳ Vary Q (via rotameter)
↳ Measure pressure loss → calculate head loss (H_{loss} or $H_{net} = H_d - H_s$)
↳ Non-dimensionalize variables & then plot them!

PROBLEM: 8.30 Calculate the extra stress tensor $\underline{\underline{\tau}}$ for potential flow around a sphere of an inviscid, Newtonian fluid. Calculate the total stress tensor $\underline{\underline{\Pi}}$. Comment on what is obtained.

PROBLEM: 8.40 What is the force (i.e., drag) on a thin plate given the following conditions? The fluid is water, the plate is 0.52 m long and 6.3 m wide, and the free-stream velocity is 1.3×10^{-2} m/s.

Boundary layer flow:

$\nabla \cdot \underline{v} = 0$

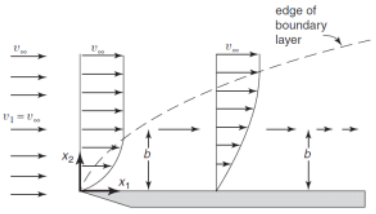
Navier Stokes

} v varying w/ two dimensions → Assumptions → Velocity profile for laminar flow past a flat plate

$$\frac{v_1}{v_\infty} = f'(\zeta) = 1 - (0.5434)\log\left[1 + (68.3)10^{-0.6247\zeta}\right]$$

$$\underline{F} = \iint [\hat{n} \cdot \underline{\underline{\pi}}]_{@surface} \, dS$$

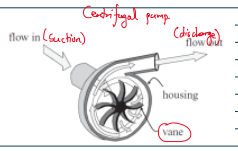
$$\tau_w = 0.332 \mu v_\infty \left(\frac{\rho v_\infty}{\mu x_1}\right)^{1/2} \leftarrow$$ Drag is the force component in flow direction



Pump Characteristics Curves

Pumps → mechanical devices that allow for fluid transportation via pipes (Section 9.2.4.8)

- Centrifugal pumps are a common type → transfer rotational energy (from electric motor) to fluid movement
- Working principle is energy balance! Electric energy → shaft work (by pump to fluid) → affects speed, pressure & other fluid characteristics
- Pump manufacturers need to characterise a pump → Pump operation under various conditions [Pump curves!]



$\Delta E = Q + W$ — total energy of system; Heat from environment to the system via boundary; Work done on system by environment

Energy balance on CV: $\frac{dE_{CV}}{dt} + \iint_{CS} (\hat{n}\cdot\vec{v})\rho\hat{E}\,dS = \dot{Q}_{net} - \dot{W}_{net,CV}$ $\quad [E_{CV} = U + E_K + E_P]$

Closed system; Open system (mass flow in & out)

Some assumptions

$\Delta E_P + \Delta E_K + \Delta U = Q_{in,CV} - W_{sh,CV}$ $\quad (W_f = F\cdot d = (P\cdot A)\cdot d)$

W_s Shaft work (work performed by fluid due to moving parts); W_f Flow work (work done by fluid as it enters & leaves the system)

$H = U + PV$

SS Main Energy balance: $\boxed{\Delta E_P + \Delta E_K + \Delta H = Q_{in,CV} - W_{s,by\,CV}}$

Single in, single out, incompressible fluid

$$\dot{m}\left(\frac{\langle v\rangle_2^2}{2} - \frac{\langle v\rangle_1^2}{2}\right) + \dot{m}g(z_2 - z_1) + \dot{m}\left(\hat{U}_2 - \hat{U}_1 + \frac{P_2}{\rho} - \frac{P_1}{\rho}\right) = \dot{Q}_{in,CV} - \dot{W}_{s,by\,CV}$$

Mechanical energy Balance (MEB)

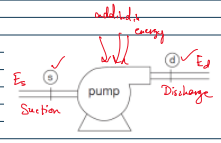
Temp constant, no phase changes/chemical reaction, negligible heat transfer

$$\frac{\Delta P}{\rho} + \frac{\Delta\langle v\rangle^2}{2\alpha} + g\Delta z + F = -\frac{W_{s,by}}{m}$$

ρ density; $\langle v\rangle$ average velocity; $\alpha = 1$ (turbulent flow); F friction term; $[\Delta W - Q]$

Laminar; turbulent

Pump can add energy via:
- Increase velocity of fluid
- create pressure differential
- Pump can change fluid height (raise its potential energy)



η = Pump efficiency = $\frac{\text{(useful energy)}}{\text{(total energy)}}$ (energy converted to work (rather than friction))

$$= \frac{\left(-\frac{W_{s,by}}{m} - F\right)}{\left(-\frac{W_{s,by}}{m}\right)}$$

Pump brake horsepower: $bhp = -W_{s,by}$ (Total work delivered by pump)

$bhp \cdot \eta$

Dividing by g:

$$\frac{\Delta P}{\rho g} + \frac{\Delta\langle v\rangle^2}{2\alpha g} + \Delta z = \eta\left(-\frac{W_{s,by}}{m}\right) = \eta\,(bhp)$$

Δ = out − in; outlet; inlet

$$\frac{\Delta p}{\rho g} + \frac{\Delta\langle v\rangle^2}{2\alpha g} + \Delta z = \eta\left(\frac{-W_{s,by}}{mg}\right) = \eta\,(bhp)$$

$$\left(\frac{p_d}{\rho g} + \frac{\langle v\rangle_d^2}{2g} + z_d\right) - \left(\frac{p_s}{\rho g} + \frac{\langle v\rangle_s^2}{2g} + z_s\right) = \eta\left(\frac{-W_{s,by}}{mg}\right)$$

outlet, inlet, Δ = out − in

Output $= \eta \cdot bhp$

Discharge head (H_d) [=] length

Suction head (H_s) [=] length

$$(H_d - H_s) = \eta\left(\frac{-W_{s,by}}{mg}\right) = \eta\left(\frac{-bhp}{mg}\right)$$

Head loss

give info about $\dot{m}$ & bhp

Head loss (ft) → Q (Capacity) ↑ related to $\langle v\rangle$?

@ fixed RPM

Pumping Ability of a CP:
+ Shaft rotational speed (RPM)
+ Shape of vanes
+ flow path

Lab: Characterize relationships b/w discharge flow rate, net head, bhp, efficiency in a CP

↳ Vary Q (via rotameter)
↳ Measure pressure loss → calculate head loss (H_{loss} or $H_{net} = H_d - H_s$)
↳ Non-dimensionalize variables & then plot them!

ΔH = Head loss; $\frac{\Delta p}{\rho g} + \Delta z + \frac{\Delta\langle v\rangle^2}{2g}$

discharge, Pump, Suction

PROBLEM: 8.30 Calculate the extra stress tensor $\underline{\underline{\tau}}$ for potential flow around a sphere of an inviscid, Newtonian fluid. Calculate the total stress tensor $\underline{\underline{\Pi}}$. Comment on what is obtained.

inviscid → $\mu = 0$ → Shear stress = 0

$$\underline{\underline{\tau}} = 0$$

$$\rho\left(\frac{\partial \underline{v}}{\partial t} + \underline{v}\cdot\nabla\underline{v}\right) = \mu\nabla^2\underline{v} + \rho\underline{g} - \nabla p$$

$$\underline{\underline{\Pi}} = -p\underline{\underline{\delta}} + \underline{\underline{\tau}}$$

dynamic pressure $\hat{p}$

Addison Cycle

Fluid 1 (μ_1)

Interface

Fluid 2 (μ_2) $\tau_1 = \tau_2$

Air ($\mu = 0$)

Fluid 1 (μ_1) → τ@interface $= 0$

PROBLEM: 8.40 What is the force (i.e., drag) on a thin plate given the following conditions? The fluid is water, the plate is 0.52 m long and 6.3 m wide, and the free-stream velocity is 1.3×10^{-2} m/s

$v_{inlet} = v_\infty$ (μ)

$F_r = \iint (\hat{n}\cdot\underline{\underline{\pi}})\big|_\omega \, dS$

6.3 m

0.52 m

$v \propto x_1$

Boundary layer flow:

$\nabla\cdot\underline{v} = 0$

Navier Stokes

v varying w/ two dimensions

Assumptions

PDE

Velocity profile for laminar flow past a flat plate

non-dime x_1

$$\frac{v_1}{v_\infty} = f'(\zeta) = 1 - (0.5434)\log\left[1 + (68.3)10^{-0.6247\zeta}\right]$$

Navier Stokes ✓

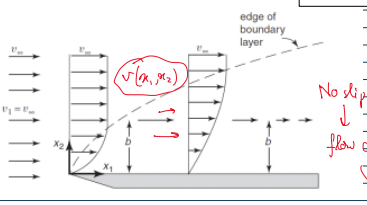
dimensions

laminar flow past a flat plate

$$\vec{F} = \iint_{Surface} [\hat{n} \cdot \vec{\pi}] \, dS$$

$$\tau_w = 0.332 \mu v_\infty \left(\frac{\rho v_\infty}{\mu x} \right)^{1/2}$$

Drag is the force component in flow direction



No slip → flow changes → v

$$\left(\frac{\tau_w}{0.332} \right)^2 = \mu v_\infty^2 \left(\frac{\rho v_\infty}{\mu x_1} \right)$$

$$[\tau] \quad \mu v_\infty^3 \rho \, x_1$$

$$\frac{kg}{m s} \cdot \frac{m^3}{s^3} \cdot \frac{kg}{m^3} \cdot m = \frac{kg^2}{m^2 s^4}$$

$$\tau^2 \quad [\tau] \quad p_a^2$$

$$\frac{kg^2}{m^2 s^4}$$

$$\sum F = m \cdot a \quad \left[\frac{kg \cdot m}{s^2} \right]$$

$$p_a = \frac{F}{A} = \frac{N}{m^2} = \frac{kg \cdot m}{m^2 s^2} = \frac{kg}{m s^2}$$

$$Stress = \frac{F}{A}$$

$$F = \tau_w \cdot A$$

Week of 7th Nov.

Tuesday, 7 November 2023 9:43 AM

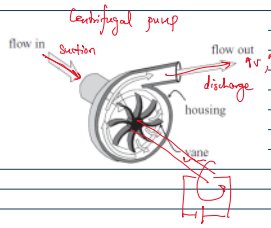
Pump Characteristics Curves

Pumps → mechanical devices that allow for fluid transportation via pipes (Section 9.2.4.1)

↳ Centrifugal pumps are a common type → transfer rotational energy (from electric motor) to fluid movement

+ Working principle is energy balance! Electric energy → Shaft work (by pump to fluid) → affects speed, pressure & other fluid characteristics

+ Pump manufacturers need to characterise a pump → Pump operation under various conditions [Pump curves!]



$\Delta E = Q + W$ (total energy of system; Heat from environment to the system via boundary; Work done on system by environment)

Energy balance on CV: $\frac{dE_{CV}}{dt} + \iint_{CS} (\hat{n}\cdot\vec{v})\rho\hat{E}\,dS = \dot{Q}_{in,CV} - \dot{W}_{s,CV}$ (In, out) $\quad [E_{CV} = U + E_K + E_P]$

Closed system

Open system (mass flow in & out)

Some assumptions

$\Delta E_p + \Delta E_K + \Delta U = Q_{in,CV} - W_{s,by\,CV}$ (W_s = Shaft work (work performed by fluid due to moving parts); $W_s = F\cdot d = (P\cdot A)\cdot d$; flow work ($W_f$) (work done by fluid as it enters & leaves the system))

$H = U + PV$

SS Macro Energy balance

$$\Delta E_p + \Delta E_K + \Delta H = Q_{in,CV} - W_{s,by\,CV}$$

Single in, single out, Incompressible fluid

$$m\left(\frac{\langle v\rangle_2^2}{2} - \frac{\langle v\rangle_1^2}{2}\right) + mg(z_2 - z_1) + m\left(\hat{U}_2 - \hat{U}_1 + \frac{P_2}{\rho} - \frac{P_1}{\rho}\right) = \dot{Q}_{in,CV} - \dot{W}_{s,by\,CV}$$

Temp constant, no phase changes/chemical reaction, negligible heat transfer

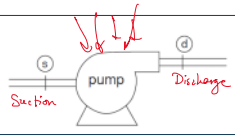
Mechanical energy Balance (MEB)

$$\frac{\Delta P}{\rho} + \frac{\Delta\langle v\rangle^2}{2\alpha} + g\Delta z + F = -\frac{W_{s,by}}{m}$$

(ρ: density; $\langle v\rangle$: average velocity; $\alpha = 1$ (turbulent flow); friction term $[\Delta U - Q]$)

Pump can add energy via:
+ Increase velocity of fluid
+ Create pressure differential
+ Pump can change fluid height (raise its potential energy)

PFD



η = Pump efficiency = (useful energy) / (total energy)

(energy converted to W_s (rather than friction))

$$\eta = \frac{\left(\frac{-W_{s,by}}{m} - F\right)}{\left(\frac{-W_{s,by}}{m}\right)}$$

η $\left(\frac{-W_{s,by}}{m}\right)$

$\left(\frac{-W_{s,by}}{m}\right)$

Pump brake horsepower: $bhp = -W_{s,by}$ (Total work delivered by pump)

↓ Dividing by g

$$\frac{\Delta p}{\rho g} + \frac{\Delta \langle v \rangle^2}{2\alpha g} + \Delta z = \eta \left(\frac{-W_{s,by}}{mg}\right) = \eta \cdot (bhp)$$

out in $\eta \cdot bhp$

$$\left(\frac{P_d}{\rho g} + \frac{\langle v \rangle_d^2}{2g} + z_d\right) - \left(\frac{P_s}{\rho g} + \frac{\langle v \rangle_s^2}{2g} + z_s\right) = \eta \left(\frac{-W_{s,by}}{mg}\right)$$

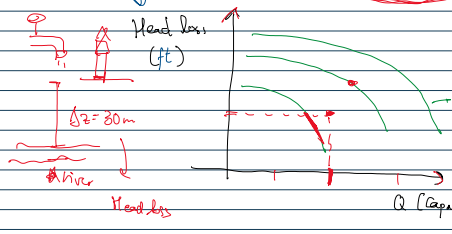
Discharge head (H_d) [=] length

Suction head (H_s) [=] length

$$(H_d - H_s) = \eta \left(\frac{-W_{s,by}}{mg}\right) = \eta \left(\frac{-bhp}{mg}\right)$$

Head loss

give info about η & bhp



Pumping Ability of a CP:
+ Shaft rotational speed (RPM)
↑ Shape of vanes
+ flow path

Lab: Characterize relationships b/w discharge flow rate, net head, bhp, efficiency in a CP

↳ Vary Q (via rotameter)
↳ Measure pressure loss → calculate head loss (H_{loss} or $H_{net} = H_d - H_s$)
↳ Non-dimensionalize variables & then plot them!

Δz, $\langle v \rangle$

PROBLEM: 8.30 Calculate the extra stress tensor $\underline{\underline{\tau}}$ for potential flow around a sphere of an inviscid, Newtonian fluid. Calculate the total stress tensor $\underline{\underline{\Pi}}$. Comment on what is obtained.

$\mu = 0$; $Re \to \infty$

relativistic

$\tau_{xy} = 0$ air

liquid/interface

flow

∇P

$$\tau = \mu \left((\nabla \underline{v}) + (\nabla \underline{v})^T \right)$$

$$\rho \left(\frac{\partial \underline{v}}{\partial t} + (\nabla \underline{v}) \cdot \underline{v} \right) = -\nabla P + \mu (\nabla^2 \underline{v}) + \rho \underline{g}$$

$\tau \approx 0$ ✓

nondim

$\frac{1}{Re} (\nabla^2 \underline{v})$

$Re \to \infty$

$\underline{\underline{\Pi}} = \underline{\underline{\tau}} - p \underline{\underline{I}}$

Dynamic pressure

$\nabla P = \nabla p + \rho g$

PROBLEM: 8.40 What is the force (i.e., drag) on a thin plate given the following conditions? The fluid is water, the plate is 0.52 m long and 6.3 m wide, and the free-stream velocity is 1.3×10^{-2} m/s.

Boundary layer flow:

$\nabla \cdot \underline{v} = 0$

Navier stokes

v varying in two dimensions

Assumptions

Velocity profile for laminar flow past a flat plate

$$\frac{v_1}{v_\infty} = f(x_2) = 1 - (0.5434)\log\left[1 + (68.3)10^{-0.624 x_2}\right]$$

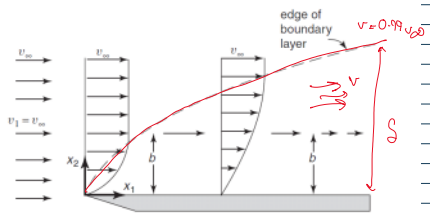
$$\underline{F} = \iint_{\text{surface}} [\hat{n} \cdot \underline{\Pi}]\, dS$$

$\underline{\Pi} = -p\underline{I} + \underline{\tau}$

Drag is the force component in flow direction

$\underline{\tau} = \left(\frac{\partial v}{\partial n}\right)$

$\tau_w = 0.332 \mu v_\infty \left(\frac{\rho v_\infty}{\mu x_1}\right)^{1/2}$



Week of 27th Nov.

Tuesday, 14 November 2023 10:35 AM

Shift in perspective : Flow profiles (microscopic look) to Large scale Info
→ Total force on wall
→ Overall flow rate
→ Total work performed by a pump/turbine miner

Macroscopic Mass Balance

Mass balance on a CV : $\frac{dm_{cv}}{dt} + \iint_{cs} \rho (\hat{n}\cdot\underline{v})\, dS = 0$

→ Multiple inlet & outlet
→ constant density

$\sum \rho_i \iint_{A_i} (\hat{n}\cdot\underline{v})\, dS$

$\underline{v} = v\,\hat{e}_v$ (mag. × "velocity" direction [constant @ each inlet/outlet] (or unit vector))

$\sum \rho_i \iint_{A_i} v \cos\theta\, dS = \sum \rho_i \cos\theta \underbrace{\iint_{A_i} v\, dS}$

$\langle v \rangle = \frac{\iint_{A_i} v\cdot dS}{\iint_{A_i} dS} = \frac{1}{A_i}\iint_{A_i} v\cdot dS$

$\therefore \iint_{A_i} v\cdot ds = \langle v \rangle A_i$

$$\boxed{\frac{dm_{cv}}{dt} + \sum_{i=1}^{\#\text{ streams}} [\rho A \cos\theta \langle v \rangle] = 0}$$

θ_i = Angle b/w $\hat{n}_i$ & $\hat{v}_i$ (will carry the negative sign depending on inlet & outlet)

Macroscopic Momentum Balance

$\frac{dP}{dt} + \iint_{cs} (\hat{n}\cdot\underline{v})\rho\underline{v}\, dS = \sum_{\text{on cv}} \underline{f}$

$\sum \rho_i \cos\theta\, \hat{v} \iint v^2 \cdot dS = \sum_{i=1}^{\#\text{ streams}} \rho A \cos\theta \langle v^2 \rangle \hat{v}$

$\beta = \frac{\langle v \rangle^2}{\langle v^2 \rangle}$; $\beta = 1$ (turbulent) ; $\beta = 0.75$ (laminar)

$\langle v^2 \rangle$ somewhat hard to compute

$$\boxed{\frac{dP}{dt} + \sum_{i=1}^{\#} \left[\frac{\rho A \cos\theta \langle v \rangle^2}{\beta}\hat{v}\right] = \sum_{\text{on cv}} \underline{f}}$$

$\hat{v}$: unit vector!

$= \underline{f}_g + \underline{f}_{inlet} + \underline{f}_{outlet} + \underline{f}_{walls}$

$\underline{f}_g$: mg

$\iint [\hat{n}\cdot\Pi]\, ds = \iint \hat{n}\cdot[-p\underline{I} + \underline{\tau}]\, ds$

$= \iint -p\hat{n}\, dS + \iint \hat{n}\cdot\underline{\tau}\, ds$ ($\underline{\tau} \to 0$)

pressure dominated

(Viscous terms accounted for in $\underline{f}_{walls}$!)

Macroscopic Energy Balance

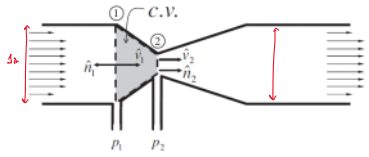
$\frac{dE}{dt} + \iint_{cs} (\hat{n}\cdot\underline{v})\rho\hat{E}\, dS = Q_{in} - W_{by}$

$\alpha = \frac{\langle v \rangle^3}{\langle v^3 \rangle}$; $\alpha \approx 1$ (turbulent)

→ Open sys,
→ $E = E_K + E_P + U$
→ constant velocity across inlet/outlet
→ $\hat{H} = \hat{U} + P\hat{V}$
→ Flow work / shaft work
→ No phase change
→ SS, single input/single output
→ $\Delta T \sim 0$, no reaction

$$\boxed{\frac{\Delta p}{\rho} + \frac{\Delta\langle v \rangle^2}{2\alpha} + g\Delta z + F = -\frac{W_{s,by}}{m}}$$

PROBLEM: 1.21 A Venturi meter with a 1.00 *in* diameter throat is to be installed into a 2 *in* line (Schedule 40 piping, smooth) with water flowing at 25°C. If the flow is turbulent, and the range of expected flow rates is 0 – 200 *gpm*, what is the expected range of pressure drop in the Venturi meter? You may neglect friction.



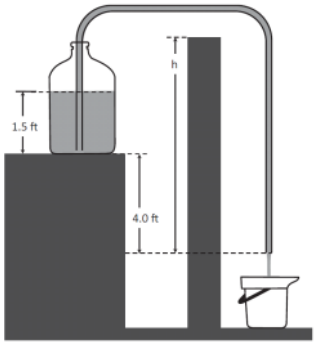
In a Venturi meter, flow is directed through a gently tapering tube. Pressure is measured before the contraction (1) and at the point of smallest diameter (2) (i.e., the throat). The relationship between the measured pressures and the fluid velocity may be deduced from the mechanical energy balance with assistance of the macroscopic mass balance.

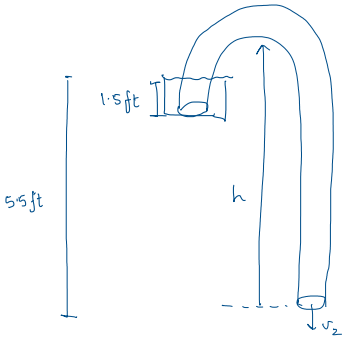
2" Schedule 40 pipe; ID = 2.067"

Pg 772 [Ex. 9.5]

(1) Velocity relationships (v_1 vs. v_2)?

PROBLEM: 1.23 A gasoline tank is connected to a 25 ft hose (ID = 1.50 cm) as shown in Figure 1.49. The ambient temperature is 38°C. What is the maximum height of the barrier over which the gasoline may be siphoned? You may neglect frictional losses. Note the following physical property data: density of gasoline=5.6 lb_m/gal, vapor pressure at 38°C is 12.3 $psia$.





(1) What is the exit velocity?

Siphon flow

Can treat the fluid as a continuum (no breaks) (free to choose the pressure that is the most convinient)

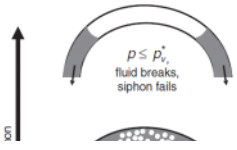
↓

As liquid level is falling (the pressure @ bottom will be changing)

↳ hard to use that

(2) Max height: fluid will stop flowing when suction breaks (i.e. gas/air is introduced somehow)

↳ This can happen when the pressure equals vapor pressure (to "boil" the liquid)



↳ This can happen when the pressure equals vapor pressure (to "boil" the liquid)

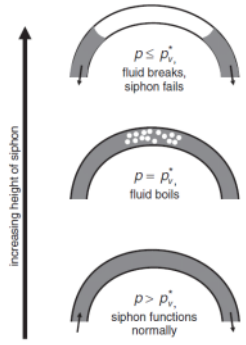
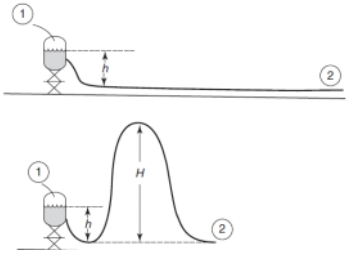


Figure 1.16 Schematic of the formation of a vapor gap in a siphon. This occurs when the barrier over which the fluid travels rises above a critical height; the critical height may be calculated as shown in the example.

PROBLEM: 1.24 A water tank is connected to a 100 ft hose (ID= 1.50 cm) as shown in the top of Figure 1.50. The height h is 1.8 m. Calculate the average velocity of water in the hose. Do not neglect friction; you may assume turbulent flow.



Week of 27th Nov.

Tuesday, 14 November 2023 10:35 AM

Shift in perspective: Flow profiles (microscopic look) to Large scale Info (Practical)
→ Total force on wall
→ Overall flow rate
→ Total work performed by a pump/turbine mixer

Navier Stoke → $v(r,\theta,z)$

Macroscopic Mass Balance

Mass balance on a CV: $\frac{dm_{cv}}{dt} + \iint_{CS} \rho(\hat{n}\cdot\underline{v})\,dS = 0$ → CS_1, CS_2

→ Multiple inlet & outlet
→ constant density

$\sum \rho_i \iint_{A_i} (\hat{n}\cdot\underline{v})\,dS$

$(\hat{n}\cdot\underline{v}) = \|\hat{n}\|\,\|\underline{v}\|\cos\theta$ ($\|\hat{n}\| = 1$, $\|\underline{v}\| = v$)

$\underline{v} = v\,\hat{e}_v$ (mag. × "velocity" direction [constant @ each inlet/outlet] (or unit vector))

$\sum \rho_i \iint_{A_i} v\cos\theta\,dS = \sum \rho_i \cos\theta \iint_{A_i} v\,dS$

$\langle v\rangle = \frac{\iint_{A_i} v\cdot dS}{\iint_{A_i} dS} = \frac{1}{A_i}\iint_{A_i} v\cdot dS$

$\therefore \iint_{A_i} v\cdot ds = \langle v\rangle A_i$

Macroscopic Mass Balance

$\frac{dm_{cv}}{dt} + \sum_{i=1}^{\#\text{ streams}} [\rho A \cos\theta \langle v\rangle] = 0$

θ_i = Angle b/w $\hat{n}_i$ & $\hat{v}_i$ (will carry the negative sign depending on inlet & outlet)

$\theta = 0$, $\cos(0) = 1$

$\theta = 180$, $\cos\theta < 0$

Macroscopic Momentum Balance

$\frac{dP}{dt} + \iint_{CS} (\hat{n}\cdot\underline{v})\,\rho\underline{v}\,dS = \sum_{\text{on CV}} \underline{f}$ (vector)

$\sum \rho_i \cos\theta\,\hat{v} \iint v\cdot ds = \sum_{i=1}^{\#\text{ streams}} \rho A \cos\theta \langle v^2\rangle \hat{v}$ (somewhat hard to compute)

$\hat{e}_v$

$\beta = 1$ (turbulent); $\beta = 0.75$ (Laminar)

$\beta = \frac{\langle v\rangle^2}{\langle v^2\rangle}$

$\langle v^2\rangle \neq \langle v\rangle^2$

$\langle v\rangle^2 = \left(\frac{\iint v\,ds}{\iint ds}\right)^2$; $\langle v^2\rangle = \frac{\iint v^2\,ds}{\iint ds}$

$\langle v^2\rangle \to \langle v\rangle^2$

$\frac{dP}{dt} + \sum_{i=1}^{\#} \left[\frac{\rho A \cos\theta \langle v\rangle^2}{\beta}\hat{v}\right] = \sum_{\text{on CV}} \underline{f}$ (unit vector!)

$= \underline{f}_g + \underline{f}_{\text{inlet}} + \underline{f}_{\text{outlet}} + \underline{f}_{\text{walls}}$

$\underline{f}_g$ = mg

$\iint [\hat{n}\cdot\Pi]\,ds = \iint \hat{n}\cdot[-p\underline{I} + \underline{\tau}]\,ds$

$= \iint -p\hat{n}\,dS + \iint \hat{n}\cdot\underline{\tau}\,ds$ (pressure dominated; viscous terms accounted for in f_{wall}!)

f = Force = (Pressure)·A

$= -P_1 A_1$; $f_2 = P_2 A_2$

Macroscopic Energy Balance

$\frac{dE}{dt} + \iint_{CS} (\hat{n}\cdot\underline{v})\,\rho\hat{E}\,dS = Q_{in} - W_{by}$

→ Open sys.
→ $E = E_K + E_P + U$
→ constant velocity across inlet/outlet
→ $\hat{H} = \hat{U} + P\hat{V}$
→ Flow work / shaft work
→ No phase change
→ SS, single input/single output
→ $\Delta T = 0$, no reaction

$\alpha = \frac{\langle v\rangle^3}{\langle v^3\rangle}$; $\alpha \approx 1$ (turbulent) ; $\alpha = 0.99$

$\frac{\Delta P}{\rho g} + \frac{\Delta\langle v\rangle^2}{g2\alpha} + \frac{\Delta z}{} + \frac{F}{g} = \frac{-W_{s,by}}{mg}$

Net head = H_d

$= H_d + h_f + h_p + h_{vf} + \sum \left[\frac{}{}\right]$ length

MEB → Mechanical Energy Balance

h + z

$f_f \frac{L}{D}\frac{\Delta P}{}$

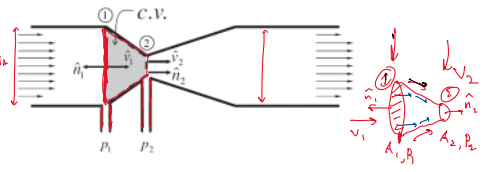
closed form

$f = \frac{A}{\sqrt{f}}$

PROBLEM: 1.21 A Venturi meter with a 1.00 *in* diameter throat is to be installed into a 2 *in* line (Schedule 40 piping, smooth) with water flowing at 25°C. If the flow is turbulent and the range of expected flow rates is 0 – 200 *gpm*, what is the expected range of pressure drop in the Venturi meter? You may neglect friction.

0 gpm 200 gpm

$\alpha = \beta = 1$



In a Venturi meter, flow is directed through a gently tapering tube. Pressure is measured before the contraction (1) and at the point of smallest diameter (2) (i.e., the throat). The relationship between the measured pressures and the fluid velocity may be deduced from the mechanical energy balance with assistance of the macroscopic mass balance.

2" Schedule 40 pipe: ID = 2.067"

Pg 792 [Ex. 9.5]

(1) Velocity relationships (v_1 vs. v_2)?

$$\frac{dm}{dt} + \sum_i \rho \cos\theta_i \langle v \rangle_i A_i = 0$$

Mass Balance:

$$\cancel{\frac{dm}{dt}} + \rho_1 A_1 \cos\theta_1 \langle v_1 \rangle + \rho_2 A_2 \cos\theta_2 \langle v_2 \rangle = 0$$

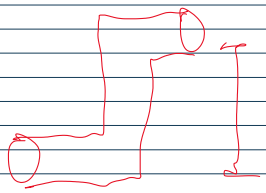
$$\Rightarrow \rho_1 A_1 (-1) \langle v_1 \rangle + \rho_2 A_2 (1) \langle v_2 \rangle = 0$$

$$\rho_1 A_1 \langle v_1 \rangle = \rho_2 A_2 \langle v_2 \rangle \quad \checkmark$$

$$\boxed{Q_1 = Q_2} \quad (\rho_1 = \rho_2)$$

$$\frac{\pi D_1^2}{4} \langle v_1 \rangle = \frac{\pi D_2^2}{4} \langle v_2 \rangle$$

$$\langle v_1 \rangle = \left(\frac{D_2}{D_1}\right)^2 \langle v_2 \rangle$$



(2) Pressure drop? [MEB]

$$\frac{\Delta P}{\rho} + \frac{\Delta \langle v \rangle^2}{2\alpha} + g\Delta z + F = -\frac{W_s}{m}$$

$\alpha = 1$:

$$\frac{\Delta P}{\rho} + \frac{\Delta \langle v \rangle^2}{2} = 0$$

(Out - in)

$$\frac{P_2 - P_1}{\rho} = -\frac{(\langle v_2 \rangle^2 - \langle v_1 \rangle^2)}{2} \quad ; \quad \langle v_1 \rangle = \left(\frac{D_2}{D_1}\right)^2 \langle v_2 \rangle$$

$$\frac{P_1 - P_2}{1} = \frac{1}{2}\left(\langle v_2 \rangle^2 - \left[\left(\frac{D_2}{D_1}\right)^2 \langle v_2 \rangle\right]^2\right)$$

$\langle v_2 \rangle \approx v_2$

$$\boxed{-\Delta P = \frac{\rho \langle v_2 \rangle^2}{2}\left(1 - \left(\frac{D_2}{D_1}\right)^4\right)}$$

$$v_2 = \frac{Q}{A_2}$$

$$\rightarrow \left[\text{Pressure drop} = -\Delta P = -(P_2 - P_1)\right]$$

"eng."

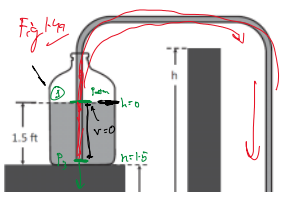
PROBLEM: 1.23 A gasoline tank is connected to a 25 *ft* hose (ID = 1.50 *cm*) as shown in Figure 1.49. The ambient temperature is 38°C. What is the maximum height of the barrier over which the gasoline may be siphoned? You may neglect frictional losses. Note the following physical property data: density of gasoline=5.6 lb_m/gal, vapor pressure at 38°C is 12.3 *psia*.

Siphon flow

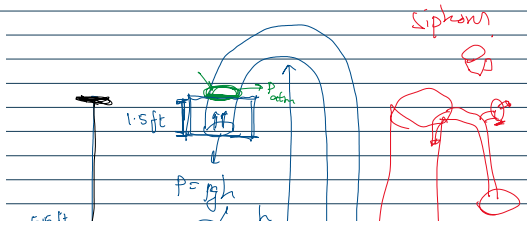
You treat the fluid as a continuum (no breaks) (free to choose the pressure that is the most convinient)

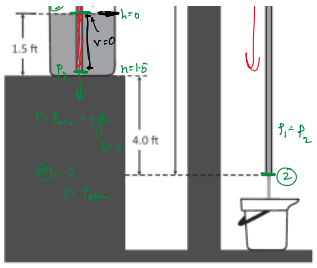
As liquid level is falling (the pressure @ bottom still be changing) → hard to use that

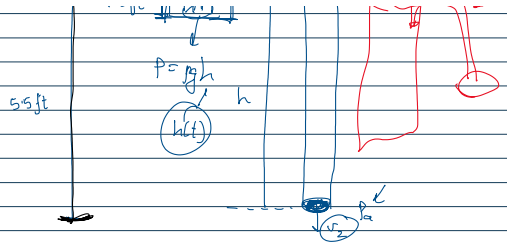
Fig 1.49



Siphon







(1) What is the exit velocity?

Mechanical energy balance:

$$\frac{\Delta P}{\rho} + \frac{\Delta(v^2)}{2\alpha} + g\Delta z + \hat{F} = -\frac{W_{s,m}}{m}$$

W_s shaft; $\hat{F}$ friction; $\alpha = 1$ (turbulent)

$$\frac{(P_2 - P_1)}{\rho} + \frac{v_2^2 - v_1^2}{2} + g(z_2 - z_1) = 0$$

$$\frac{v_2^2}{2} = -g(z_2 - z_1)$$

$$v_2 = \sqrt{2g(z_1 - z_2)}$$

$z_1 - z_2 = 5.5$

(2) Max height: fluid will stop flowing when suction breaks (i.e. gas/air is introduced somehow)

↳ This can happen when the pressure equals vapor pressure (to "boil" the liquid)

Find the height @ $P = P_{vap}$

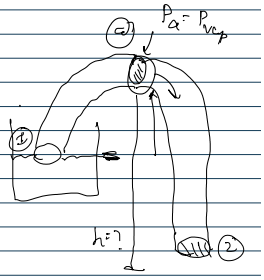
$$\frac{\Delta P}{\rho} + \frac{\Delta v^2}{2\alpha} + g\Delta z + \hat{F} = -\frac{W_s}{m}$$

$P_{vap} = 12.3$ $\quad z_2 - z_1$

$$\frac{P_2 - P_a}{\rho} + \frac{v_2 - v_a}{2\alpha} + g(+h) = 0$$

atm = 14.19

$h = 0$



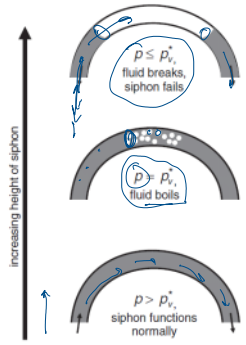
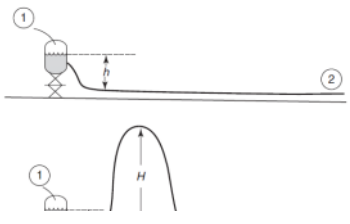
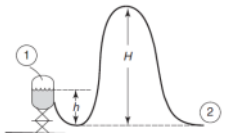


Figure 1.16 Schematic of the formation of a vapor gap in a siphon. This occurs when the barrier over which the fluid travels rises above a critical height; the critical height may be calculated as shown in the example.

PROBLEM: 1.24 A water tank is connected to a 100 *ft* hose (ID= 1.50 *cm*) as shown in the top of Figure 1.50. The height h is 1.8 *m*. Calculate the average velocity of water in the hose. Do not neglect friction; you may assume turbulent flow.





ζL_{32}

Shift in perspective : Flow profiles (microscopic look) → Large scale Info
- → Total force on wall
- → Overall flow rate
- → Total work performed by a pump/turbine motor

Hard to model $v(r, \theta, z)$

Macroscopic Mass Balance

Macro / Micro / Meso

$v = v_i \hat{e}_v$

Mass balance on a CV : $\frac{dm_{CV}}{dt} + \iint_{CS} \rho (\hat{n} \cdot \underline{v})\, dS = 0$

- → Multiple inlet & outlet : $\sum \rho_i \iint_{A_i} (\hat{n} \cdot \underline{v})\, dS$
- → constant density

$\underline{v} = v \hat{e}_v$ (mag. × "velocity" direction [constant @ each inlet/outlet] (or unit vector))

$(\hat{n} \cdot \underline{v}) = v(\hat{n} \cdot \hat{e}_v) \Rightarrow v(1)(1)\cos\theta$

$\underline{a} \cdot \underline{b} = \|a\| \|b\| \cos\theta$ — dot product

$\sum \rho_i \iint_{A_i} v \cos\theta\, dS = \sum \rho_i \cos\theta \iint_{A_i} v\, dS$

$\langle v \rangle = \frac{\iint_{A_i} v \cdot dS}{\iint_{A_i} dS} = \frac{1}{A_i} \iint_{A_i} v \cdot dS$

$\therefore \iint_{A_i} v \cdot ds = \langle v \rangle A_i$

$\frac{dm_{CV}}{dt} + \sum_{i=1}^{\#\text{ streams}} [\rho A \cos\theta \langle v \rangle] = 0$

θ_i = Angle b/w $\hat{n}_i$ & $\hat{v}_i$ (will carry the negative sign depending on inlet & outlet)

Macroscopic Momentum Balance

$F = \frac{dP}{dt} = \frac{d(mv)}{dt}$

$\frac{dP}{dt} + \iint_{CS} (\hat{n} \cdot \underline{v}) \rho \underline{v}\, dS = \sum_{\text{on CV}} \underline{f}$

$\underline{v} = v \hat{e}_v$

$\sum \rho_i \cos\theta\, \hat{v} \iint v^2 \cdot dS = \sum_{i=1}^{\#\text{ streams}} \rho A \cos\theta \langle v^2 \rangle \hat{v}$ (somewhat hard to compute)

$\langle v^2 \rangle = \frac{\iint v^3 \cdot d_s}{A_i}$

$\beta = 1$ (turbulent) ; $\beta = 0.75$ (laminar)

$\beta = \frac{\langle v^2 \rangle}{\langle v \rangle^2}$

$\beta = 1$, $\beta = 0.75$, $\langle v^2 \rangle$, $\langle v \rangle^2 \rightarrow \beta$

$\frac{dP}{dt} + \sum_{i=1}^{\#} \left[\rho A \cos\theta \langle v \rangle^2 \beta \hat{v} \right] = \sum_{\text{on CV}} \underline{f}$ (unit vector!)

$= \underline{f}_g + \underline{f}_{\text{inlet}} + \underline{f}_{\text{outlet}} + \underline{f}_{\text{walls}}$

mg

$\iint [\hat{n} \cdot \Pi]\, ds = \iint \hat{n} \cdot [-p\underline{\underline{I}} + \underline{\underline{\tau}}]\, dS$

$= \iint -p \hat{n}\, dS + \iint \hat{n} \cdot \underline{\underline{\tau}}\, ds$ (pressure dominated; $\underline{\underline{\tau}} \rightarrow 0$; viscous terms accounted for in $\underline{f}_{\text{walls}}$!)

$\underline{f}_{in} = -P_1 A_1 \hat{n}_{A_1}$

$\underline{f}_{out} = -P_2 A_2 \hat{n}_{A_2}$

Macroscopic Energy Balance

$\frac{d\hat{E}}{dt} + \iint_{CS} (\hat{n} \cdot \underline{v}) \rho \hat{E}\, dS = Q_{in} - W_{by}$

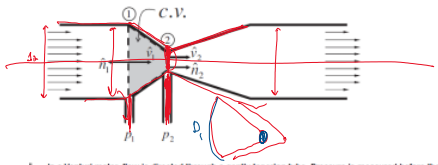
$\alpha = \frac{\langle v \rangle^2}{\langle v^2 \rangle}$; $\alpha \approx 1$ (turbulent)

$\frac{\Delta P}{\rho} + \frac{\Delta \langle v \rangle^2}{2\alpha} + g\Delta z + F = -\frac{W_{s,by}}{m}$

MEB

- → Open sys,
- → $E = E_K + E_p + U$
- → constant velocity across inlet/outlet
- → $\hat{H} = \hat{U} + P\hat{V}$
- → Flow work / shaft work
- → No phase change
- → SS, single input/single output
- → $\Delta T = 0$, no reaction

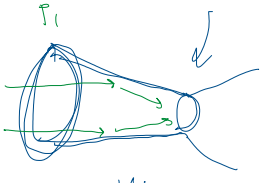
PROBLEM: 1.21 A Venturi meter with a 1.00 *in* diameter throat is to be installed into a 2 *in* line (Schedule 40 piping, smooth) with water flowing at 25°C. If the flow is turbulent and the range of expected flow rates is 0 – 200 gpm, what is the expected range of pressure drop in the Venturi meter? You may neglect friction.



In a Venturi meter, flow is directed through a gently tapering tube. Pressure is measured before the contraction (1) and at the point of smallest diameter (2) (i.e., the throat). The relationship between the measured pressures and the fluid velocity may be deduced from the mechanical energy balance with assistance of the macroscopic mass balance.

2" Schedule 40 pipe: ID = 2.067"

Pg 792 [Ex. 9.8]



(1) Velocity relationships (v_1 vs. v_2) ; v_1, D_1 ; v_2, D_2 — Mass balance

balance.

(1) Velocity relationships (v_1 vs. v_2)?

Mass balance

$$\frac{dm_{cv}}{dt}^{0} + \sum_i \rho_i A_i \cos\theta_i \langle v \rangle_i = 0$$

$$\rho_1 A_1 \cos\theta_1 \langle v_1 \rangle + \rho_2 A_2 \cos\theta_2 \langle v_2 \rangle = 0$$

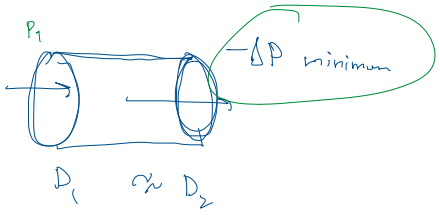
$$\rho_1 \frac{\pi D_1^2}{4} (-1) \langle v_1 \rangle + \rho_2 A_2 (1) \langle v_2 \rangle = 0$$

$$\rho_1 \frac{\pi D_1^2}{4} \langle v_1 \rangle = \rho_2 \frac{\pi D_2^2}{4} \langle v_2 \rangle$$

$$Q_1 = Q_2 \ (@SS)$$

$$\boxed{\langle v_1 \rangle = \left(\frac{D_2}{D_1}\right)^2 \langle v_2 \rangle}$$

vs.



$1 - 1 \to 0$

$$\frac{\Delta P}{\rho}^{\checkmark} + \Delta z^{0} + \frac{\Delta \langle v \rangle^2}{2\alpha} + F^{0} = -\frac{W_{s,by}}{m}^{0}$$

Ignore your $\langle v \rangle$

$$\frac{P_2 - P_1}{\rho} = -\frac{(\langle v_2 \rangle^2 - \langle v_1 \rangle^2)}{2} \quad ; \quad \alpha = 1 \ (\text{turbulent})$$

$$\text{Pressure drop} = -\Delta P = P_1 - P_2 = \frac{\rho V_2^2}{2}\left(1 - \left(\frac{D_2}{D_1}\right)^4\right)$$

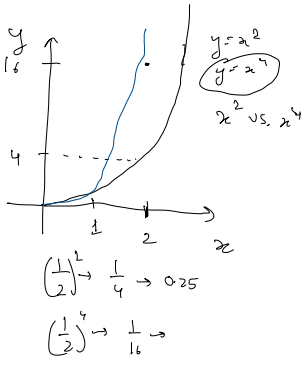
D_1^2 vs. D_2^4

D_2 ↑↑

$\simeq 0$

$P_1 - P_2 \simeq 0$

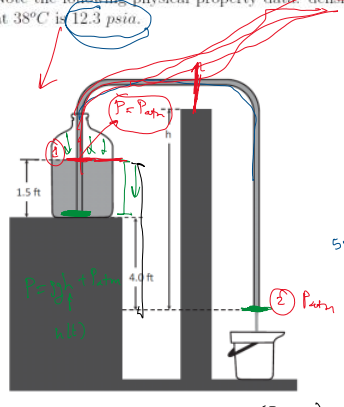
$P_1 = P_2$

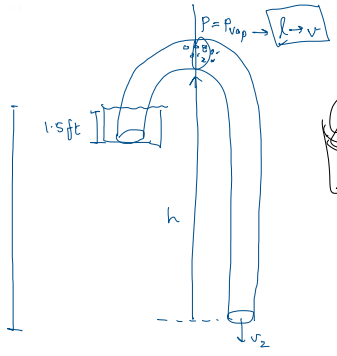


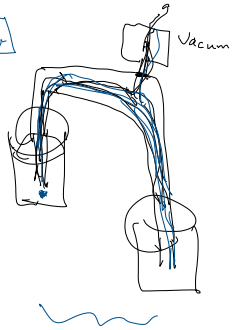
$\left(\frac{1}{2}\right)^2 \to \frac{1}{4} \to 0.25$

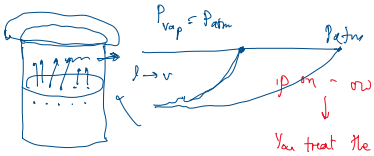
$\left(\frac{1}{2}\right)^4 \to \frac{1}{16} \to$

PROBLEM: 1.23 A gasoline tank is connected to a 25 *ft* hose (ID = 1.50 *cm*) as shown in Figure 1.49. The ambient temperature is 38°C. What is the maximum height of the barrier over which the gasoline may be siphoned? You may neglect frictional losses. Note the following physical property data: density of gasoline=5.6 lb_m/gal, vapor pressure at 38°C is 12.3 *psia*.









You treat the fluid as a continuum (no breaks) (free to choose the pressure that is the most convinient)

↓

As liquid level is falling (the pressure @ bottom still be changing)

↓

Hard to use that

(1) What is the exit velocity?

$$\frac{\Delta P}{\rho} + \frac{\Delta \langle v \rangle^2}{2\alpha} + g\Delta z + F^{0} = \frac{W_s}{m}^{0}$$

$(z_2 - z_1)$

$$\langle v_2 \rangle^2 - 0^2 + g(66.5) = 0$$

$\frac{P_1}{\rho} + \frac{v_1^2}{2\alpha} + gz_1 + h_f = \frac{v_2^2}{2\alpha}$

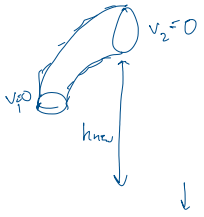
$\frac{v_2^2 - 0}{2} + g(66.5) = 0$

$\langle v_2 \rangle = \sqrt{-2g(-5.5)}$

$v_2 = ___$

② Max height: fluid will stop flowing when suction breaks (i.e. gas/air is introduced somehow)

↳ This can happen when the pressure equals vapor pressure (to "boil" the liquid)



$P_{vap} = 12.3$

$\frac{P_2 - P_1}{\rho} + \frac{v_2^2 - v_1^2}{2} + g(-h) = 0$

Solve for h

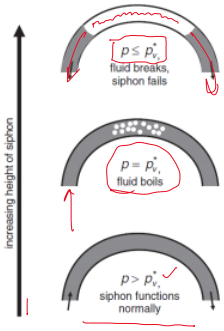
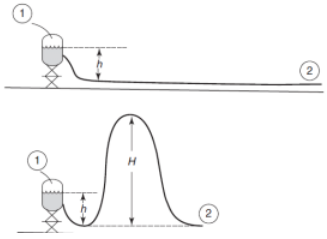


Figure 1.16 Schematic of the formation of a vapor gap in a siphon. This occurs when the barrier over which the fluid travels rises above a critical height; the critical height may be calculated as shown in the example.

PROBLEM: 1.24 A water tank is connected to a 100 *ft* hose (ID= 1.50 *cm*) as shown in the top of Figure 1.50. The height h is 1.8 *m*. Calculate the average velocity of water in the hose. Do not neglect friction; you may assume turbulent flow.

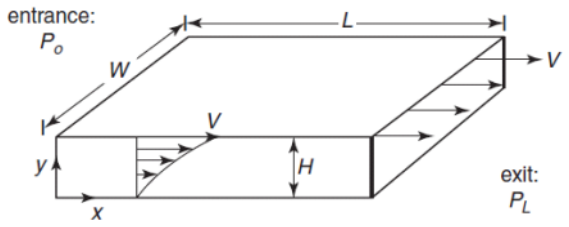


Week of 5th Dec.

Tuesday, December 5, 2023 10:43 AM

PROBLEM: 3.42 An incompressible fluid enters a converging bend flowing at a steady flow rate of Q gpm (Figure 3.42). The bend makes a 20^o turn and has a circular cross-section all along its length. At the inlet to the bend the radius of the conduit is R_1 and at the exit the radius is a smaller value, R_2. What is the average velocity of the fluid entering the conduit in terms of these variables? What is the average velocity of the fluid exiting the conduit?

PROBLEM: 6.41 Flow Problem: Combined backward pressure and forward drag of a Newtonian fluid in a slit. Calculate the velocity profile for the flow shown in Figure 6.21. The flow is steady flow of an incompressible, Newtonian fluid between two wide plates. The flow is driven forward by the motion of the top plate (the top plate moves in the x-direction at speed V), and the flow is opposed by the pressure, which is slightly higher at the exit,P_L, than at the entrance, P_0, $P_L > P_0$. You may neglect the effect of gravity. Use the coordinate system given in Figure 6.21.



PROBLEM: 9.19 What is the direction and magnitude of the force needed to support the 90^o expanding pipe bend shown in Figure 9.47? The water flow is steady and turbulent, and the cross-section of the inlet of the pipe bend is πR_1^2 and the cross-section of the outlet of the bend is πR_2^2, where $R_2 > R_1$. Evaluate your answer for $R_1 = 0.545\ in$ and $R_2 = 0.834\ in$ and various values of flow parameters.

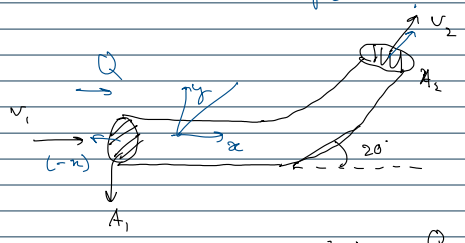
Week of 5th Dec.

Tuesday, December 5, 2023 10:43 AM

PROBLEM: 3.42 An incompressible fluid enters a converging bend flowing at a steady flow rate of Q gpm, (Figure 3.42). The bend makes a 20° turn and has a circular cross-section all along its length. At the inlet to the bend the radius of the conduit is R_1 and at the exit the radius is a smaller value R_2. What is the average velocity of the fluid entering the conduit in terms of these variables? What is the average velocity of the fluid exiting the conduit?

1) Draw diagram
2) Set up the info (w/ the coordinates)
3) knowns, unknowns

How can you solve?

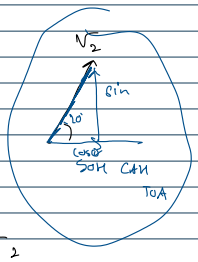


$Q_1 = Q_2$ ✓

$\langle v_1 \rangle = \frac{1}{\quad} \langle v_2 \rangle$ ✓

Q ✓ , ρ ✓

↓ Area



$\langle v_1 \rangle = \frac{Q}{A_1}$

$\langle v_1 \rangle = \frac{Q}{\pi R_1^2}$; $\langle v_2 \rangle = \frac{Q}{\pi R_2^2}$

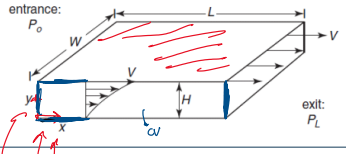
$\rho \cos\theta \langle v \rangle^2 \hat{e}_v$

$\hat{e}_{v_1} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}_{xy}$

$\hat{e}_{v_2} = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}_{xy}$

dt ρ Navier ✓

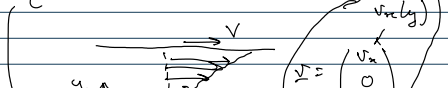
PROBLEM: 6.41 Flow Problem: Combined backward pressure and forward drag of a Newtonian fluid in a slit. Calculate the velocity profile for the flow shown in Figure 6.21. The flow is steady flow of an incompressible, Newtonian fluid between two wide plates. The flow is driven forward by the motion of the top plate (the top plate moves in the x-direction at speed V), and the flow is opposed by the pressure, which is slightly higher at the exit (P_L) than at the entrance, P_0. $P_L > P_0$. You may neglect the effect of gravity. Use the coordinate system given in Figure 6.21.



1) ✓

1) Mass balance ✓ Cartesian x, y

$\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = 0$



$\vec{v} = \begin{pmatrix} v_x \\ 0 \\ 0 \end{pmatrix}$

Equation of Continuity (Rectangular Coordinates)

$$\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot (\rho \vec{v}) = \frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}(\rho v_x) + \frac{\partial}{\partial y}(\rho v_y) + \frac{\partial}{\partial z}(\rho v_z) = 0$$

Equation of Continuity (Cylindrical Coordinates)

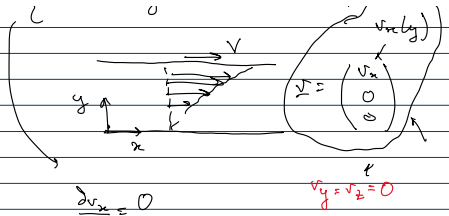
$$\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot (\rho \vec{v}) = \frac{\partial \rho}{\partial t} + \frac{1}{r}\frac{\partial}{\partial r}(\rho r v_r) + \frac{1}{r}\frac{\partial}{\partial \theta}(\rho v_\theta) + \frac{\partial}{\partial z}(\rho v_z) = 0$$

Equation of Continuity (Spherical Coordinates)

$$\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot (\rho \vec{v}) = \frac{\partial \rho}{\partial t} + \frac{1}{r^2}\frac{\partial}{\partial r}(\rho r^2 v_r) + \frac{1}{r \sin\theta}\frac{\partial}{\partial \theta}(\rho v_\theta \sin\theta) + \frac{1}{r \sin\theta}\frac{\partial}{\partial \phi}(\rho v_\phi) = 0$$

Cartesian coordinates:

$\rho\left(\frac{\partial v_x}{\partial t} + v_x\frac{\partial v_x}{\partial x} + v_y\frac{\partial v_x}{\partial y} + v_z\frac{\partial v_x}{\partial z}\right) = -\frac{\partial p}{\partial x} + \mu\left(\frac{\partial^2 v_x}{\partial x^2} + \frac{\partial^2 v_x}{\partial y^2} + \frac{\partial^2 v_x}{\partial z^2}\right) + \rho g_x$ B.7-1



$\frac{\partial v_x}{\partial x} = 0$ (Developed flow) ✓

Cartesian coordinates:

$$\rho\left(\frac{\partial v_x}{\partial t} + v_x\frac{\partial v_x}{\partial x} + v_y\frac{\partial v_x}{\partial y} + v_z\frac{\partial v_x}{\partial z}\right) = -\frac{\partial p}{\partial x} + \mu\left(\frac{\partial^2 v_x}{\partial x^2} + \frac{\partial^2 v_x}{\partial y^2} + \frac{\partial^2 v_x}{\partial z^2}\right) + \rho g_x \quad \text{B.7-1}$$

$$\rho\left(\frac{\partial v_y}{\partial t} + v_x\frac{\partial v_y}{\partial x} + v_y\frac{\partial v_y}{\partial y} + v_z\frac{\partial v_y}{\partial z}\right) = -\frac{\partial p}{\partial y} + \mu\left(\frac{\partial^2 v_y}{\partial x^2} + \frac{\partial^2 v_y}{\partial y^2} + \frac{\partial^2 v_y}{\partial z^2}\right) + \rho g_y \quad \text{B.7-2}$$

$$\rho\left(\frac{\partial v_z}{\partial t} + v_x\frac{\partial v_z}{\partial x} + v_y\frac{\partial v_z}{\partial y} + v_z\frac{\partial v_z}{\partial z}\right) = -\frac{\partial p}{\partial z} + \mu\left(\frac{\partial^2 v_z}{\partial x^2} + \frac{\partial^2 v_z}{\partial y^2} + \frac{\partial^2 v_z}{\partial z^2}\right) + \rho g_z \quad \text{B.7-3}$$

(2) Navier stokes:

x: $0 = -\frac{\partial p}{\partial x} + \mu\frac{\partial^2 v_x}{\partial y^2}$

$0 = -\frac{\partial p}{\partial y}$

$0 = -\frac{\partial p}{\partial z}$

P is not changing w/ y & z

$P \neq f(y,z)$

$\frac{\partial p}{\partial x} = \mu\frac{\partial^2 v_x}{\partial y^2} = \lambda$

P has no y depen — v_x has no x depen

$\frac{\partial p}{\partial x} = \lambda$

$\frac{\partial^2 v_x}{\partial y^2} = \frac{\lambda}{\mu}$

BC's:

@ $y=0$, $v_x = 0$, @ $x=0$, $P = P_0$

@ $y=h$, $v_x = V$, @ $x=L$, $P = P_L$

2nd order

$\frac{\partial^2 v_x}{\partial y^2} = \frac{\lambda}{\mu}$

$\frac{\partial v_x}{\partial y} = \Phi$

$\frac{\partial \Phi}{\partial y} =$

Integrate ↓

$\frac{\partial v_x}{\partial y} = \frac{\lambda}{\mu}y + C_1$

Integrate ↓

$v_x = \frac{\lambda}{2\mu}y^2 + C_1 y + C_2$

@ $y=0$, $v_x = 0 \rightarrow C_2 = 0$

@ $y=h$, $v_x = V \rightarrow V = \frac{\lambda}{2\mu}h^2 + C_1 h$

$\rightarrow C_1 = \left(V - \frac{\lambda h^2}{2\mu}\right)\frac{1}{h} = \left(\frac{V}{h} - \frac{\lambda h}{2\mu}\right)$

$$\boxed{v_x = \frac{\lambda}{2\mu}y^2 + \left(\frac{V}{h} - \frac{\lambda h}{2\mu}\right)y}$$

↳ input λ

1st order

$\frac{\partial P}{\partial x} = \frac{dP}{dx} = \lambda$

$P = \lambda x + C$

@ $x=0$, $C = P_0$

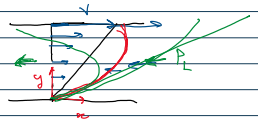
@ $x=L$, $P_L = \lambda L + P_0$

$\boxed{\lambda = \frac{P_L - P_0}{L}}$ ✓✓

$P = \lambda x + P_0$

$V = V_{max} \Rightarrow$??

@ max v $\rightarrow \frac{\partial v_x}{\partial y} = 0$

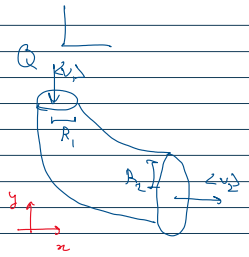


$\frac{\partial}{\partial y}(m) = \frac{\lambda}{\mu}y + \left(\frac{V}{h} - \frac{\lambda h}{2\mu}\right) = 0$

$= y = \left(\frac{\lambda h}{2\mu} - \frac{V}{h}\right)\cdot\frac{\mu}{\lambda}$

$$\boxed{y = \frac{h}{2} - \frac{V\mu}{\lambda h}} \rightarrow \neq h$$ ✓✓

PROBLEM: 9.19 What is the direction and magnitude of the force needed to support the 90^o expanding pipe bend shown in Figure 9.47? The water flow is steady and turbulent, and the cross-section of the inlet of the pipe bend is πR_1^2 and the cross-section of the outlet of the bend is πR_2^2, where $R_2 > R_1$. Evaluate your answer for $R_1 = 0.545\ in$ and $R_2 = 0.834\ in$ and various values of flow parameters.



(1) Macro

(2) force → momentum

velocity → Mass ($Q_1 = Q_2$)

$\langle v_2 \rangle$, A

CS

$$\frac{d\vec{P}}{dt} = \rho_1 A_1 \cos\theta \langle v_1 \rangle^2 \hat{e}_{v_1} + \rho_2 A_2 \cos\theta \langle v_2 \rangle^2 \hat{e}_{v_2} = \sum f_{on,CV} \quad \begin{pmatrix} 0 \\ 0 \\ -M_{cv} g \end{pmatrix}_{xyz}$$

$$\vec{f}_{in} + \vec{f}_{out} + \vec{f}_g + \vec{f}_w$$

$$\vec{f} = \iint_S (\hat{n} \cdot \bar{\bar{\pi}})_{comp} \, dS$$

$$\bar{\bar{\pi}} = -p\bar{\bar{I}} + \bar{\bar{\tau}}^{\,0}$$

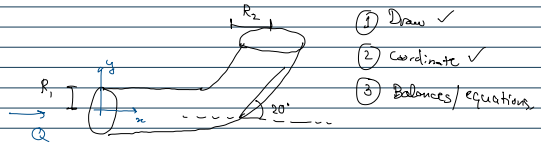
$$\vec{f}_{in} = -P_1 A_1 \hat{n}_1$$

$$\vec{f}_{out} = -P_2 A_2 \hat{n}_2$$

Week of 5th Dec.

Tuesday, December 5, 2023 10:43 AM

PROBLEM: 3.42 An incompressible fluid enters a converging bend flowing at a steady flow rate of Q gpm (Figure 3.42). The bend makes a 20° turn and has a circular cross-section all along its length. At the inlet to the bend the radius of the conduit is R_1 and at the exit the radius is a smaller value, R_2. What is the average velocity of the fluid entering the conduit in terms of these variables? What is the average velocity of the fluid exiting the conduit?



1. Draw ✓
2. Coordinate ✓
3. Balances / equations

$Q = Q_1 = Q_2$ ✓

$\langle v_1 \rangle \cdot A_1 = \langle v_2 \rangle A_2$

$Q = \langle v \rangle \cdot A$

$\langle v_1 \rangle = \frac{Q}{A_1} = \frac{Q}{\pi R_1^2}$ ✓

$\langle v_2 \rangle = \frac{Q}{A_2} = \frac{Q}{\pi R_2^2}$ ✓

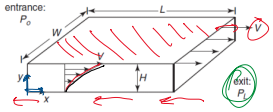
$\langle v \rangle = \frac{\iint v \cdot dA}{\iint dA}$

$v = (ax + bx^2 + c)\, e_x$

Cauchy = τ → $\tau = \mu \dot{\gamma}^{n-1}$

Navier stokes

PROBLEM: 6.41 Flow Problem: Combined backward pressure and forward drag of a Newtonian fluid in a slit. Calculate the velocity profile for the flow shown in Figure 6.21. The flow is steady flow of an incompressible Newtonian fluid between two wide plates. The flow is driven forward by the motion of the top plate (the top plate moves in the x-direction at speed V), and the flow is opposed by the pressure, which is slightly higher at the exit, P_L, than at the entrance, P_0, $P_L > P_0$. You may neglect the effect of gravity. Use the coordinate system given in Figure 6.21.



Equation of Continuity (Rectangular Coordinates)

$$\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot (\rho \vec{v}) = \frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}(\rho v_x) + \frac{\partial}{\partial y}(\rho v_y) + \frac{\partial}{\partial z}(\rho v_z) = 0$$

Equation of Continuity (Cylindrical Coordinates)

$$\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot (\rho \vec{v}) = \frac{\partial \rho}{\partial t} + \frac{1}{r}\frac{\partial}{\partial r}(\rho r v_r) + \frac{1}{r}\frac{\partial}{\partial \theta}(\rho v_\theta) + \frac{\partial}{\partial z}(\rho v_z) = 0$$

Equation of Continuity (Spherical Coordinates)

$$\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot (\rho \vec{v}) = \frac{\partial \rho}{\partial t} + \frac{1}{r^2}\frac{\partial}{\partial r}(\rho r^2 v_r) + \frac{1}{r \sin\theta}\frac{\partial}{\partial \theta}(\rho v_\theta \sin\theta) + \frac{1}{r \sin\theta}\frac{\partial}{\partial \phi}(\rho v_\phi) = 0$$

1. Draw / study diagram
2. Coordinates ✓
3. Cont & flow Navier ✓
   - v, p

$v_y = v_z = 0$

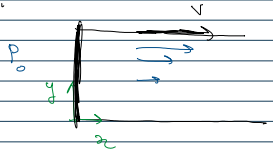
Cartesian coordinates

$$\rho\left(\frac{\partial v_x}{\partial t} + v_x\frac{\partial v_x}{\partial x} + v_y\frac{\partial v_x}{\partial y} + v_z\frac{\partial v_x}{\partial z}\right) = -\frac{\partial p}{\partial x} + \mu\left(\frac{\partial^2 v_x}{\partial x^2} + \frac{\partial^2 v_x}{\partial y^2} + \frac{\partial^2 v_x}{\partial z^2}\right) + \rho g_x \quad \text{B.7-1}$$

$$\rho\left(\frac{\partial v_y}{\partial t} + v_x\frac{\partial v_y}{\partial x} + v_y\frac{\partial v_y}{\partial y} + v_z\frac{\partial v_y}{\partial z}\right) = -\frac{\partial p}{\partial y} + \mu\left(\frac{\partial^2 v_y}{\partial x^2} + \frac{\partial^2 v_y}{\partial y^2} + \frac{\partial^2 v_y}{\partial z^2}\right) + \rho g_y \quad \text{B.7-2}$$

$$\rho\left(\frac{\partial v_z}{\partial t} + v_x\frac{\partial v_z}{\partial x} + v_y\frac{\partial v_z}{\partial y} + v_z\frac{\partial v_z}{\partial z}\right) = -\frac{\partial p}{\partial z} + \mu\left(\frac{\partial^2 v_z}{\partial x^2} + \frac{\partial^2 v_z}{\partial y^2} + \frac{\partial^2 v_z}{\partial z^2}\right) + \rho g_z \quad \text{B.7-3}$$

velocity ??



$\underline{v} = \begin{pmatrix} v_x \\ 0 \\ 0 \end{pmatrix}$ $v_x = f(y)$

$v_y = v_z = 0$

(1) Mass: $\nabla \cdot \underline{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = 0 \rightarrow \frac{\partial v_x}{\partial x} = 0$ (Developed flow) ✓

(2) Navier: $-\frac{\partial P}{\partial x} + \mu \frac{\partial^2 v_x}{\partial y^2} = 0$

PROBLEM: 9.19 What is the direction and magnitude of the force needed to support the 90° expanding pipe bend shown in Figure 9.47? The water flow is steady and turbulent, and the cross-section of the inlet of the pipe bend is πR_1^2 and the cross-section of the outlet of the bend is πR_2^2, where $R_2 > R_1$. Evaluate your answer for $R_1 = 0.545\ in$ and $R_2 = 0.834\ in$ and various values of flow parameters.